

nPM1300

Product Specification

v1.3

nPM1300

nPM1300 is a highly integrated Power Management IC (PMIC) for rechargeable applications. It is design compatible with an nRF52, nRF53, or nRF54 Series System on Chip (SoC), nRF91 Series System in Package (SiP) and third party host devices for developing low-power wireless solutions.

nPM1300 has several power and system management features that can be implemented with dedicated components. Power management is achieved through flexible power regulation and a linear-mode lithium-ion (Li-ion), lithium-polymer (Li-poly), and lithium iron phosphate (LiFePO₄) battery charger in a compact 3.1x2.4 mm WLCSP or 5x5 mm QFN32 package. A minimum of 5 passive components are required.

nPM1300 supports charging up to 800 mA and delivers up to 500 mA of adjustable regulated voltage. Power is supplied to external components from two configurable, dual mode 200 mA BUCK regulators, and two dual purpose 50 mA LDO/100 mA load switches. In addition, an unregulated power rail, VSYS, delivers up to 1 A when powered from battery, or up to 1.5 A when powered from a USB port configured as DCP. The maximum total current on VSYS is 1.5 A.

The host can read battery temperature, voltage, and current, which are utilized by a fuel gauge algorithm in the nRF Connect Software Development Kit. The fuel gauge provides the application with a battery state-of-charge estimate comparable to Coulomb counters at a significantly lower power consumption.

Low quiescent current (IQ) extends battery life during shipping and storage with Ship mode. Battery life can also be extended during operation with auto-controlled Hysteretic mode for high efficiency down to 1 μ A load currents.

The integrated system management features reduce the cost and size of applications. The following integrated features are found in the device:

- System-level watchdog
- Intelligent power-loss warning
- Ship and Hibernate modes for increased battery life
- Up to 5 GPIO pins and 3 LED drivers
- System Monitor
- Ultra-low power, high accuracy fuel gauge tailored for embedded IoT applications

System management features and I/Os are configured through an I²C compatible two-wire Interface (TWI).

The nPM1300 Evaluation Kit allows for simple evaluation and code-free configuration of nPM1300. By connecting to the nPM PowerUP app found in nRF Connect for Desktop, the nPM1300 settings can easily be configured through an intuitive GUI and exported as code to be implemented in your MCU's application.

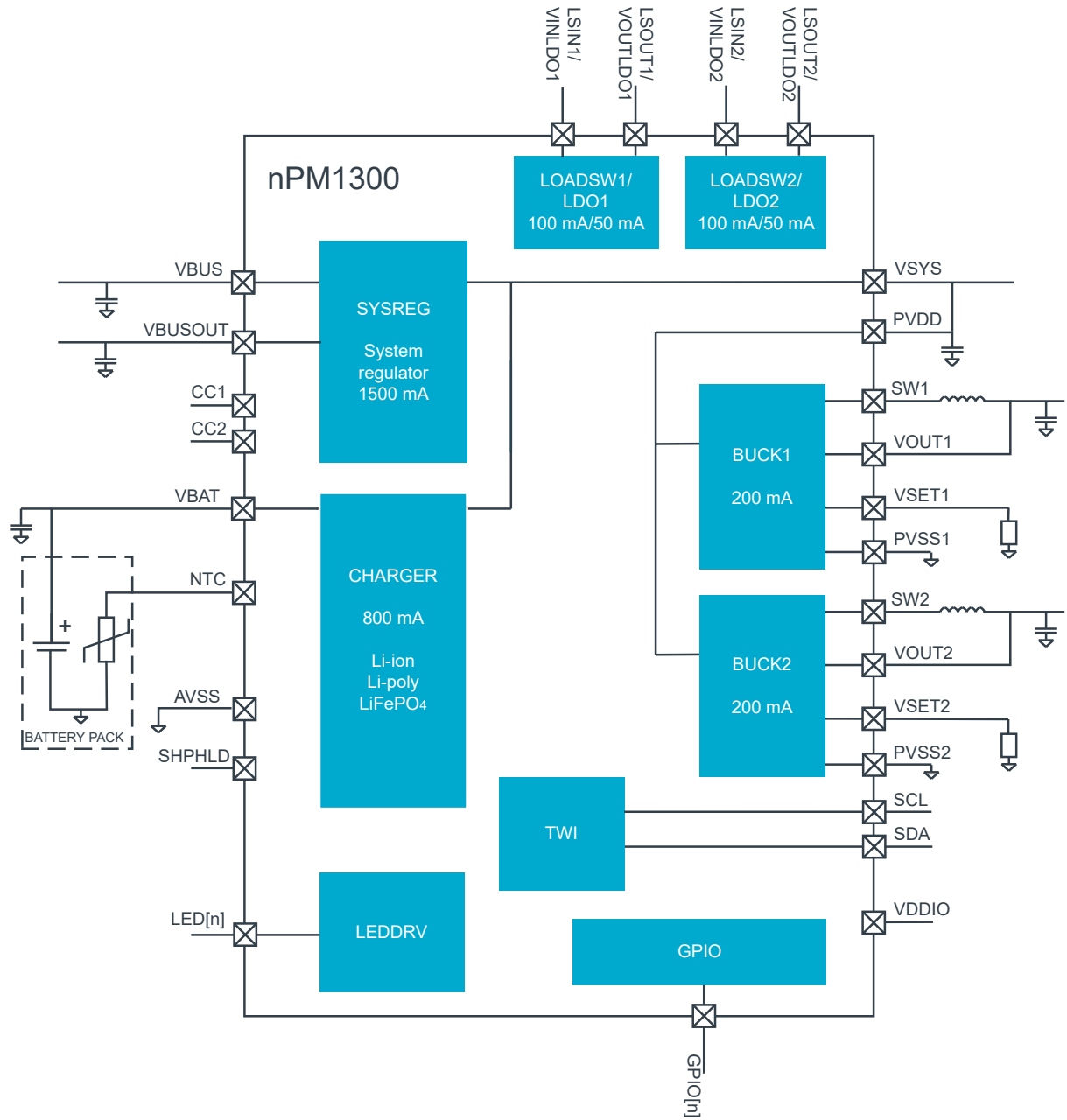


Figure 1: nPM1300

Key features

Features:

- 800 mA linear battery charger
 - Linear charger for lithium-ion, lithium-polymer, and lithium iron phosphate batteries
 - Configurable charge current from 32 mA to 800 mA
 - Charging termination voltage from 3.5 V to 4.45 V
 - Configurable thermal regulation
 - JEITA compliant
 - Dynamic power-path management
- Input current limiter
 - USB Type-C compliant
 - 4.0 V to 5.5 V operational input voltage range
 - 22 V tolerant
- Two 200 mA buck regulators
 - Automatic transition between Hysteretic and pulse width modulation (PWM) modes
 - Forced PWM mode for low-ripple operation
 - Pin-selectable output voltage
- Ultra-low power, high accuracy fuel gauge tailored for embedded IoT applications
 - Battery voltage, current and temperature monitored by nPM1300
 - Nordic Fuel Gauge algorithm running on host System-on-Chip/MCU
- Two 50 mA LDO/100 mA load switches
- I²C compatible TWI for control and monitoring
- 10-bit ADC for system monitoring
 - Measures VBUS voltage, battery voltage, current, and die temperature
- Three pre-configured and programmable 5 mA low-side LED drivers
- Configurable timer
 - Boot monitor
 - Watchdog timer with selectable reset or power cycling
 - Wake-up timer
 - General purpose timer
- Power-fail warning (POF)
- Configurable hard reset
- General purpose GPIOs that can control BUCKs, load switches, interrupt output, reset, power fail warning, or as a general purpose I/O
- Seamless integration and code free configuration with the nPM1300 Evaluation Kit and nPM PowerUp desktop app
- Package options available:
 - 3.1x2.4 mm WLCSP package
 - 5.0x5.0 mm QFN package

Applications:

- Wearables
 - Health/fitness sensor and monitoring devices
- Computer peripherals and I/O devices
 - Mouse
 - Keyboard
 - Multi-touch trackpad
- Asset trackers
- Interactive entertainment devices
 - Remote controls
 - Gaming controllers
- IoT applications
 - Smart/low-energy sensors
 - Loggers
 - Actuator controls

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1 Revision history

Date	Version	Description
March 2026	1.3	The following has been updated: • Product overview – Maximum current for VSYS (unregulated power rail); System electrical specification • LOADSW/LDO – Electrical specification • CHARGER – Values for IBATLIM_{LOW} and IBATLIM_{HIGH} • Dynamic power-path management – Description and graphs in Electrical characteristics • Registers • BCHGVTERM • BCHGVTERM_R • ADCIBATMEASSTATUS • GPIOPDEN[n] • Editorial
December 2025	1.2.1	LOADSW/LDO – Corrected to reflect functionality
March 2025	1.2	The following has been added or updated: • CHARGER – Added register BCHGVBATLOWCHARGE; added information for enabling charging for low voltage batteries • LOADSW/LDO – Added value to LDSWSTATUS register • POF – Updated signal polarity; electrical specification • System Monitor – Updated equations • Battery detection has been disabled • Editorial
July 2024	1.1	The following has been added or updated: • Battery current measurement on page 104 • Battery discharge current limit on page 32
October 2023	1.0	First release

2 About this document

This document is organized into chapters that are based on the modules available in the IC.

2.1 Document status

The document status reflects the level of maturity of the document.

Document name	Description
Objective Product Specification (OPS)	Applies to document versions up to 1.0. This document contains target specifications for product development.
Product Specification (PS)	Applies to document versions 1.0 and higher. This document contains final product specifications. Nordic Semiconductor ASA reserves the right to make changes at any time without notice in order to improve design and supply the best possible product.

Table 1: Defined document names

2.2 Core component chapters

Every core component has a unique capitalized name or an abbreviation of its name, e.g. LED, used for identification and reference. This name is used in chapter headings and references, and it will appear in the C-code header file to identify the component.

The core component instance name, which is different from the core component name, is constructed using the core component name followed by a numbered postfix, starting with 0, for example, LED0. A postfix is normally only used if a core component can be instantiated more than once. The core component instance name is also used in the C-code header file to identify the core component instance.

The chapters describing core components may include the following information:

- A detailed functional description of the core component
- Register configuration for the core component
- Electrical specification tables, containing performance data which apply for the operating conditions described in [Recommended operating conditions](#) on page 18.

3 Product overview

nPM1300 is a highly integrated Power Management IC (PMIC) for rechargeable applications. It is design compatible with an nRF52, nRF53, or nRF54 Series System on Chip (SoC), nRF91 Series System in Package (SiP), and third party host devices for developing low-power wireless solutions.

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Low quiescent current (IQ) extends battery life during shipping and storage with Ship mode. Battery life can also be extended during operation with auto-controlled Hysteretic mode for high efficiency down to 1 µA load currents.

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System management features and I/Os are configured through an I²C compatible two-wire Interface (TWI).

The nPM1300 Evaluation Kit provides simple evaluation and code-free configuration of nPM1300. Connecting to the nPM PowerUP app found in nRF Connect for Desktop enables the nPM1300 settings to be easily configured through an intuitive GUI and exported as code to be implemented in your MCU's application.

3.1 Block diagram

The block diagram illustrates the overall system.

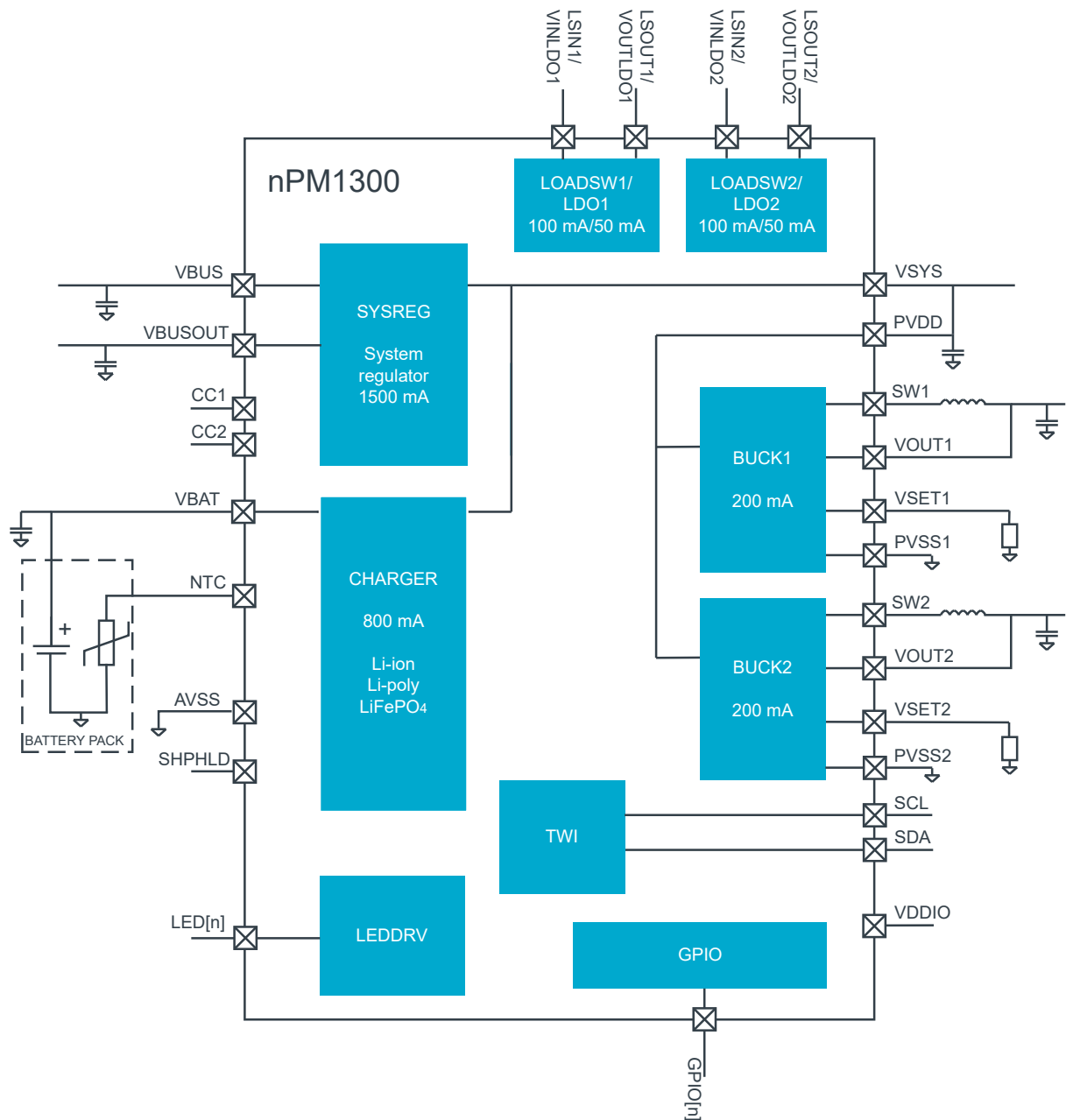


Figure 2: Block diagram

3.1.1 In-circuit configurations

The device is configurable for different applications and battery characteristics through input pins.

The following pins must be configured before power-on reset. For the full pin list, see [Pin assignments](#) on page 154.

Pin	Function	Reference
VDDIO	Supply for the TWI control interface and GPIOs	Interface supply on page 125, GPIO — General purpose input/output on page 89
VSET1	BUCK1 enable and VOUT1 voltage level selection at power-on reset	BUCK — Buck regulators on page 49
VSET2	BUCK2 enable and VOUT2 voltage level selection at power-on reset	BUCK — Buck regulators on page 49
CC1, CC2	USB charger detection (USB Type-C)	USB port detection on page 20

Table 2: In-circuit configurations

3.2 System description

The device has the following core components that are described in detail in their respective chapters.

- [SYSREG — System regulator](#) on page 20
- [CHARGER — Battery charger](#) on page 26
- [BUCK — Buck regulators](#) on page 49
- [LOADSW/LDO — Load switches/LDO regulators](#) on page 74
- [LEDDRV — LED drivers](#) on page 85
- [GPIO — General purpose input/output](#) on page 89

The system regulator (SYSREG) is supplied by VBUS. It supports 4.0 V to 5.5 V for internal functions and tolerates transient voltages up to 22 V. Overvoltage protection is implemented for both internal and external circuitry. SYSREG also implements current limiting for VBUS to comply with the USB Type-C specification. SYSREG supports Type-C charger detection.

The battery charger (CHARGER) is a JEITA compliant linear battery charger for lithium-ion (Li-ion), lithium-polymer (Li-poly), and lithium iron phosphate (LiFePO₄) batteries. CHARGER controls the charge cycle using a standard Li-ion charge profile. CHARGER implements dynamic power-path management regulating current in and out of the battery, depending on system requirements, to ensure immediate system operation from **VBUS** if the battery is depleted. Safety features, such as [battery temperature monitoring](#) and [charger thermal regulation](#) are supported.

Two independent, highly efficient buck regulators (BUCK) supply the application circuitry and offer several output voltage options. BUCK is controlled through registers or GPIO pins. Default output voltage can be set with external resistors.

The two load switches (LOADSW/LDO) can function as switches or linear voltage regulators to complement the power distribution network. LOADSW/LDO is controlled through registers or GPIO pins.

The System Monitor provides measurements for battery voltage, battery current, VBUS, battery, and die temperature.

GPIO has the following configurable features:

- General purpose input
- Control input
- Output
- BUCK[n] control
- LOADSW[n] control

The device also features [Ship and Hibernate modes](#), the lowest quiescent current states. They disconnect the battery from the system and reduce the quiescent current of the device to extend battery life. Hibernate mode can be utilized during normal operation as the device can autonomously wake-up after a preconfigured timeout. This makes it possible to extend battery life to the maximum capacity.

3.3 Power-on reset (POR) and brownout reset (BOR)

The device is supplied by **VBUS** or **VBAT**.

When one of the following conditions are met, a power-on reset (POR) occurs.

- **VBUS** > **VBUS_{POR}**
- **VBAT** > **VBAT_{POR}**

When both of the following conditions are met, a brownout reset (BOR) occurs.

- **VBUS** < **VBUS_{BOR}**
- **VBAT** < **VBAT_{BOR}**

3.4 Supported battery types

The charger supports rechargeable Li-ion, Li-polymer, or LiFePO₄ batteries.

Battery packs connected to the **VBAT** pin must contain the following protection circuitry:

- Overvoltage protection
- Undervoltage protection
- Overcurrent discharge protection
- Thermal fuse to protect from overtemperature (if NTC thermistor is not present)

3.5 Thermal protection

A global thermal shutdown is triggered when the die temperature exceeds the operating temperature range, see [TSD](#). All device functions are disabled in thermal shutdown. The device functions are re-enabled when the temperature is sufficiently reduced according to a hysteresis [TSD_{HYST}](#). The die temperature limit is only monitored when charging is enabled or when a BUCK is enabled and is in PWM mode.

A secondary mechanism disables the charger when the die reaches the host software programmable temperature of [DIETEMPSTOP](#) on page 45 . Once this temperature is reached, charging stops but all other functionality remains active. Charging restarts when the die temperature reaches the host software programmable temperature of [DIETEMPRESUME](#) on page 46.

3.6 System efficiency

Shown here is the characterization of the power path system efficiency under different load current conditions.

In the following figure, the load current is swept from 100 nA to 200 mA and back to capture mode change hysteresis.

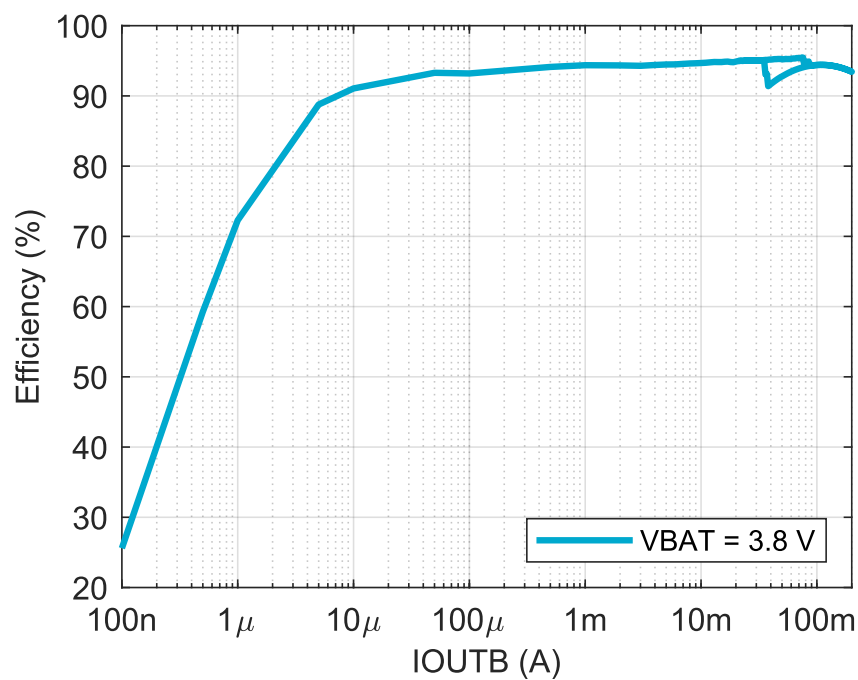


Figure 3: $V_{OUT} = 3.3\text{ V}$ system efficiency, $MODE = AUTO$, $V_{IN} = 3.8\text{ V}$

3.7 Electrical characteristics

The following graphs show quiescent current characteristics.

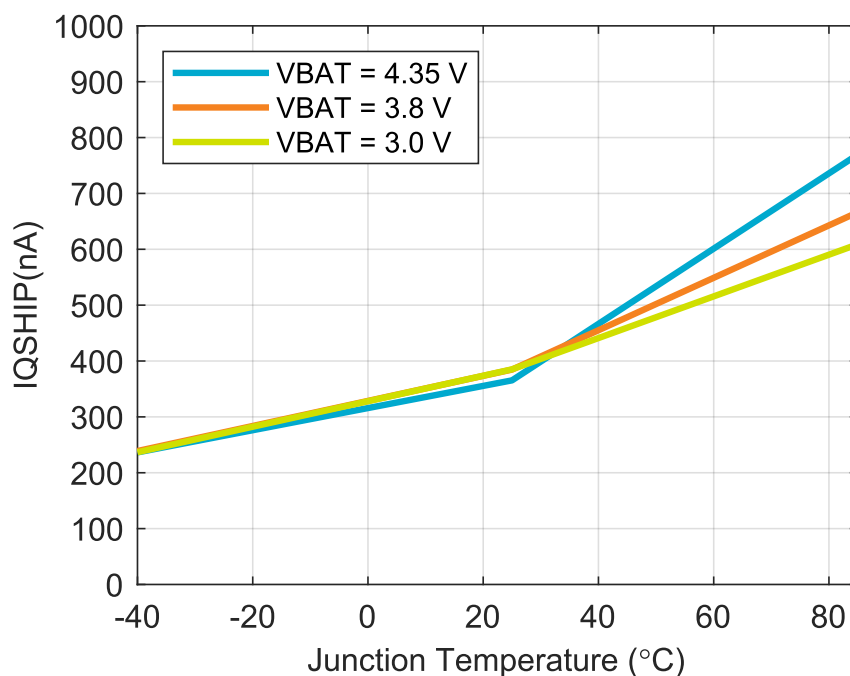


Figure 4: SHIP mode current vs. junction temperature

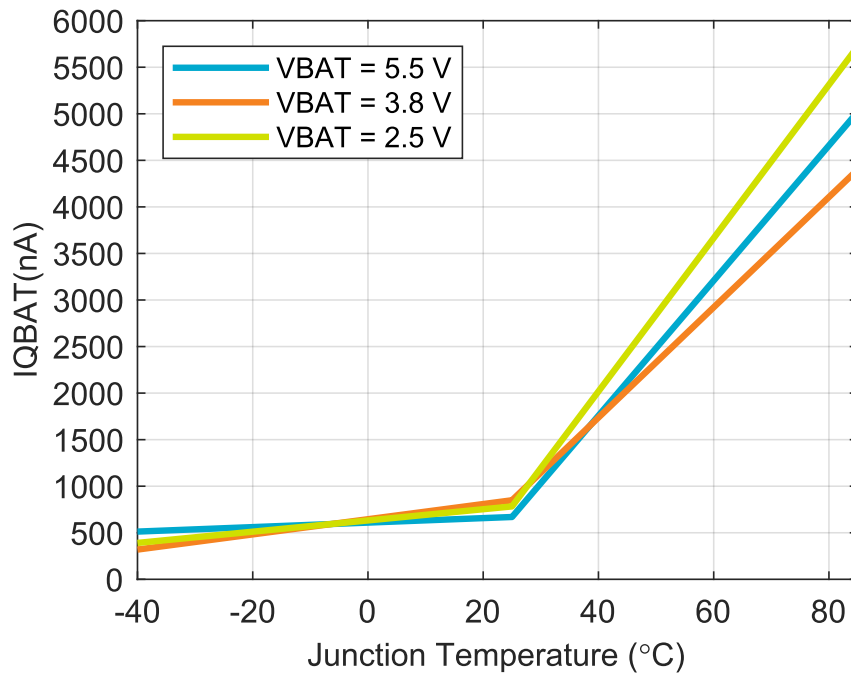


Figure 5: Discharge mode current vs. junction temperature

3.8 System electrical specification

Symbol	Description	Min.	Typ.	Max.	Unit
I_{Q_SHIP}	Ship mode quiescent current		370		nA
I_{Q_SHIPT}	Hibernate mode quiescent current		500		nA
I_{Q_BAT}	Quiescent current, battery operation, all BUCKs and LDOs disabled, VBUS disconnected		600		nA
	Quiescent current, battery operation, one BUCK enabled in Auto mode, no load, VBUS disconnected		800		nA
	Quiescent current, battery operation, both BUCKs enabled in Auto mode, no load, VBUS disconnected		1100		nA
	Quiescent current, battery operation, one BUCK enabled in PWM mode, VOUT=1.8 V, no load, VBUS disconnected		4.0		mA
	Quiescent current, battery operation, both BUCKs enabled in PWM mode, VOUT=1.8 V, no load, VBUS disconnected		7.1		mA
TSD	Thermal shutdown threshold		120		°C
TSD _{HYST}	Thermal shutdown hysteresis		20		°C
IOUT _{VSYS}	Maximum VSYS DC load current when powered from battery (VBUS=0 V, VSYS>VSYS _{POF})			1.0	A
	Maximum VSYS DC load current when powered from VBUS, or from both VBUS and battery (VBUS=5 V, VSYS>VSYS _{POF})			1.5	A

Table 3: System electrical specification

4 Absolute maximum ratings

Maximum ratings are the extreme limits to which the device can be exposed for a limited amount of time without permanently damaging it. Exposure to absolute maximum ratings for prolonged periods of time may affect the reliability of the device.

Pin	Note	Min.	Max.	Unit
VBUS	Power (relative to pins AVSS , PVSS1 , and PVSS2)	-0.3	22	V
VBAT , VSYS , PVDD , VDDIO	Power (relative to pins AVSS , PVSS1 , and PVSS2)	-0.3	5.5	V
NTC , CC1 , CC2 , SHPHLD , LED0 , LED1 , LED2 , LSIN1/VINLDO1 , LSOUT1/VOU TLDO1 , LSIN2/VINLDO2 , LSOUT2/VOU TLDO2 , VSET1 , VSET2 , VBUSOUT , VOUT1 , VOUT2 , SW1 , SW2	Analog pins (relative to pins AVSS , PVSS1 , and PVSS2)	-0.3	5.5	V
GPIO [0 . . 4] , SDA , SCL	Digital pins (relative to pins AVSS , PVSS1 , and PVSS2)	-0.3	VDDIO+0.3	V

Table 4: Absolute maximum ratings

	Note	Min.	Max.	Unit
Storage temperature		-40	+125	°C
MSL QFN	Moisture sensitivity level		2	
MSL CSP	Moisture sensitivity level		1	
ESD HBM	Human Body Model Class 2		2	kV
ESD CDM	Charged Device Model		500	V

Table 5: Environmental ratings



5 Recommended operating conditions

The operating conditions are the physical parameters that the chip can operate within.

Symbol	Description	Min.	Max	Unit
V _{BUS} _{OP}	Supply voltage	4.0	5.5	V
V _{BAT} _{OP}	Battery voltage	2.3	4.45	V
V _{DDIO}	I/O supply voltage	1.7	V _{SYS}	V
T _J	Junction temperature	-40	+125	°C
T _A	Ambient temperature	-40	+85	°C

Table 6: Recommended operating conditions

Note: Any system features powered by V_{SYS} will only operate when the V_{SYS} voltage > V_{SYS}_{POF}.

5.1 Dissipation ratings

Thermal resistances and thermal characterization parameters as defined by JESD51-7 are shown in the following tables.

Symbol	Parameter	QFN 32 pins	Units
R _{θJA}	Junction-to-ambient thermal resistance	24.2	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	10.7	°C/W
R _{θJB}	Junction-to-board thermal resistance	8.8	°C/W
Ψ _{JT}	Junction-to-top characterization parameter	0.15	°C/W
Ψ _{JB}	Junction-to-board characterization parameter	8.9	°C/W

Table 7: QFN32 thermal resistance and characterization parameters

Symbol	Parameter	CSP 35 pins	Units
R _{θJA}	Junction-to-ambient thermal resistance	48.3	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	6.0	°C/W
R _{θJB}	Junction-to-board thermal resistance	23.0	°C/W
Ψ _{JT}	Junction-to-top characterization parameter	0.5	°C/W
Ψ _{JB}	Junction-to-board characterization parameter	23.4	°C/W

Table 8: CSP thermal resistance and characterization parameters

5.2 CSP light sensitivity

The CSP package is sensitive to light.

All CSP package variants are sensitive to visible and close-range infrared light. The final product design must shield the chip through encapsulation or shielding/coating the CSP device.

CSP package variant CAAA has a backside coating that covers the marking side of the device with a light absorbing film. The side edges and the ball side of the device are still exposed and need to be protected.

6 Core Components

6.1 SYSREG — System regulator

VBUS supplies the input voltage to the system regulator (SYSREG). VBUS voltage is supplied by an AC wall adapter or a USB port.

SYSREG supplies VSYS.

Features of SYSREG are the following:

- Operating voltage up to 5.5 V
- Overvoltage protection to 22 V
- Undervoltage detection
- USB port detection and a current limiter to comply with the USB specification
- Provides **VBUSOUT** voltage for host devices

6.1.1 VBUS input current limiter

The VBUS input current limiter manages VBUS current limitation and charger detection for USB Type-C compatible chargers.

It supplies **VSYS** but does not regulate its voltage. VBUS voltage is seen at **VSYS** as a supply, if the VBUS voltage is within specified limits and higher than VBAT by at least **VBUS_{VALID}**.

There are two USB compliant, accurate current limits: **IBUS_{100MA}** (100 mA) and **IBUS_{500MA}** (500 mA).

In addition, there are current limits in 100 mA steps from 600 mA to 1500 mA. The 1500 mA limit is compatible with USB Type-C.

The default current limit is **IBUS_{100MA}** (100 mA). Host software can configure the current in register **VBUSINLIMO** on page 24. When **TASK.UPDATE.ILIMSW** is written, **VBUSIN.LIMO** takes effect.

6.1.2 VBUS overvoltage protection

The overvoltage threshold for **VBUS** is **VBUS_{OVP}**. The undervoltage threshold for **VBUS** is **VBUS_{MIN}**.

SYSREG is disabled when **VBUS_{voltage}** is above the overvoltage threshold **VBUS_{OVP}**, or below the undervoltage threshold **VBUS_{MIN}**. This isolates **VBUS** and prevents current flowing from **VSYS** to **VBUS**.

6.1.3 USB port detection

USB charger detection is performed through pins **CC1** and **CC2**. These pins must be connected directly to the USB connector for detection to happen.

These pins have internal pull-downs with resistance equal to **R_d**.

When the device is plugged into a wall adaptor or USB power source, USB port detection runs automatically. One of the CC lines is connected to a pull-up at the source. The other CC line stays pulled down. The voltage over the corresponding **R_d** determines if a connection was made and if SYSREG can deliver 500 mA or higher current.

Comparators with thresholds at **V_{RDCONN}**, **V_{RD1A5}**, and **V_{RD3A}** monitor CC line voltage when VBUS is present. All comparator output is debounced with **t_{RDDEB}** and available to host software through register **USBCDETECTSTATUS** on page 25.

If enabled, an interrupt is issued to the host whenever a threshold is crossed (when voltage decreases or increases). The events are visible in register **EVENTSVBUSIN1SET** on page 144.

The USB power source capability is detected by one CC line at a time, depending on the orientation of the USB plug on the device. The other CC line remains at 0 V. The charger type is defined in the **VBUSIN.CC1CMP** or **VBUSIN.CC2CMP** field, depending on which pin is used for connection.

The default VBUS current limit of 100 mA is used until the power source capability is detected. Host software can update the VBUS current limit in **VBUSINILIM0** on page 24 after device detection. When a USB cable is unplugged and plugged back in, or a reset occurs, the default current limit is used.

When **TASK.UPDATE.ILIMSW** is written, **VBUSIN.LIM0** takes effect. The VBUS current limit reverts to its default value (100 mA) when the following occur:

- A reset
- The USB cable is unplugged and plugged back in

If USB Type-C configuration is not used, **CC1** and **CC2** can be left floating or connected to ground. The default VBUS current limit will remain at 100 mA until the host negotiates and configures a higher current.

Note: Overvoltage or undervoltage events may occur when connecting or removing a supply to **VBUS**.

6.1.4 USB2.0 Selective Suspend

The device can satisfy USB2.0 Selective Suspend mode current consumption through configuration. It must be informed by host software through the TWI in register **VBUSSUSPEND** on page 25 to minimize current consumption from VBUS to **I_{SUSP}**.

The current consumed through pin **VBUSOUT** is not included. VBUS is disconnected from VSYS but **VBUSOUT** remains active. As a consequence, charging is paused. The device exits this mode only when instructed by the host software through a TWI command. Charging resumes automatically.

6.1.5 VBUSOUT

The device supplies **VBUSOUT** voltage when **VBUS** voltage is present and higher than VBAT by at least **VBUS_{VALID}**.

VBUSOUT provides overvoltage and undervoltage protection for safe connection to the nRF device. Designs using the **VBUSOUT** pin as a supply must make sure the voltage level complies with the nRF device due to output resistance **R_{VBUSOUT}**. When USB is suspended, the combined current for nPM1300 and the **VBUSOUT** pin must be within the allowed USB suspend current.

VBUSOUT must have a decoupling capacitor. The capacitor is discharged by **R_{PDVBUSOUT}** once VBUS has been disconnected.

6.1.6 Electrical specification

Symbol	Description	Min.	Typ.	Max.	Unit
VBUS _{OVP}	Overvoltage protection threshold		5.5		V
VBUS _{VALIDR}	VBUS valid threshold, VBUS - VBAT, VBUS rising		210		mV
VBUS _{VALIDF}	VBUS valid threshold, VBUS - VBAT, VBUS falling or VBAT rising		160		mV
VBUS _{POR}	Power-on reset release voltage for VBUS		3.9		V
VBUS _{BOR}	Brownout reset trigger for VBUS with VBAT not present		3.8		V
VBUS _{MIN}	Undervoltage threshold with VBAT present		3.6		V
IBUS _{100MA}	VBUS input current limit, 100 mA ¹			95	mA
IBUS _{500MA}	VBUS input current limit, 500 mA ¹			495	mA
IBUS _{LIMACC}	Accuracy of IBUS current limit (steps from 600 to 1500 mA) ¹	-10		+10	%
I _{SUSP}	VBUS current consumption in suspend mode Current from VBUSOUT is excluded		1.8		mA
R _{ON}	Resistance between VBUS and VSYS VBUSINLIM0 = 15 (1.5 A) VBUS = 5 V		300		mΩ
R _{VBUSOUT}	On resistance of the VBUSOUT switch VBUS = 5.0 V		7.5		Ω
R _d	Pull-down resistance on pins CC1 and CC2		5.1		kΩ
V _{RDCONN}	Threshold to detect connection		0.2		V
V _{RD1A5}	Threshold to detect charger type on CC1 or CC2 pins		0.66		V
V _{RD3A}	Threshold for 3 A current limit		1.23		V
R _{PDVBUSOUT}	Pull-down resistor on VBUSOUT pin when VBUS is disconnected.		35		kΩ
t _{RDDEB}	Debounce time for CC voltage level detection		15		ms

Table 9: SYSREG electrical specification

¹Includes internal device consumption and current flowing through pin VBUSOUT.

6.1.7 Electrical characteristics

The following graphs show typical electrical characteristics for VBUSIN.

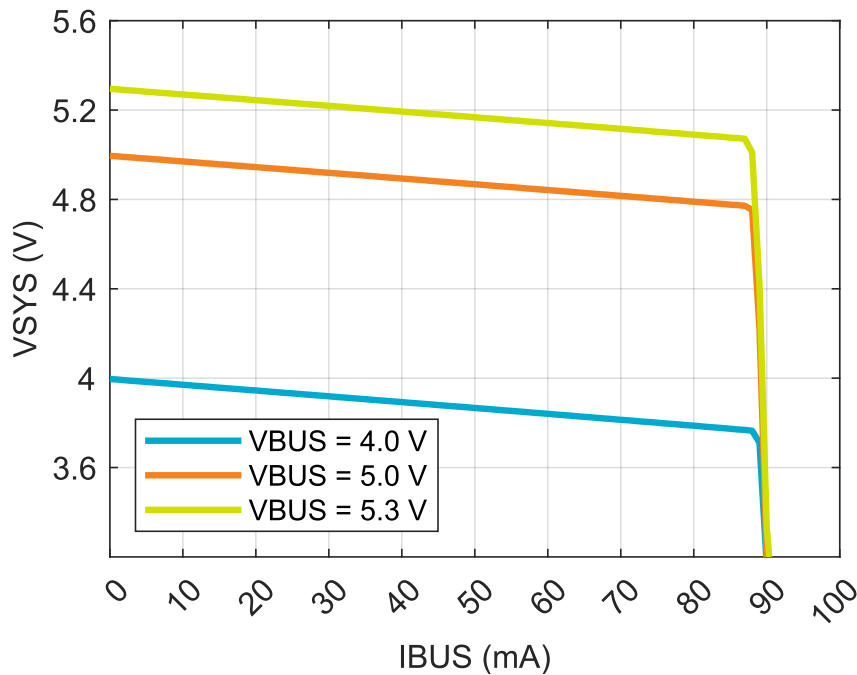


Figure 6: VSYS voltage vs. VBUS current, ILIM = 100 mA

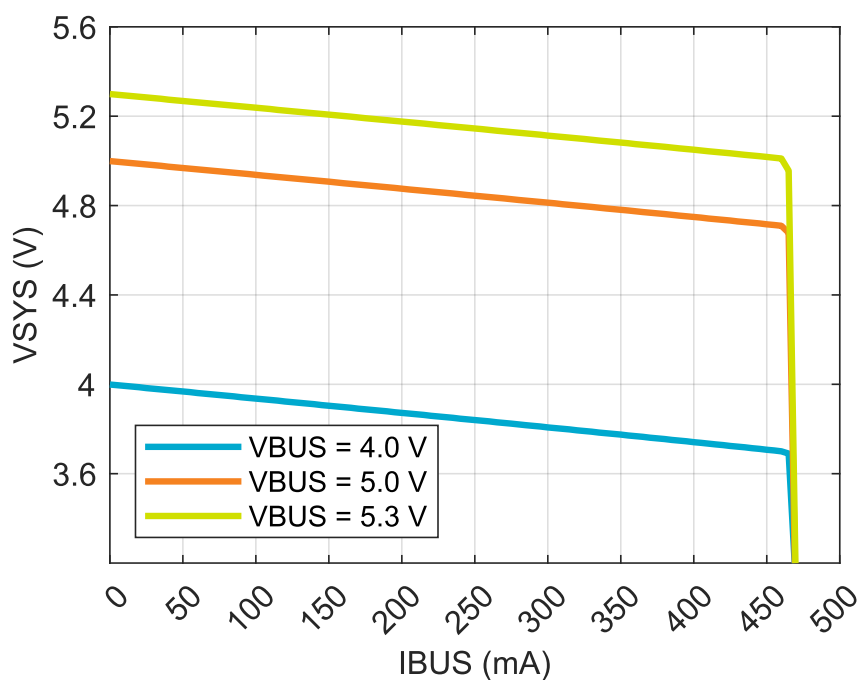


Figure 7: VSYS voltage vs. VBUS current, ILIM = 500 mA

6.1.8 Registers

Instances

Instance	Base address	Description
VBUSIN	0x00000200	VBUSIN registers
		VBUSIN register map

Register overview

Register	Offset	Description
TASKUPDATEILIMSW	0x0	Select input current limit
VBUSINILIMO	0x1	Configure input current limit
VBUSINILIMSTARTUP	0x2	Configure input current limit for startup
VBUSPEND	0x3	Enable suspend mode
USBCDETECTSTATUS	0x5	Status of charger detection
VBUSINSTATUS	0x7	Status register

6.1.8.1 TASKUPDATEILIMSW

Address offset: 0x0

Select input current limit

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKUPDATEILIM			Switch from VBUSINILIMSTARTUP to VBUSINILIMO								
			NOEFFECT	0	No effect								
			SELVBUSILIMO	1	Use VBUSINILIMO. VBUS removal switches back to VBUSINILIMSTARTUP								

6.1.8.2 VBUSINILIMO

Address offset: 0x1

Configure input current limit

Bit number						7	6	5	4	3	2	1	0
ID										A	A	A	A
Reset 0x01						0	0	0	0	0	0	0	1
ID	R/W	Field	Value ID	Value	Description								
A	RW	VBUSINILIMO			Input current limit for VBUS selected by host								
			500MA0	0	500 mA								
			100MA	1	100 mA								
			200MA	2	200 mA								
			300MA	3	300 mA								
			400MA	4	400 mA								
			500MA	5	500 mA								
			600MA	6	600 mA								
			700MA	7	700 mA								
			800MA	8	800 mA								
			900MA	9	900 mA								
			1000MA	10	1000 mA								
			1100MA	11	1100 mA								
			1200MA	12	1200 mA								
			1300MA	13	1300 mA								
			1400MA	14	1400 mA								
			1500MA	15	1500 mA								

6.1.8.3 VBUSINILIMSTARTUP

Address offset: 0x2

Configure input current limit for startup

Bit number	7	6	5	4	3	2	1	0
ID					A	A	A	A
Reset 0x01	0	0	0	0	0	0	0	1

ID	R/W	Field	Value ID	Value	Description
A	RW	VBUSINILIMSTARTUP			Default input current limit for VBUS
			500MA0	0	500 mA
			100MA	1	100 mA
			200MA	2	200 mA
			300MA	3	300 mA
			400MA	4	400 mA
			500MA	5	500 mA
			600MA	6	600 mA
			700MA	7	700 mA
			800MA	8	800 mA
			900MA	9	900 mA
			1000MA	10	1000 mA
			1100MA	11	1100 mA
			1200MA	12	1200 mA
			1300MA	13	1300 mA
			1400MA	14	1400 mA
			1500MA	15	1500 mA

6.1.8.4 VBUSSUSPEND

Address offset: 0x3

Enable suspend mode

Bit number	7	6	5	4	3	2	1	0
ID								A
Reset 0x00	0	0	0	0	0	0	0	0

ID	R/W	Field	Value ID	Value	Description
A	RW	VBUSSUSPENDENA			Control suspend mode
			NORMAL	0	Normal mode
			SUSPENDMODE	1	Suspend mode

6.1.8.5 USBDETECTSTATUS

Address offset: 0x5

Status of charger detection

Bit number						7	6	5	4	3	2	1	0
ID						B B A A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	VBUSINCC1CMP			CC1 charger detection								
			NOCONNECTION	0	No connection								
			DEFAULTUSB	1	Default USB 100/500 mA								
			1ASHIGHPOWER	2	1.5 A high power								
			3AHIGHPOWER	3	3 A high power								
B	R	VBUSINCC2CMP			CC2 charger detection								
			NOCONNECTION	0	No connection								
			DEFAULTUSB	1	Default USB 100/500 mA								
			1ASHIGHPOWER	2	1.5 A high power								
			3AHIGHPOWER	3	3 A high power								

6.1.8.6 VBUSINSTATUS

Address offset: 0x7

Status register

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	VBUSINPRESENT			VBUS present								
			NOTDETECTED	0	Not present								
			DETECTED	1	Present								
B	R	VBUSINCURRLIMACTIVE			VBUS current limit								
			NOTDETECTED	0	Not active								
			DETECTED	1	Active								
C	R	VBUSINOVPROTACTIVE			VBUS overvoltage								
			NOTACTIVE	0	Not active								
			ACTIVE	1	Active								
D	R	VBUSINUNDERVOLTAGE			VBUS undervoltage								
			NOTDETECTED	0	Not detected								
			DETECTED	1	Detected								
E	R	VBUSINSUSPENDMODEACTIVE			Suspend mode								
			NORMAL	0	Normal								
			SUSPEND	1	Suspend								
F	R	VBUSINVBUSOUTACTIVE			VBUSOUT pin								
			NOTACTIVE	0	Not active								
			ACTIVE	1	Active								

6.2 CHARGER — Battery charger

The battery charger is suitable for general purpose applications with lithium-ion (Li-ion), lithium-polymer (Li-poly), and lithium iron phosphate (LiFePO₄) batteries. The following sections describe how to configure CHARGER to match the battery type.

The main features of the battery charger are the following:

- Linear charger for Li-ion, Li-poly, and LiFePO₄ battery chemistries
- Bidirectional power FET for dynamic power-path management

- Automatic trickle, constant current, constant voltage, and end-of-charge/recharge cycle
- Maintains **VBUS** current below programmed limit
- JEITA compliant with a configurable battery charging temperature profile

Charging is configured and enabled through host software. The voltage and charging current are configurable and the device manages the charging cycle after the charging parameters are defined.

V_{TERM} must be set to a lower voltage than the battery overvoltage protection.

6.2.1 Charging cycle

Host software enables charging through register **BCHGENABLESET** on page 40. Battery charging starts when **VBUS** voltage is present and higher than **VBAT** by at least $VBUS_{\text{VALIDR}}$. Charging stops in case **VBUS**-**VBAT** is less than $VBUS_{\text{VALIDF}}$ either because **VBUS** drops or **VBAT** rises. Once charging has started, host software must use register **EVENTSBCHARGER0CLR** on page 132 to initialize battery charger events.

When charging is enabled, charging will not start if the battery voltage is less than $VBAT_{\text{LOW}}$. Charging batteries with a voltage lower than $VBAT_{\text{LOW}}$ is enabled by setting register **BCHGVBATLOWCHARGE** on page 48. When charging starts, it enters trickle charging. Constant current charging starts when the battery voltage is above $V_{\text{TRICKLE_FAST}}$. After the battery voltage reaches V_{TERM} , the charger enters constant voltage charging. The battery voltage is maintained while monitoring current flow into the battery. When the current into the battery drops below I_{TERM} , charging is complete. Charging is disabled using register **BCHGENABLECLR** on page 40.

If charging is enabled on a fully charged battery, up to 100 ms of trickle charge and up to 4 ms of constant current charging is applied.

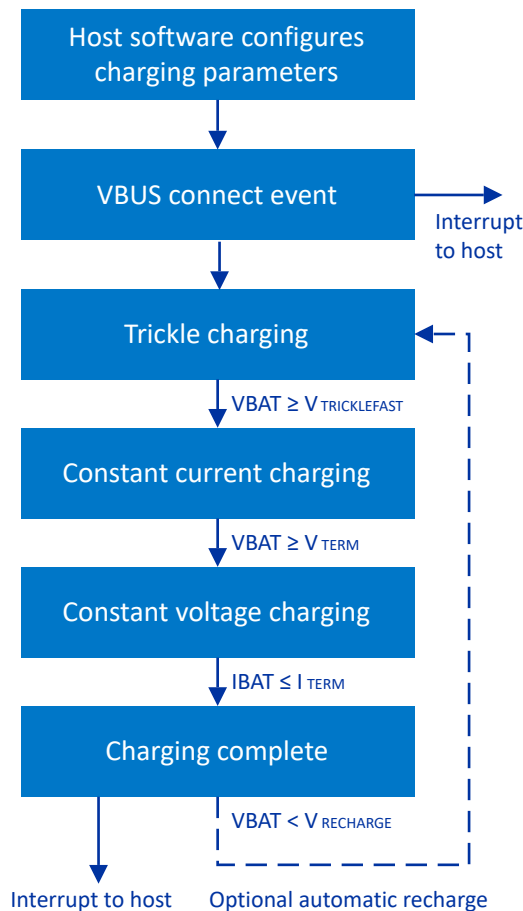


Figure 8: Charging cycle flow chart

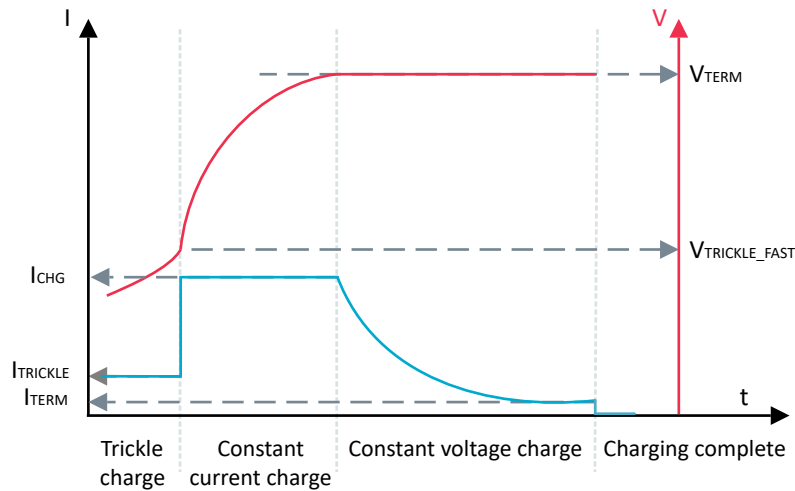


Figure 9: Charging cycle

Note: Events `EVENTBATDETECTED` and `EVENTBATLOST` and status bit `BATTERYDETECTED` are not available.

Note: When attempting to start charging when `VBUS` is present but no battery is connected, the host software will see repeated charger events (such as `EVENTCHGCOMPLETED`).

6.2.2 Termination voltage (`VTERMSET`)

`VTERM` is configured through TWI according to the battery type in use, see register `BCHGVTERM` on page 42.

For a higher battery temperature range, a lower termination voltage (`VTERMR`) is available and configured separately in register `BCHGVTERMR` on page 43. `VTERM` and `VTERMR` can be set to the same value.

6.2.3 Charge current limit (`ICHG`)

The charge current limit is set between 32 mA and 800 mA in 2 mA steps. Charging current `ICHG` is configured with TWI with a default value of 32 mA.

Note: Do not configure the charge current outside this range as it can lead to device instability.

CHARGER must be disabled before changing the current setting in registers `BCHGSETMSB` on page 41 and `BCHGSETLSB` on page 41. The setting takes effect when charging is enabled.

The charge current is configured using a 9-bit value. The following shows how the register value for `BCHGSETMSB` on page 41 can be calculated, where `ICHG` is the charge current in mA:

$$I_{SETMSB} = \text{floor}\left(\frac{I_{CHG}}{4}\right)$$

The following shows how the register value for `BCHGSETLSB` on page 41 can be calculated, where `ICHG` is the charge current in mA:

$$I_{SETLSB} = \begin{cases} 1, & \frac{I_{CHG}}{2} \text{ is odd} \\ 0, & \frac{I_{CHG}}{2} \text{ is even} \end{cases}$$

Some example charging currents are found in the following table.

I _{CHG}	BCHGISETMSB	BCHGISETLSB
32 mA	8	0
34 mA	8	1
400 mA	100	0
800 mA	200	0

Table 10: Charging current

Trickle charging current, I_{TRICKLE}, is 10% of I_{CHG}. Trickle charging is active when V_{BAT} < V_{TRICKLE_FAST} (default 2.9 V).

Termination current, I_{TERM}, is programmable to 10% (default) or 20% of I_{CHG}. Termination current is active in the constant voltage phase of charging. Charging stops when the charging current reaches this value.

These parameters are configured in registers [BCHGVTRICKLESEL](#) on page 43 and [BCHGITERMSEL](#) on page 43.

6.2.4 Monitor battery temperature

CHARGER supports three types of NTC thermistors for battery temperature (T_{BAT}) monitoring. Only one can be enabled at a time.

The host software must select the corresponding setting that matches the battery thermistor before enabling charging in register [ADCNTCSRSEL](#) on page 108. The following thermistor resistors are supported.

Nominal resistance	Resistance accuracy	B25/50	Beta accuracy	B25/85
10 kΩ	1%	3380 K	1%	3434/3435 K
47 kΩ	1%	4050 K	1%	4108 K
100 kΩ	1%	4250 K	1%	4311 K

Table 11: Supported thermistor resistors

Note: If a capacitor is placed in parallel with the thermistor, the max capacitance is 100 pF.

If a thermistor is not used, the **NTC** pin must be tied directly to ground or through a resistor. The functionality must be disabled in register [BCHGDISABLESET](#) on page 40. This does not impact [Charger thermal regulation](#) on page 30.

The following battery temperature thresholds can be set: T_{COLD} ≤ T_{COOL} ≤ T_{WARM} ≤ T_{HOT}.

These limits can be set between -20°C and +60°C, and setting adjacent thresholds to identical values is allowed. For example, setting T_{WARM} = T_{HOT} means that there is no warm region. Charging does not happen below T_{COLD} or above T_{HOT}. Charging can be paused at T_{WARM} instead of T_{HOT} by setting register [BCHGCONFIG](#) on page 48.

The thresholds are written into corresponding registers. The battery temperature variable, K_{NTCTEMP}, is calculated using the following equation:

$$K_{NTCTEMP} = \text{round}\left(1024 \cdot \frac{R_T}{R_T + R_B}\right)$$

Here, R_T is the thermistor resistance at a desired temperature and R_B (internal bias resistor) equals the thermistor resistance at 25°C. See [NTCCOLD](#) on page 44, [NTCCOOL](#) on page 44, [NTCWARM](#) on page 45, and [NTCHOT](#) on page 45 for more information. Default values in the registers match the JEITA guideline and are intended for the 10 kΩ thermistor defined in [Supported thermistor resistors](#) on page 29.

Temp.	10 kΩ	47 kΩ	100 kΩ	Register
0°C	749	787	799	NTCCOLD
10°C	658	684	693	NTCCOOL
45°C	337	306	297	NTCWARM
60°C	237	197	186	NTCHOT

Table 12: Battery temperature default values

The charging current can be reduced by 50% between NTCCOLD and NTCCOOL. The termination voltage can be configured independently between NTCWARM and NTCHOT. Default is I_{COOL} (50% of I_{CHG}), but this can be disabled.

Temperature region	Temperature limits, default setting	Charge current	Termination voltage
Cold	$T_{BAT} < T_{COLD}$ $T_{COLD} = 0^\circ\text{C}$	0 (OFF)	N/A
Cool	$T_{COLD} < T_{BAT} < T_{COOL}$ $T_{COOL} = 10^\circ\text{C}$	I_{COOL} / I_{CHG}	V_{TERM}
Nominal	$T_{COOL} < T_{BAT} < T_{WARM}$ $T_{WARM} = 45^\circ\text{C}$	I_{CHG}	V_{TERM}
Warm	$T_{WARM} < T_{BAT} < T_{HOT}$ $T_{HOT} = 60^\circ\text{C}$	I_{CHG}	V_{TERMR}
Hot	$T_{BAT} > T_{HOT}$	0 (OFF)	N/A

Table 13: Battery temperature regions

Battery temperature is measured by the on-chip System Monitor at regular intervals during charging. The latest result is available in registers [ADCNTCRESULTMSB](#) on page 110 and [ADCGPORESULTLSBS](#) on page 110.

When the battery temperature rises over T_{WARM} or T_{HOT} , or falls below T_{COOL} or T_{COLD} , an interrupt is sent.

6.2.5 Charger thermal regulation

Heat dissipation from the linear charger is managed by setting a maximum temperature limit for the die. This limit must not exceed device and PCB temperature requirements.

To enable automatic thermistor and die temperature monitoring during charging, set register [TASKAUTOTIMUPDATE](#) on page 109. This should also be set after setting the automated period.

Die temperature monitoring has a default limit of T_{CHGSTOP} . Charging stops when the die temperature reaches the limit. It resumes when the die cools down to $T_{\text{CHGRESUME}}$.

T_{CHGSTOP} controls the junction temperature rise and limits the temperature rise on the PCB and device mechanics. The device can be configured to send an interrupt when the limit is met.

The die temperature variable, K_{DIETEMP} , is calculated with the following equation:

$$K_{\text{DIETEMP}} = \text{round}\left(\frac{394.67^{\circ}\text{C} - T_{\text{D}}}{0.7926}\right)$$

Here, T_{D} represents the die temperature limit in degrees Celsius.

Registers [DIETEMPSTOP](#) on page 45 and [DIETEMPSTOPLSB](#) on page 46 are concatenated to create a 10-bit value that defines the charging stop temperature T_{CHGSTOP} . Registers [DIETEMPRESUME](#) on page 46 and [DIETEMPRESUMELSB](#) on page 46 are concatenated to create a 10-bit value that defines the charging resume temperature $T_{\text{CHGRESUME}}$. The host software reads register [DIETEMPSTATUS](#) on page 47 to determine if the die temperature is above T_{CHGSTOP} .

The following table consists of die temperature value examples.

K_{DIETEMP}	T_{D}
435	50°C
422	60°C
410	70°C
397	80°C
384	90°C
372	100°C
359	110°C

Table 14: Die temperature example

6.2.6 Charger error conditions

A CHARGER error condition occurs when one of the following are present:

- Trickle charge timeout, see [t_{OUTTRICKLE}](#)
- Safety timer expires, see [t_{OUTCHARGE}](#)

After an error is detected, charging is disabled. The charging error indication is activated and the charging indication is deactivated. Error conditions are cleared when **VBUS** is disconnected and reconnected again.

Errors are reported in register [BCHGERRREASON](#) on page 47 and [BCHGERRSENSOR](#) on page 48.

Host software clears errors with register [TASKCLEARCHGERR](#) on page 39 and releases the charger from the error state with register [TASKRELEASEERR](#) on page 39.

When the safety timer expires, the host must make sure it is safe to charge before resetting register [TASKCLEARSAFETYTIMER](#) on page 39.

6.2.7 Charging status and error indication

When CHARGER is enabled and the LEDs are configured, the LEDs indicate the charging status.

The **LED[n]** pins sink 5 mA of current when active. They are high impedance when disabled. This is suitable for driving LEDs or connecting to host GPIOs in a weak pull-up configuration. The LED anode must be connected to a voltage rail that allows forward bias. If a general purpose open drain output is needed,

the LED pins can be used with a pull-up resistor connected to a voltage rail. See [LEDDRV — LED drivers](#) on page 85 for more information.

Charging status

Charging status is available in register [BCHGCHARGESTATUS](#) on page 47.

LED drivers are configured through TWI to indicate if charging is active or charging is complete.

The charging indication turns off when charging is complete. It turns on when charging starts. The charging indication is off when CHARGER is disabled temporarily due to die temperature exceeding the configured limit.

The charging indication is off when battery temperature is below cold or above hot thresholds. No error is indicated in these cases. The charging indication is off when $VBUS > VBUS_{OVP}$ and no error is indicated.

Error indications

Errors are reported in register [BCHGERRREASON](#) on page 47 and [BCHGERRSENSOR](#) on page 48.

6.2.8 End-of-charge and recharge

Charging terminates automatically when the battery voltage reaches V_{TERM} and charging current is less than I_{TERM} . An interrupt is issued to the host.

Unless disabled through bit [DISABLE.RECHARGE](#), charging restarts automatically when VBAT is less than $V_{RECHARGE}$ and an interrupt is sent to the host.

6.2.9 Dynamic power-path management

CHARGER manages battery current flow to maintain $VSYS$ voltage.

The battery is isolated when $VBUS$ is connected and the battery is fully charged. Under this condition, $VBUS$ supplies $VSYS$. When $VBUS$ is disconnected, $VSYS$ is supplied by its decoupling capacitors until it is connected to $VBAT$. The connection is triggered by $VSYS$ reaching $VSYS_{BELOW_VBAT}$ level and is completed within t_{SWITCH} .

After $VBUS$ has been connected (and debounced), $VSYS$ is disconnected from $VBAT$ and supplied by its decoupling capacitors until supplied by $VBUS$, which happens within t_{SWITCH} .

The system load requirements are prioritized over battery charge current when $VBUS$ is connected and the battery is charging. During charging, if the system current load exceeds $IBUS_{LIM}$, the battery charge current decreases to maintain the $VSYS$ voltage. CHARGER reduces the current to maintain an internal voltage of $V_{CHDROPOUT}$ above the $VBAT$ voltage. If more current is required, CHARGER enters the SUPPLEMENT state to provide current from the battery, up to $IBAT_{LIM}$.

Note: $VSYS$ must not be supplied from an external source.

6.2.10 Battery discharge current limit

There are two selectable levels, $IBATLIM_{LOW}$ and $IBATLIM_{HIGH}$, to limit battery discharge current and optimize current measurement and fuel gauge performance.

The discharge current limit is configured through registers [BCHGSETDISCHARGEMSB](#) on page 42 and [BCHGSETDISCHARGELSB](#) on page 42.

If the system load exceeds the discharge current limit, $VSYS$ voltage drops below $VSYS_{POF}$ causing the device to reset, as described in [POF — Power-fail comparator](#) on page 112.

Note: The discharge current limit has been primarily designed to protect the device from overcurrent conditions, such as a short circuit on VSYS. It has not been designed for accurate current limiting.

The two discharge current settings are found in the following table.

Discharge current limit selection	Purpose	Battery discharge current limit	BCHGISETDISCHARGEMSE	BCHGISETDISCHARGELSB
IBATLIM _{LOW}	Increase current measurement accuracy and optimize fuel gauge performance at lower discharge currents	200 mA ¹	42	0
IBATLIM _{HIGH}	Maximum range of current measurement	1 A ¹	207	1

Table 15: Discharge current limit selection recommendations

¹ Validated through simulation.

6.2.11 Electrical specification

Symbol	Description	Min.	Typ.	Max.	Unit
V_{BAT_POR}	VBAT power-on reset release voltage		2.75		V
V_{BAT_BOR}	VBAT brownout reset trigger		2.4		V
V_{BAT_LOW}	Minimum VBAT voltage for charging		2.1		V
$V_{RECHARGE}$	Battery voltage level needed to restart charging, % of V_{TERM}		95		%
$V_{TERMACC}$	Accuracy of termination voltage VBUS = 5.0 V $T_J > 0^\circ\text{C}$	-1		+1	%
V_{TERM}	Range of termination voltage		3.50 to 3.65 4.00 to 4.45		V
V_{TERMR}	Range of termination voltage for NTC _{HOT} > T > NTC _{WARM}		3.50 to 3.65 4.00 to 4.45		V
V_{TERM_STEP}	Termination voltage step size		50		mV
I_{CHG}	Range of constant currents		32 to 800		mA
I_{CHGACC}	Constant current accuracy	-10		10	%
$I_{CHGSTEP}$	Charging current step		2		mA
$I_{TRICKLE}$	Trickle charging current, % of I_{CHG}		10		%
I_{COOL}	Reduced charging current, % of I_{CHG}		50		%
I_{TERM}	Termination current, % of I_{CHG}		10 20		%
$V_{TRICKLE_FAST}$	Default threshold where trickle charging stops and constant current charging starts		2.9		V
$V_{CHDROPOUT}$	Charger dropout voltage		60		mV
I_{BATLIM_LOW}	Discharging battery current limit on low setting	200 ¹			mA
I_{BATLIM_HIGH}	Discharging battery current limit on high setting	1 ¹			A
$R_{ON_CHARGER}$	Resistance between pins VBAT and VSYS		160		mΩ
$V_{SYS_BELOW_VBAT}$	VSYS below VBAT comparator threshold, VBAT - VSYS, VSYS falling		75		mV
t_{SWITCH}	Dynamic power-path switching time (2.8 V < VBAT < 5.5 V)			16 ¹	μs
T_{ACC}	Temperature accuracy when using suggested NTC		±2		°C
$T_{CHGSTOP}$	Die temperature where charging stops (default)		110		°C

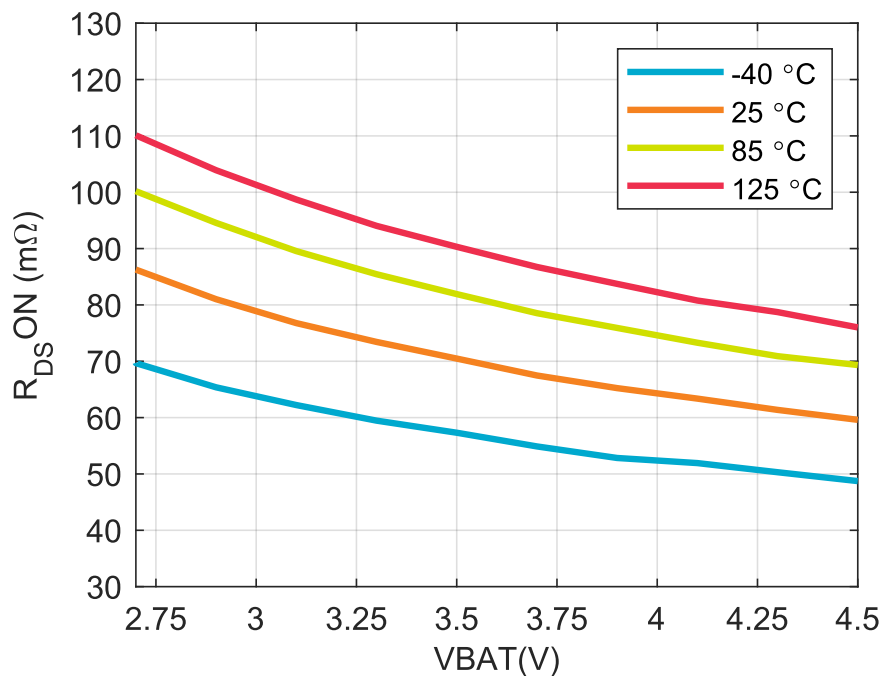
Symbol	Description	Min.	Typ.	Max.	Unit
$T_{\text{CHGRESUME}}$	Die temperature where charging resumes (default)		100		°C
$t_{\text{OUTTRICKLE}}$	Trickle charging timeout		10		min
$t_{\text{OUTCHARGE}}$	Charging timeout which covers constant current and constant voltage		7		h

Table 16: Electrical specification

¹ Validated through simulation.

6.2.12 Electrical characteristics

The following graphs show typical electrical characteristics for CHARGER.

Figure 10: WLCSP CHARGER R_{ON} vs. junction temperature

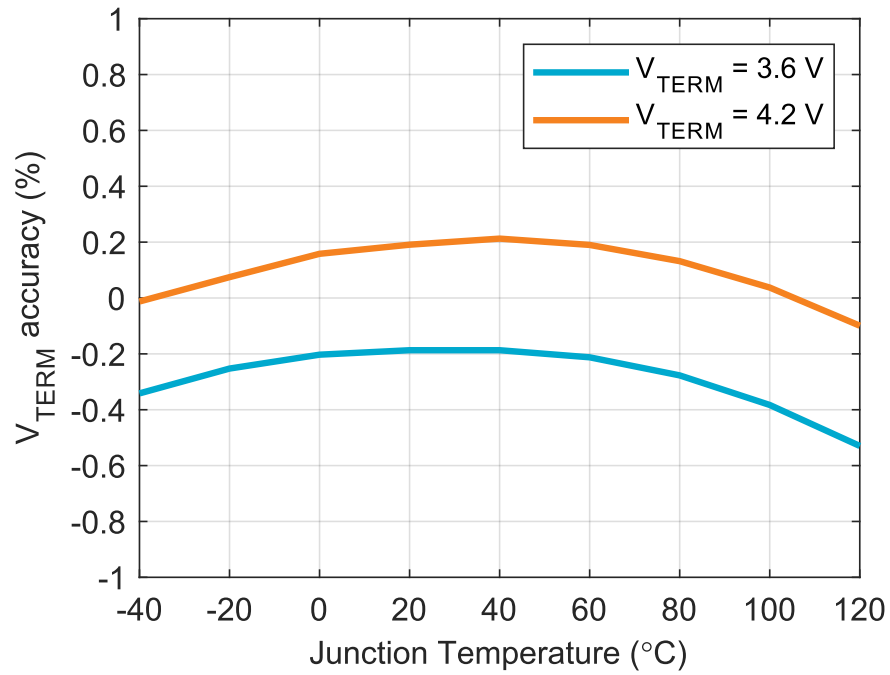


Figure 11: V_{TERM} vs. junction temperature

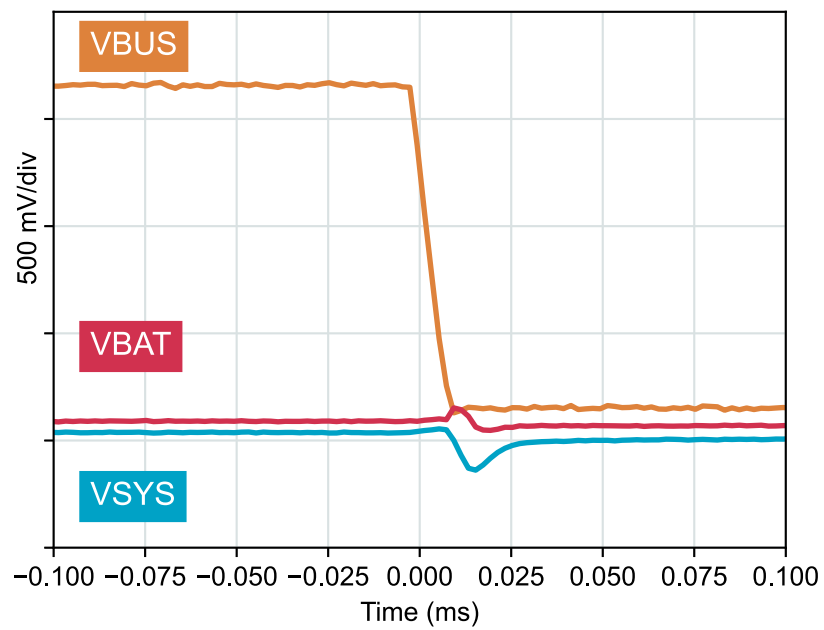


Figure 12: VBUS disconnection (350 mA load on VSYS, VBUS input current limited to 100 mA)

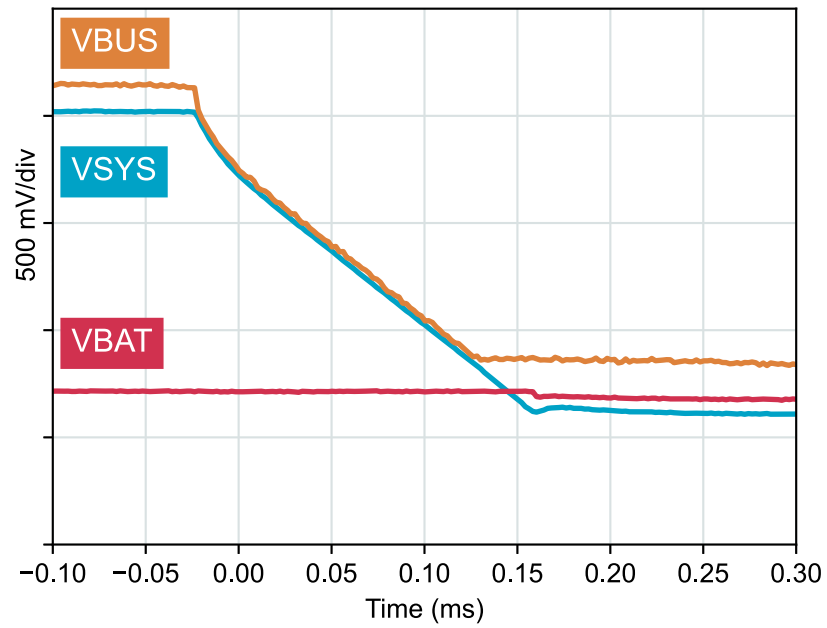


Figure 13: VBUS disconnection (150 mA load on VSYS, VBUS input current limited to 500 mA)

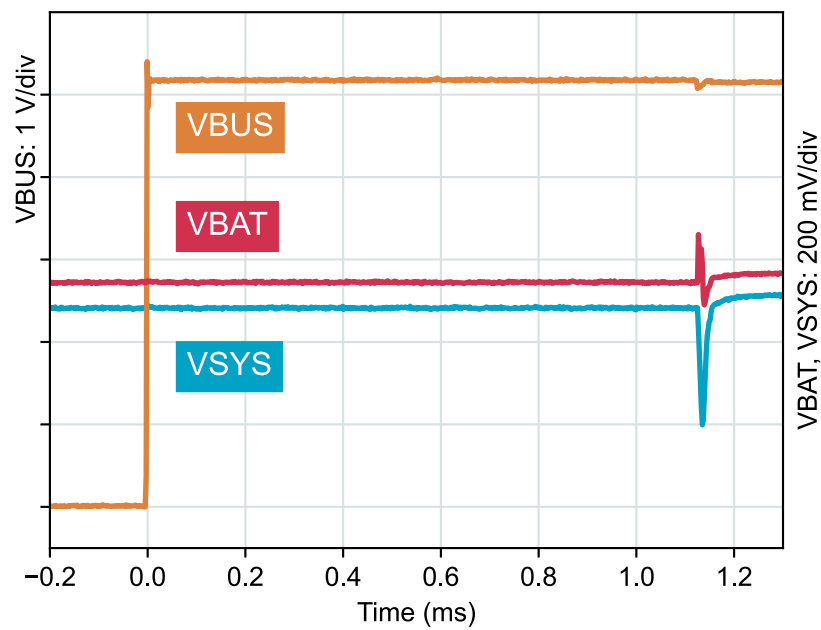


Figure 14: VBUS connection (350 mA load on VSYS, VBUS input current limited to 100 mA)

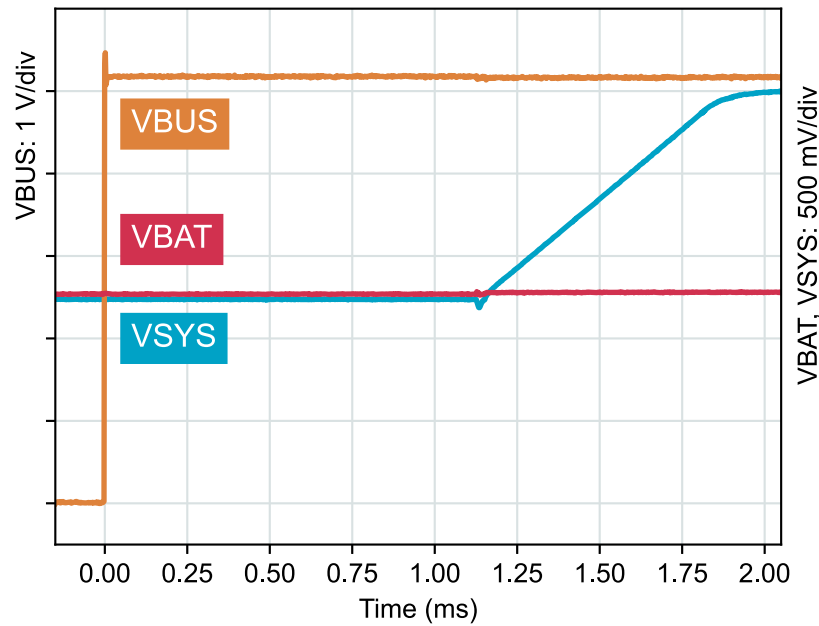


Figure 15: VBUS connection (50 mA load on VSYS, VBUS input current limited to 100 mA)

6.2.13 Registers

Instances

Instance	Base address	Description
BCHARGER	0x00000300	CHARGER registers BCHARGER register map

Register overview

Register	Offset	Description
TASKRELEASEERR	0x0	Release charger from error
TASKCLEARCHGERR	0x1	Clear error registers
TASKCLEARSAFETYTIMER	0x2	Clear safety timers
BCHGENABLESET	0x4	Enable charger
BCHGENABLECLR	0x5	Disable charger
BCHGDISABLESET	0x6	Disable automatic recharge and battery temperature monitoring
BCHGDISABLECLR	0x7	Enable automatic recharge and battery temperature monitoring
BCHGSETMSB	0x8	Configure charge current MSB
BCHGSETLSB	0x9	Configure charge current LSB
BCHGSETDISCHARGEMSB	0xA	Configure discharge current limit MSB
BCHGSETDISCHARGELSB	0xB	Configure discharge current limit LSB
BCHGVTERM	0xC	Configure termination voltage for cool and nominal temperature region
BCHGVTERMRR	0xD	Configure termination voltage for warm temperature region
BCHGVTRICKLESEL	0xE	Select trickle charge voltage threshold
BCHGITERMSEL	0xF	Select termination current
NTCCOLD	0x10	NTC thermistor threshold for cold temperature region
NTCCOLDLSB	0x11	NTC thermistor threshold for cold temperature region
NTCCOOL	0x12	NTC thermistor threshold for cool temperature region
NTCCOOLLSB	0x13	NTC thermistor threshold for cool temperature region

Register	Offset	Description
NTCWARM	0x14	NTC thermistor threshold for warm temperature region
NTCWARMLSB	0x15	NTC thermistor threshold for warm temperature region
NTCHOT	0x16	NTC thermistor threshold for hot temperature region
NTCHOTLSB	0x17	NTC thermistor threshold for hot temperature region
DIETEMPSTOP	0x18	Die temperature threshold to stop charging
DIETEMPSTOPLSB	0x19	Die temperature threshold to stop charging
DIETEMPRESUME	0x1A	Die temperature threshold to resume charging
DIETEMPRESUMELSB	0x1B	Die temperature threshold to resume charging
BCHGILIMSTATUS	0x2D	Battery discharge current limit status
NTCSTATUS	0x32	Battery temperature region status
DIETEMPSTATUS	0x33	Die temperature status
BCHGCHARGESTATUS	0x34	Charging status
BCHGERRREASON	0x36	Charge error. Latched error reasons. Cleared with TASKSCLEARCHGERR
BCHGERRSENSOR	0x37	Charger error. Latched conditions. Cleared with TASKSCLEARCHGERR
BCHGCONFIG	0x3C	Charger configuration
BCHGVBATLOWCHARGE	0x50	Enable charging at low battery voltage

6.2.13.1 TASKRELEASEERR

Address offset: 0x0

Release charger from error

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKRELEASEERROR			Release charger from error								
			NOEFFECT	0	No effect								
			TRIGGER	1	Trigger task								

6.2.13.2 TASKCLEARCHGERR

Address offset: 0x1

Clear error registers

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKCLEARCHGERR			Clear registers BCHGERRREASON and BCHGERRSENSOR								
			NOEFFECT	0	No effect								
			TRIGGER	1	Trigger task								

6.2.13.3 TASKCLEARSAFETYTIMER

Address offset: 0x2

Clear safety timers

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKCLEARSAFETYTIMER			Clear trickle and charge safety timers													
			NOEFFECT	0	No effect													
			TRIGGER	1	Trigger task													

6.2.13.4 BCHGENABLESET

Address offset: 0x4

Enable charger

Bit number											7	6	5	4	3	2	1	0
ID											B A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	RW	ENABLECHARGING W1S			Enable charger. (Read 0: Charging disabled). (Read 1: Charging enabled).													
			NOEFFECT	0	No effect													
			ENABLECHG	1	Enable battery charging													
B	RW	ENABLEFULLCHGCOOL W1S			Enable charging of cool battery with full current. (Read 0: 50% charge current value of BCHGSETMSB and BCHGSETLSB registers). (Read 1: 100% charge current value of BCHGSETMSB and BCHGSETLSB registers).													
			NOEFFECT	0	No effect													
			ENABLECOOL	1	Enable charging of cool battery													

6.2.13.5 BCHGENABLECLR

Address offset: 0x5

Disable charger

Bit number						7	6	5	4	3	2	1	0
ID						B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	ENABLECHARGING W1C			Disable charger. (Read 0: Charging disabled). (Read 1: Charging enabled).								
			NOEFFECT	0	No effect								
			DISABLECHG	1	Disable battery charging								
B	RW	ENABLEFULLCHGCOOL W1C			Disable charging of cool battery with full current. (Read 0: 50% charge current value of BCHGSETMSB and BCHGSETLSB registers). (Read 1: 100% charge current value of BCHGSETMSB and BCHGSETLSB registers).								
			NOEFFECT	0	No effect								
			DISABLECOOL	1	Disable charging of cool battery								

6.2.13.6 BCHGDISABLESET

Address offset: 0x6

Disable automatic recharge and battery temperature monitoring

Bit number						7	6	5	4	3	2	1	0
ID												B	A
Reset 0x00												0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	DISABLERECHARGE W1S			Disable automatic recharge. (Read 0: Recharge enabled). (Read 1: Recharge disabled).								
			NOEFFECT	0	No effect								
			DISABLERCHG	1	Disable recharging of battery once charged								
B	RW	DISABLENTC W1S			Disable battery temperature monitoring. (Read 0: battery temperature monitoring enabled) (Read 1: battery temperature monitoring disabled)								
			NOEFFECT	0	No effect								
			IGNORENTC	1	Disable battery temperature monitoring								

6.2.13.7 BCHGDISABLECLR

Address offset: 0x7

Enable automatic recharge and battery temperature monitoring

Bit number						7	6	5	4	3	2	1	0
ID												B	A
Reset 0x00												0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	DISABLERECHARGE W1C			Enable automatic recharge. (Read 0: Recharge enabled). (Read 1: Recharge disabled).								
			NOEFFECT	0	No effect								
			ENABLERCHG	1	Enable recharging of battery once charged								
B	RW	DISABLENTC W1C			Enable battery temperature monitoring. (Read 0: battery temperature monitoring enabled) (Read 1: battery temperature monitoring disabled)								
			NOEFFECT	0	No effect								
			USENTC	1	Enable battery temperature monitoring								

6.2.13.8 BCHGISETMSB

Address offset: 0x8

Configure charge current MSB

Bit number						7	6	5	4	3	2	1	0
ID												A	A
Reset 0x08												0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BCHGISETCHARGE MSB			Charge current (BCHGISETCHARGE MSB bits [8:1]). Default 32 mA. See chapter Charge current limit.								

6.2.13.9 BCHGISETLSB

Address offset: 0x9

Configure charge current LSB

Bit number													7	6	5	4	3	2	1	0
ID																		A		
Reset 0x00													0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description															
A	RW	BCHGISETCHARGELSB			Charge current fine tune by 2 mA (BCHGISETCHARGE LSB bit [0]). See chapter Charge current limit.															

6.2.13.10 BCHGISETDISCHARGEMSB

Address offset: 0xA

Configure discharge current limit MSB

Bit number										7	6	5	4	3	2	1	0
ID										A	A	A	A	A	A	A	A
Reset 0xCF										1	1	0	0	1	1	1	1
ID	R/W	Field	Value ID	Value	Description												
A	RW	BCHGISETDISCHARGEMSB			Discharge current limit (BCHGISETDISCHARGE MSB bits [8:1]). See more from chapter Battery discharge current limit.												

6.2.13.11 BCHGISETDISCHARGELSB

Address offset: 0xB

Configure discharge current limit LSB

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x01											0	0	0	0	0	0	0	1
ID	R/W	Field	Value ID	Value	Description													
A	RW	BCHGISETDISCHARGELSB			Discharge current limit fine tune (BCHGISETDISCHARGE LSB bit [0]). See more from chapter Battery discharge current limit.													

6.2.13.12 BCHGVTERM

Address offset: 0xC

Configure termination voltage for cool and nominal temperature region

Bit number										7	6	5	4	3	2	1	0
ID														A	A	A	A
Reset 0x02										0	0	0	0	0	0	1	0
ID	R/W	Field	Value ID	Value	Description												
A	RW	BCHGVTERMNORM			Configure termination voltage for cool and nominal temperature region.												
			3V50	0	3.50 V												
			3V55	1	3.55 V												
			3V60	2	3.60 V (default)												
			3V65	3	3.65 V												
			4V00	4	4.00 V												
			4V05	5	4.05 V												
			4V10	6	4.10 V												
			4V15	7	4.15 V												
			4V20	8	4.20 V												
			4V25	9	4.25 V												
			4V30	10	4.30 V												

Bit number						7	6	5	4	3	2	1	0
ID						A				A	A	A	A
Reset 0x02						0	0	0	0	0	0	1	0
ID	R/W	Field	Value ID	Value	Description								
			4V35	11	4.35 V								
			4V40	12	4.40 V								
			4V45	13	4.45 V								

6.2.13.13 BCHGVTERM

Address offset: 0xD

Configure termination voltage for warm temperature region

Bit number						7	6	5	4	3	2	1	0
ID						A				A	A	A	A
Reset 0x02						0	0	0	0	0	0	1	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BCHGVTERMREDUCED			Configure termination voltage for warm temperature region.								
			3V50	0	3.50 V								
			3V55	1	3.55 V								
			3V60	2	3.60 V (default)								
			3V65	3	3.65 V								
			4V00	4	4.00 V								
			4V05	5	4.05 V								
			4V10	6	4.10 V								
			4V15	7	4.15 V								
			4V20	8	4.20 V								
			4V25	9	4.25 V								
			4V30	10	4.30 V								
			4V35	11	4.35 V								
			4V40	12	4.40 V								
			4V45	13	4.45 V								

6.2.13.14 BCHGVTRICKLESEL

Address offset: 0xE

Select trickle charge voltage threshold

Bit number						7	6	5	4	3	2	1	0
ID										A			
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BCHGVTRICKLESEL			Select trickle charge voltage threshold								
			2V9	0	2.9 V (default)								
			2V5	1	2.5 V								

6.2.13.15 BCHGITERMSEL

Address offset: 0xF

Select termination current

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	RW	BCHGITERMSEL			Select terminaton current													
			SEL10	0	10% (default)													
			SEL20	1	20%													

6.2.13.16 NTCCOLD

Address offset: 0x10

NTC thermistor threshold for cold temperature region

Bit number						7	6	5	4	3	2	1	0
ID						A	A	A	A	A	A	A	A
Reset 0xBB						1	0	1	1	1	0	1	1
ID	R/W	Field	Value ID	Value	Description								
A	RW	NTCCOLDLVMSB			NTC cold level MSB								

6.2.13.17 NTCCOLDLSB

Address offset: 0x11

NTC thermistor threshold for cold temperature region

Bit number											7	6	5	4	3	2	1	0
ID											A A							
Reset 0x01											0	0	0	0	0	0	0	1
ID	R/W	Field	Value ID	Value	Description													
A	RW	NTCCOLDLVLSB			NTC cold level LSB													

6.2.13.18 NTCCOOL

Address offset: 0x12

NTC thermistor threshold for cool temperature region

Bit number						7	6	5	4	3	2	1	0
ID						A	A	A	A	A	A	A	A
Reset 0xA4						1	0	1	0	0	1	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	NTCCOOLVMSB			NTC cool level MSB								

6.2.13.19 NTCCOOLLSB

Address offset: 0x13

NTC thermistor threshold for cool temperature region

Bit number										7	6	5	4	3	2	1	0
ID										A A							
Reset 0x02										0	0	0	0	0	0	1	0
ID	R/W	Field	Value ID	Value	Description												
A	RW	NTCCOOLLVLSB			NTC cool level LSB												

6.2.13.20 NTCWARM

Address offset: 0x14

NTC thermistor threshold for warm temperature region

Bit number	7 6 5 4 3 2 1 0						
ID	A A A A A A A						
Reset 0x54	0 1 0 1 0 1 0 0						
ID	R/W	Field	Value ID	Value	Description		
A	RW	NTCWARMVLMSB			NTC warm level MSB		

6.2.13.21 NTCWARMLSB

Address offset: 0x15

NTC thermistor threshold for warm temperature region

Bit number	7 6 5 4 3 2 1 0						
ID	A A						
Reset 0x01	0 0 0 0 0 0 0 1						
ID	R/W	Field	Value ID	Value	Description		
A	RW	NTCWARMVLSB			NTC warm level LSB		

6.2.13.22 NTCHOT

Address offset: 0x16

NTC thermistor threshold for hot temperature region

Bit number	7 6 5 4 3 2 1 0						
ID	A A A A A A A						
Reset 0x3B	0 0 1 1 1 0 1 1						
ID	R/W	Field	Value ID	Value	Description		
A	RW	NTCHOTVLMSB			NTC hot level MSB		

6.2.13.23 NTCHOTLSB

Address offset: 0x17

NTC thermistor threshold for hot temperature region

Bit number	7 6 5 4 3 2 1 0						
ID	A A						
Reset 0x01	0 0 0 0 0 0 0 1						
ID	R/W	Field	Value ID	Value	Description		
A	RW	NTCHOTVLSB			NTC hot level LSB		

6.2.13.24 DIETEMPSTOP

Address offset: 0x18

Die temperature threshold to stop charging

Bit number	7 6 5 4 3 2 1 0							
ID	A A A A A A A A							
Reset 0x5A	0 1 0 1 1 0 1 0							
ID	R/W	Field	Value ID	Value	Description			
A	RW	DIETEMPSTOPCHG			Die temperature stop charging level			

6.2.13.25 DIETEMPSTOPLSB

Address offset: 0x19

Die temperature threshold to stop charging

Bit number	7 6 5 4 3 2 1 0							
ID	A A							
Reset 0x00	0 0 0 0 0 0 0 0							
ID	R/W	Field	Value ID	Value	Description			
A	RW	DIETEMPSTOPCHGLSB			Die temperature stop charging level LSB			

6.2.13.26 DIETEMPRESUME

Address offset: 0x1A

Die temperature threshold to resume charging

Bit number	7 6 5 4 3 2 1 0							
ID	A A A A A A A A							
Reset 0x5D	0 1 0 1 1 1 0 1							
ID	R/W	Field	Value ID	Value	Description			
A	RW	DIETEMPRESUMECHG			Die temperature resume charging level			

6.2.13.27 DIETEMPRESUMELSB

Address offset: 0x1B

Die temperature threshold to resume charging

Bit number	7 6 5 4 3 2 1 0							
ID	A A							
Reset 0x00	0 0 0 0 0 0 0 0							
ID	R/W	Field	Value ID	Value	Description			
A	RW	DIETEMPRESUMECHGLSB			Die temperature resume charging level LSB			

6.2.13.28 BCHGILIMSTATUS

Address offset: 0x2D

Battery discharge current limit status

Bit number	7 6 5 4 3 2 1 0							
ID	A							
Reset 0x00	0 0 0 0 0 0 0 0							
ID	R/W	Field	Value ID	Value	Description			
A	R	BCHGILIMBATACTIVE			Discharge current limiter active			
			INACTIVE	0	Battery current limiter not active			
			ACTIVE	1	Battery current limiter active			

6.2.13.29 NTCSTATUS

Address offset: 0x32

Battery temperature region status

Bit number						7	6	5	4	3	2	1	0
ID						D				C	B	A	
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	NTCCOLD			Cold								
B	R	NTCCOOL			Cool								
C	R	NTCWARM			Warm								
D	R	NTCHOT			Hot								

6.2.13.30 DIETEMPSTATUS

Address offset: 0x33

Die temperature status

Bit number						7	6	5	4	3	2	1	0
ID										A			
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	DIETEMPHIGH			Die temperature status								
			NORMAL	0	Die temperature below DIETEMPSTOP								
			HIGH	1	Die temperature above DIETEMPSTOP								

6.2.13.31 BCHGCHARGESTATUS

Address offset: 0x34

Charging status

Bit number						7	6	5	4	3	2	1	0
ID						H	G	F	E	D	C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	BATTERYDETECTED			Reserved (battery is connected)								
B	R	COMPLETED			Charging completed (battery full)								
C	R	TRICKLECHARGE			Trickle charge								
D	R	CONSTANTCURRENT			Constant current charging								
E	R	CONSTANTVOLTAGE			Constant voltage charging								
F	R	RECHARGE			Battery re-charge is needed								
G	R	DIETEMPHIGHCHGPAUSED			Charging stopped due to die temperature high								
H	R	SUPPLEMENTACTIVE			Supplement mode active								

6.2.13.32 BCHGERRREASON

Address offset: 0x36

Charge error. Latched error reasons. Cleared with TASKSCLEARHGERR

Bit number						7	6	5	4	3	2	1	0
ID						G F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	NTCSENSORERROR			NTC sensor error								
B	R	VBATSENSORERROR			Vbat sensor error								
C	R	VBATLOW			VbatLow error								
D	R	VTRICKLE			Vtrickle error								
E	R	MEASTIMEOUT			Measurement timer timeout								
F	R	CHARGETIMEOUT			Charge timer timeout								
G	R	TRICKLETIMEOUT			Trickle timer timeout								

6.2.13.33 BCHGERRSENSOR

Address offset: 0x37

Charger error. Latched conditions. Cleared with TASKSCLEARHGERR

Bit number						7	6	5	4	3	2	1	0
ID						H G F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	SENSORNTCCOLD			NTC cold region active when error occurs								
B	R	SENSORNTCCOOL			NTC cool region active when error occurs								
C	R	SENSORNTCWARM			NTC warm region active when error occurs								
D	R	SENSORNTCHOT			NTC hot region active when error occurs								
E	R	SENSORVTERM			Vterm status when error occurs								
F	R	SENSORRECHARGE			Recharge status when error occurs								
G	R	SENSORVTRICKLE			Vtrickle status when error occurs								
H	R	SENSORVBATLOW			Vbatlow status when error occurs								

6.2.13.34 BCHGCONFIG

Address offset: 0x3C

Charger configuration

Bit number						7	6	5	4	3	2	1	0
ID													
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	DISABLECHARGEWARM			Disable charging if battery is warm								
			ENABLED	0	Enable charging if battery is warm								
			DISABLED	1	Disable charging if battery is warm								

6.2.13.35 BCHGVBATLOWCHARGE

Address offset: 0x50

Enable charging at low battery voltage

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00													0 0 0 0 0 0 0 0
ID	R/W	Field	Value ID	Value	Description								
A	RW	ENABLEVBATLOWCHARGE			Enable charging when low battery is below VBATLOW								
			Disable	0	Disable								
			Enable	1	Enable								

6.3 BUCK — Buck regulators

BUCK consists of two step-down buck regulators, BUCK1 and BUCK2.

BUCK has the following features:

- Ultra-high efficiency (low IQ) and low noise operation
- PWM and Hysteretic modes with automatic switching based on load
- TWI configurable for forcing PWM mode to minimize output voltage ripple
- Configurable output voltages between 1.0 V and 3.3 V

Hysteretic mode offers efficiency at lower load currents and typically operates up to half the maximum PWM current. PWM mode provides a clean supply operation due to a constant switching frequency, F_{BUCK} . This provides optimal coexistence with RF circuits. BUCK can automatically change between Hysteretic and PWM modes.

Note: The outputs of BUCK1 and BUCK2 must not be tied together as they are not intended to operate in such configuration.

6.3.1 On/Off control

BUCK is enabled in the following ways.

- **VSET[n]** pin
- Control registers
- **GPIO[n]** pin

The **VSET1** and **VSET2** pins are enabled only at power-on. If resistor R_{VSETn} is present, BUCK is enabled with the output voltage defined by the resistor value. If the pin is grounded, BUCK is disabled. See [Default VOUT1 using an external resistor](#) on page 50 and [Default VOUT2 using an external resistor](#) on page 50.

Control registers [BUCK1ENASET](#) on page 65, [BUCK1ENACLRL](#) on page 65, [BUCK2ENASET](#) on page 66, and [BUCK2ENACLRL](#) on page 66 have enable and disable bits for each BUCK. These registers override the default BUCK state.

A GPIO can be configured in register [BUCKENCTRL](#) on page 70 to enable or disable BUCK.

If BUCK is disabled during power up, the system defaults to software control of BUCK.

6.3.2 Output voltage selection

The output voltage range for BUCK is programmable with TWI. The default output voltage selection is found on pins **VSET1** and **VSET2**, which are configured using an external resistor to **GND**. Only the output voltages shown in the tables can be selected using resistors. The **VOUT[n]** pins have two voltage configuration registers that are selectable through a GPIO pin with predefined voltage settings available.

The **VSET[n]** pins are effective only at start up. The external resistor (maximum 5% tolerance) defines the default output voltage setting as found in the following table.

Symbol	Nominal resistance	VOUT1 start up output voltage
R _{VSET1}	<100 Ω (grounded)	0 V (OFF)
	4.7 kΩ	1.0 V
	10 kΩ	1.2 V
	22 kΩ	1.5 V
	47 kΩ	1.8 V
	68 kΩ	2.0 V
	100 kΩ	2.2 V
	150 kΩ	2.5 V
	250...500 kΩ	2.7 V

Table 17: Default VOUT1 using an external resistor

Symbol	Nominal resistance	VOUT2 start up output voltage
R _{VSET2}	<100 Ω (grounded)	0 V (OFF)
	4.7 kΩ	1.8 V
	10 kΩ	2.0 V
	22 kΩ	2.2 V
	47 kΩ	2.4 V
	68 kΩ	2.5 V
	100 kΩ	2.7 V
	150 kΩ	3.0 V
	250...500 kΩ	3.3 V

Table 18: Default VOUT2 using an external resistor

Note: Do not leave **VSET [n]** floating, make sure that the **VSET [n]** pins have the correct configuration.

The output voltage range is from 1.0 V to 3.3 V in 100 mV steps and is set in the voltage configuration registers [BUCK1NORMVOUT](#) on page 67 and [BUCK2NORMVOUT](#) on page 68. Once the voltage is selected, register [BUCKSWCTRLSEL](#) on page 72 must be written to for the values to take effect.

Registers [BUCK1VOUTSTATUS](#) on page 72 and [BUCK2VOUTSTATUS](#) on page 72 indicate the status or current voltage setting.

A GPIO can be configured to select between two voltage levels. The output voltage for retention mode is configured in registers [BUCK1RETVOUT](#) on page 68 and [BUCK2RETVOUT](#) on page 69. Select a GPIO to control retention voltage in register [BUCKVRETCTRL](#) on page 71.

6.3.3 BUCK mode selection

BUCK operates in Automatic mode by default. When in Automatic mode, BUCK selects Hysteretic mode for low load currents, and PWM mode for high load currents.

In PWM mode, BUCK provides a clean supply operation due to constant switching frequency and lower voltage ripple for optimal coexistence with RF circuits.

Forced pulse width modulation (PWM) is set by the following:

- Control register bits in [BUCK\[n\]PWMSET](#)
- **GPIO[n]** pins in register [BUCKPWMCTRL](#) on page 71 overriding the register setting for one or both BUCKs

Hysteretic mode can be forced in register [BUCKCTRL0](#) on page 73 for each BUCK. This setting is not available using GPIO.

6.3.4 Active output capacitor discharge

When the converter is disabled, active discharge can be enabled or disabled in register [BUCKCTRL0](#) on page 73 using [R_{DISCH}](#) from the output capacitors. The default setting is disabled.

Capacitor discharge is forced when there is a power cycle reset. See figure [Power cycle](#) on page 117.

6.3.5 Component selection

Recommended values for the inductor are shown in the following table.

Parameter	Value	Unit
Nominal inductance	2.2	μH
Inductor tolerance	≤ 20	%
DC resistance (DCR)	≤ 400	mΩ
Saturation current (I_{sat})	> 350	mA
Rated current (I_{max})	> 200	mA

Table 19: Recommended inductor specifications

Parameter	Value	Unit
Effective capacitance	≥ 4	μF
ESR	≤ 50	mΩ

Table 20: Recommended capacitor specifications

6.3.6 Electrical specification

Symbol	Description	Min.	Typ.	Max.	Unit
$V_{OUT[n]_{ACC}}$	Output voltage accuracy	-5		+5	%
$V_{SYS_{MIN}}$	Minimum V_{SYS} voltage for enabling BUCK (dependent on POF setting)		2.7		V
I_{OUT}	Maximum BUCK current to maintain performance			200	mA
V_{DROP_OUT}	Drop-out voltage 1 mA load		100		mV

Symbol	Description	Min.	Typ.	Max.	Unit
R_{DISCH}	Active output capacitor discharge resistance		2		k Ω
$I_{PWMTHRES}$	Load current threshold from Hysteretic to PWM mode (mode = AUTO)		90		mA
$I_{HYSTTHRES}$	Load current threshold from PWM to Hysteretic mode (mode = AUTO)		40		mA
$V_{OUT_RIPPLEPWM}$	VOUT ripple in PWM mode $I_{OUT} = 200$ mA		5		mVpp
$V_{OUT_RIPPLEHYST}$	VOUT ripple in Hysteretic mode		50		mVpp
EFF_{BUCK}	Efficiency in PWM mode $V_{SYS} = 3.8$ V $V_{OUT} = 3.3$ V $I_{OUT} = 200$ mA		93		%
f_{BUCK}	Switching frequency in PWM mode		3.6		MHz
t_{STRT}	Start-up time $V_{OUT} = 3.3$ V $C = 10$ μ F		1.2		ms
$t_{PWMMODE}$	Transition time Hysteretic to PWM mode Automatic (and via TWI or GPIO)		90 (55)		μ s
t_{HYST}	Transition time Hysteretic to PWM mode Automatic (and through TWI or GPIO)		35 (25)		μ s

Table 21: BUCK electrical specification

6.3.7 Electrical characteristics

The following graphs show typical electrical characteristics for BUCK.

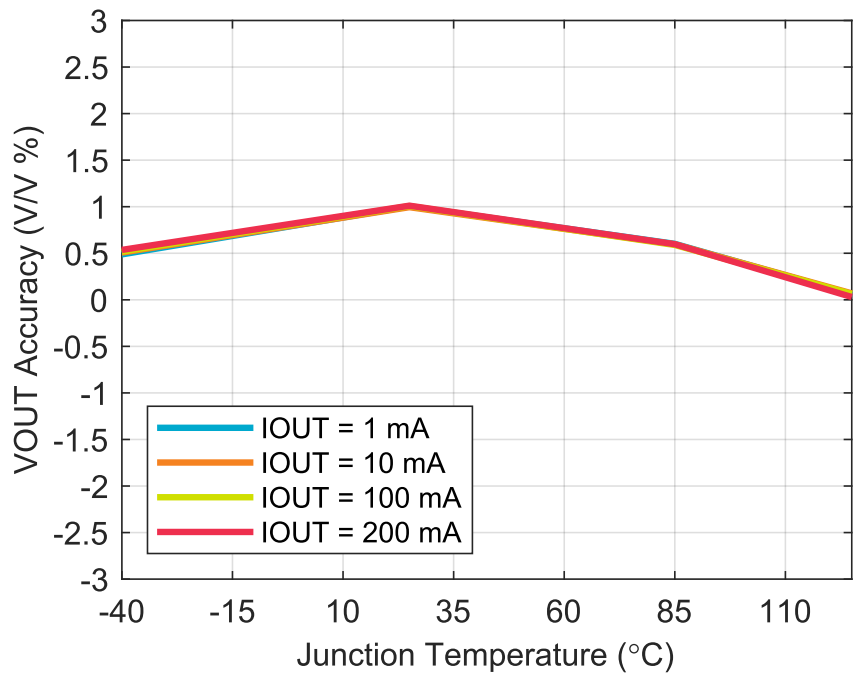


Figure 16: VBAT = 4.35 V: VOUT = 3.0 V vs. junction temperature

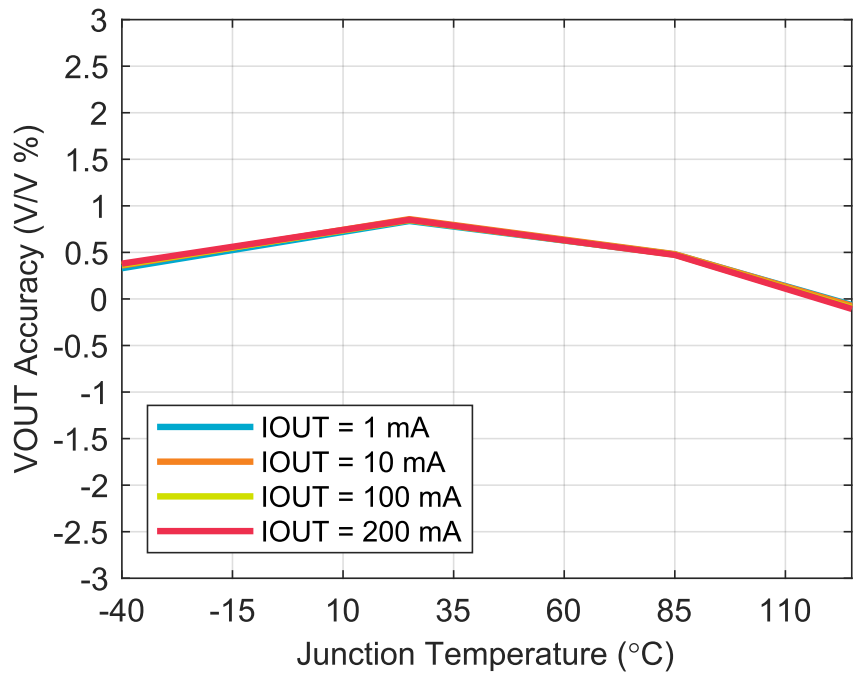


Figure 17: VBAT = 3.8: VOUT = 3.0 V vs. junction temperature

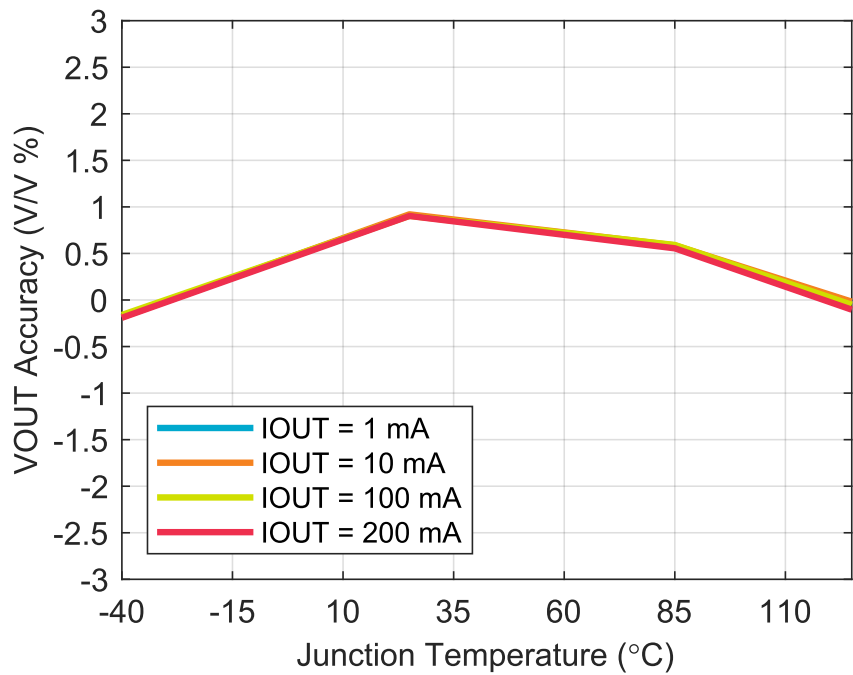


Figure 18: VBAT = 4.35 V: VOUT = 1.8 vs. junction temperature

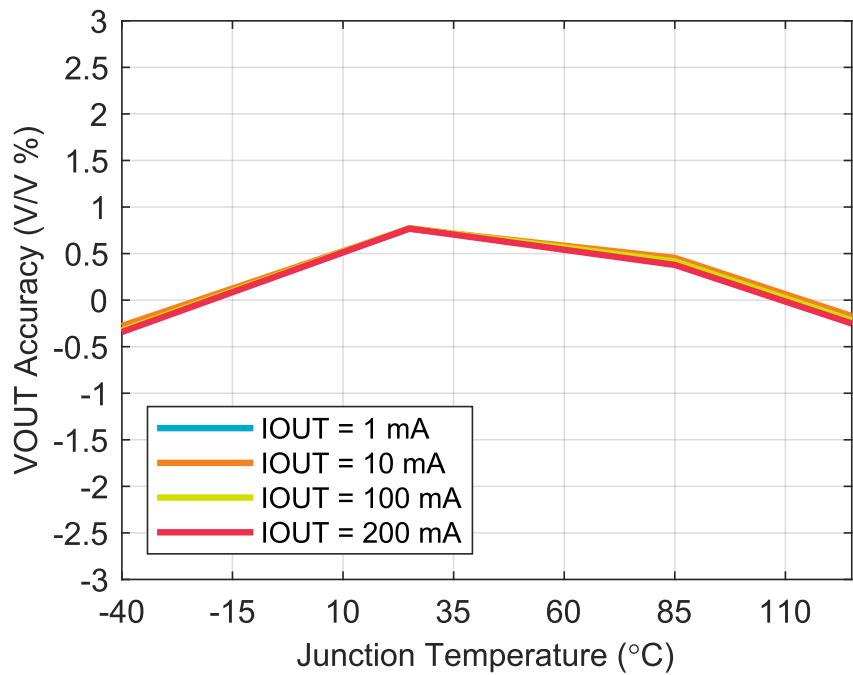


Figure 19: VBAT = 3.8 V: VOUT = 1.8 vs. junction temperature

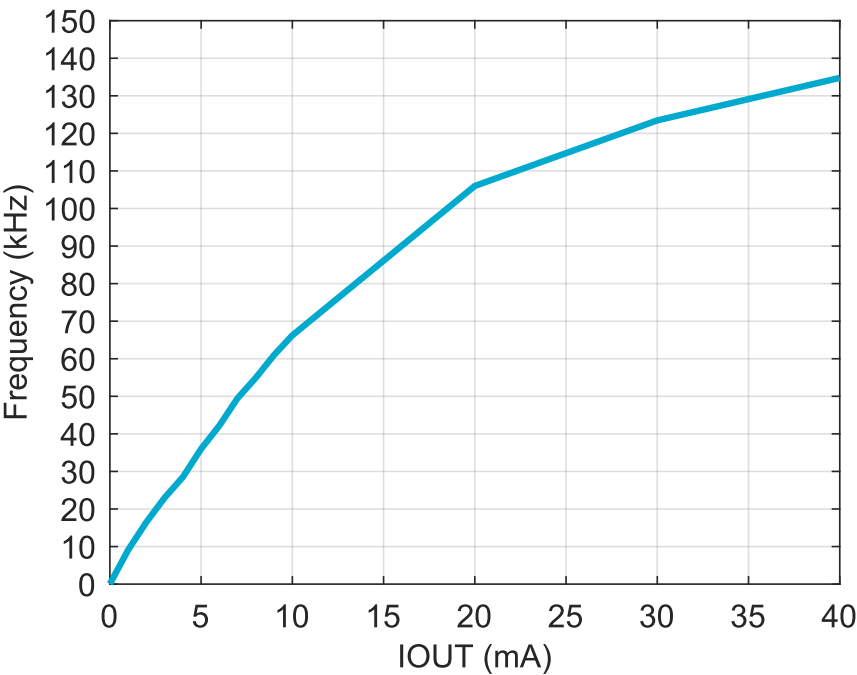


Figure 20: VBAT = 3.8 V: VOUT = 2.0 V: PFM frequency vs. current

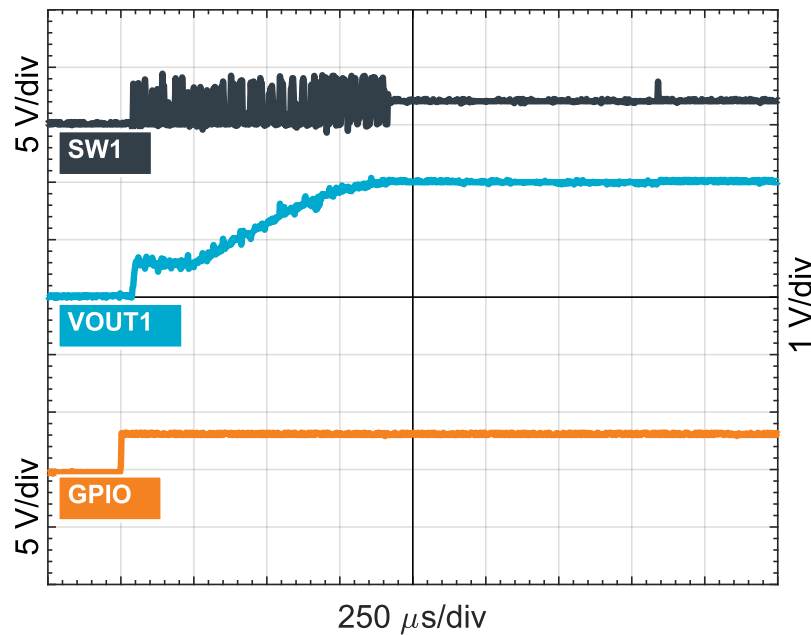


Figure 21: VBAT = 3.8 V: VOUT = 2.0 V: GPIO BUCK start

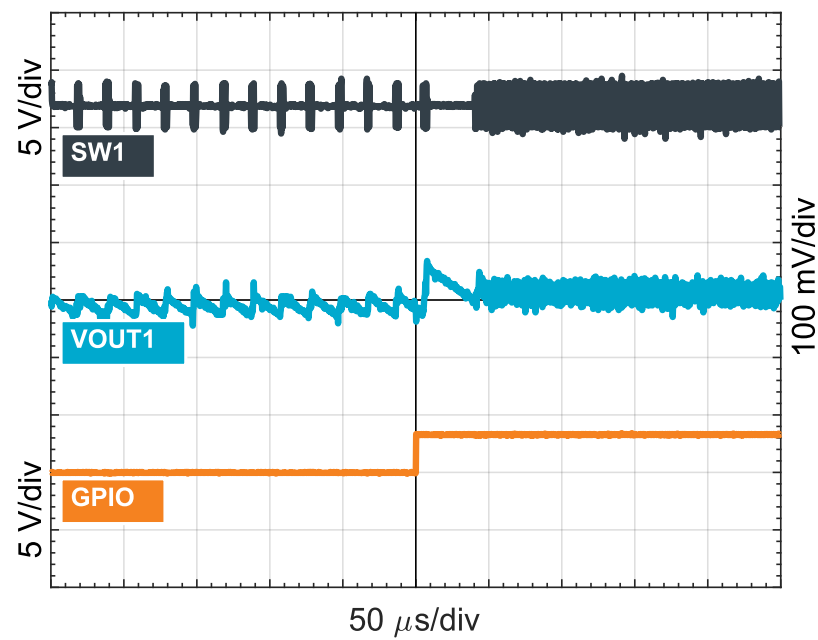


Figure 22: VBAT = 3.8 V: VOUT = 2.0 V: GPIO PWM mode selection

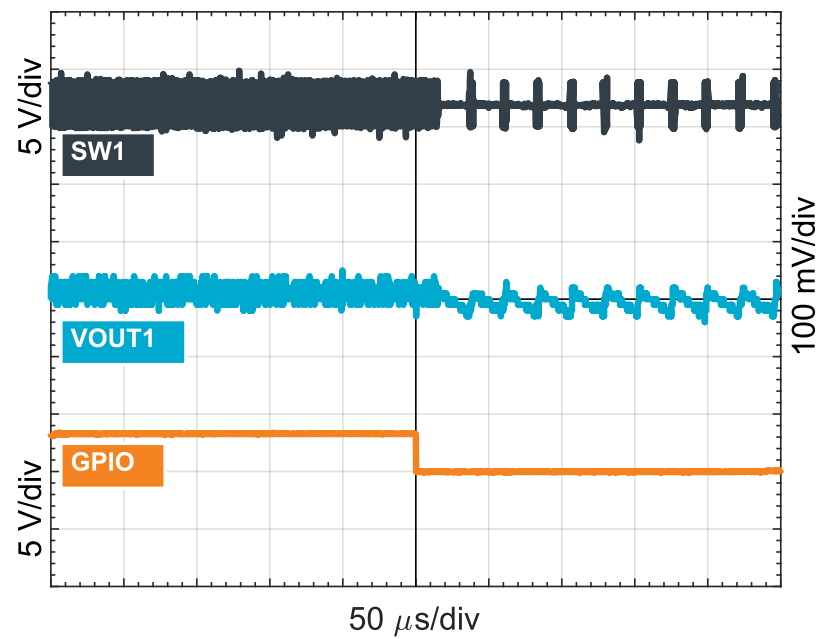


Figure 23: VBAT = 3.8 V: VOUT = 2.0 V: GPIO PFM mode selection

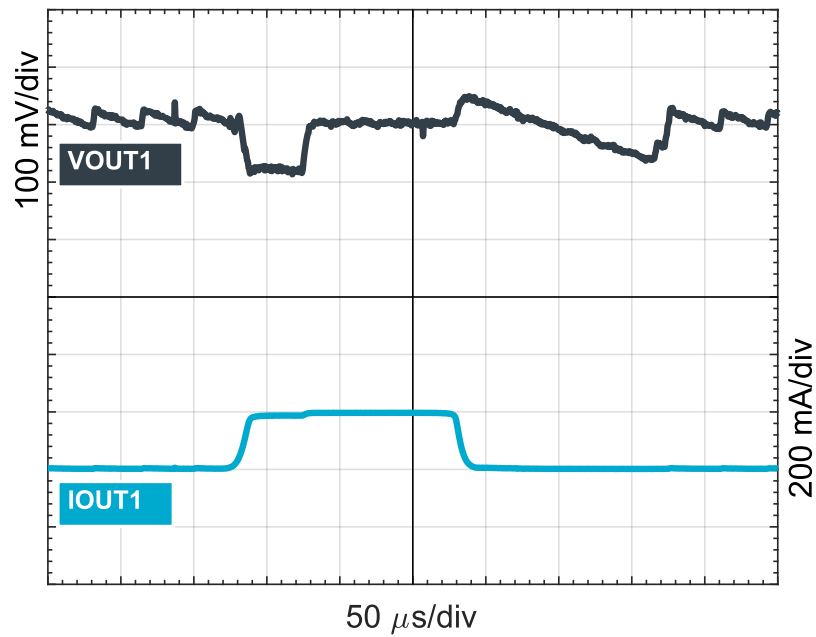


Figure 24: $V_{\text{BAT}} = 3.8 \text{ V}$: $V_{\text{OUT}} = 2.0 \text{ V}$: Auto mode extreme load transient

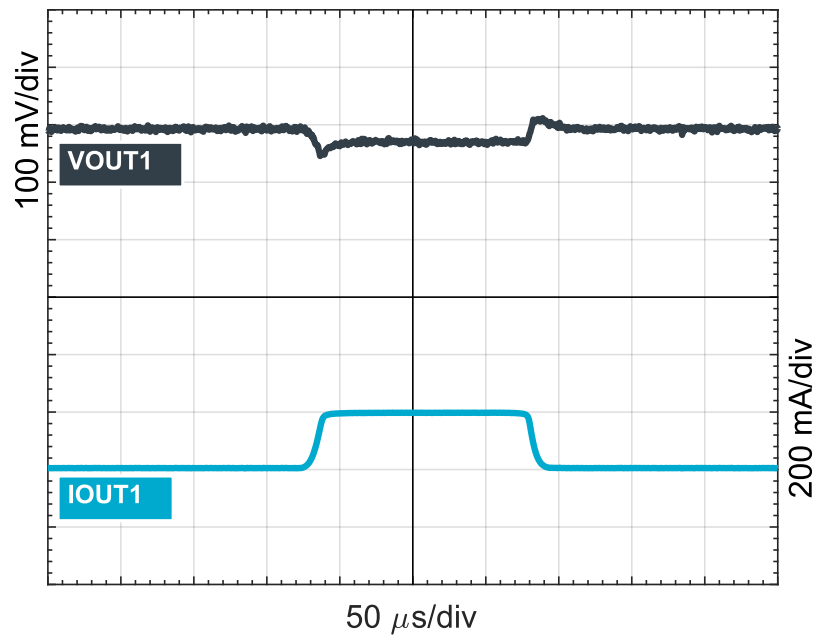


Figure 25: $V_{\text{BAT}} = 3.8 \text{ V}$: $V_{\text{OUT}} = 2.0 \text{ V}$: PWM mode extreme load transient

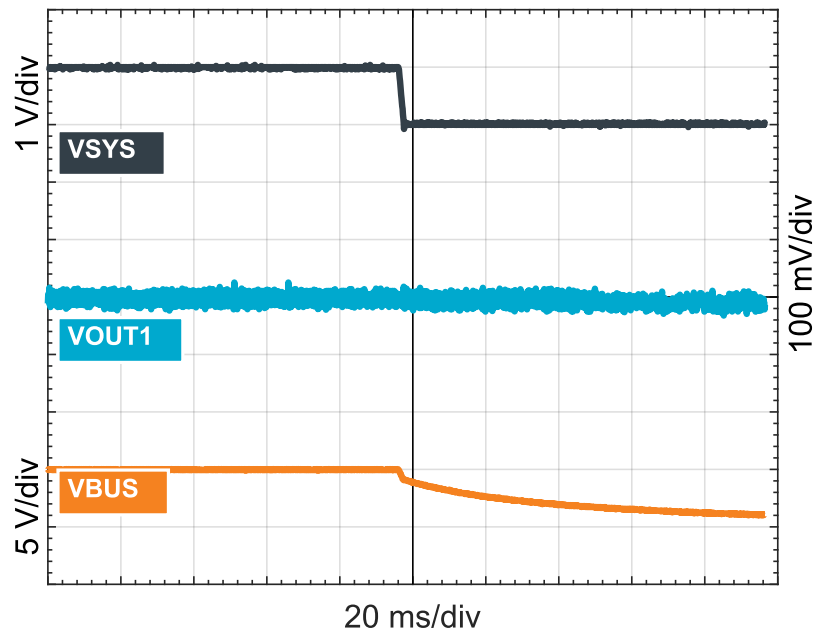


Figure 26: VBAT = 3.8 V: VOUT = 2.0 V: VBUS detach

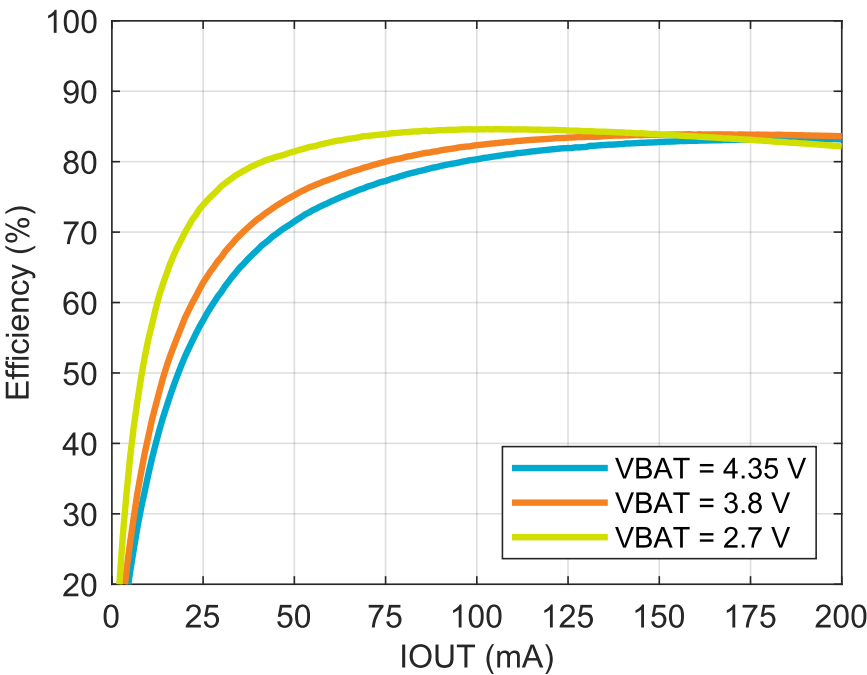


Figure 27: VOUT = 1.0 V: PWM Efficiency

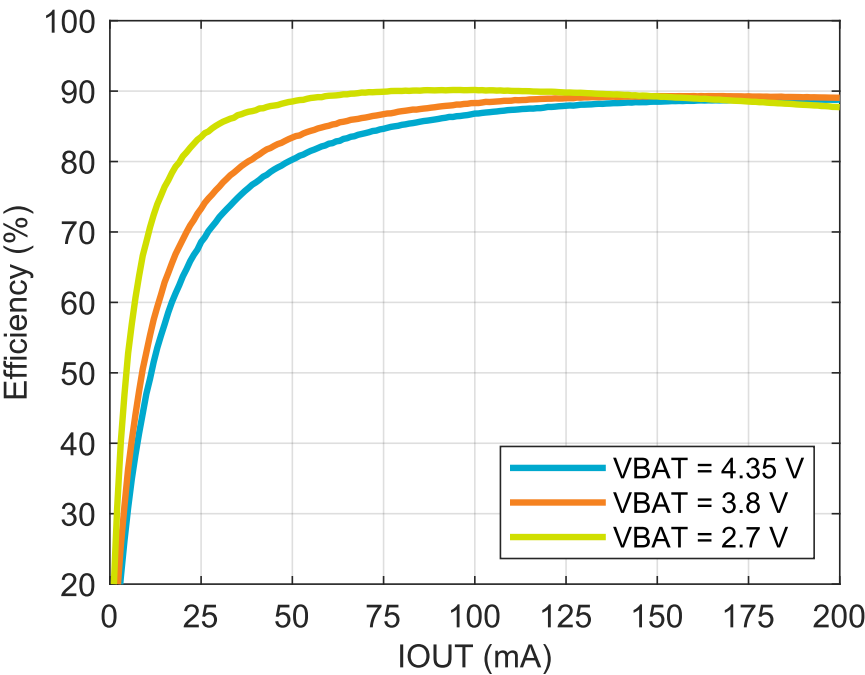


Figure 28: VOUT = 1.8 V: PWM efficiency

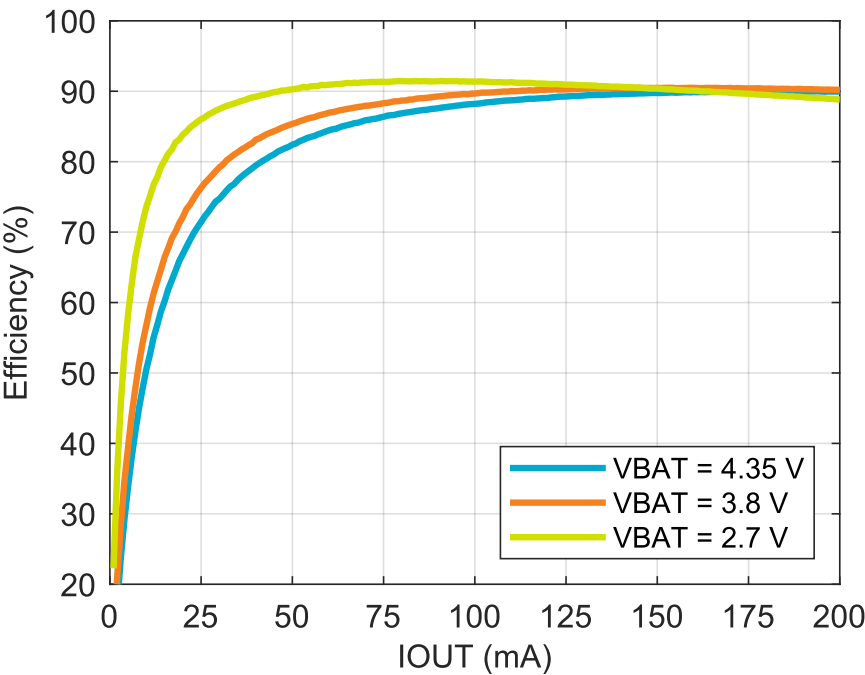


Figure 29: VOUT = 2.1 V: PWM efficiency

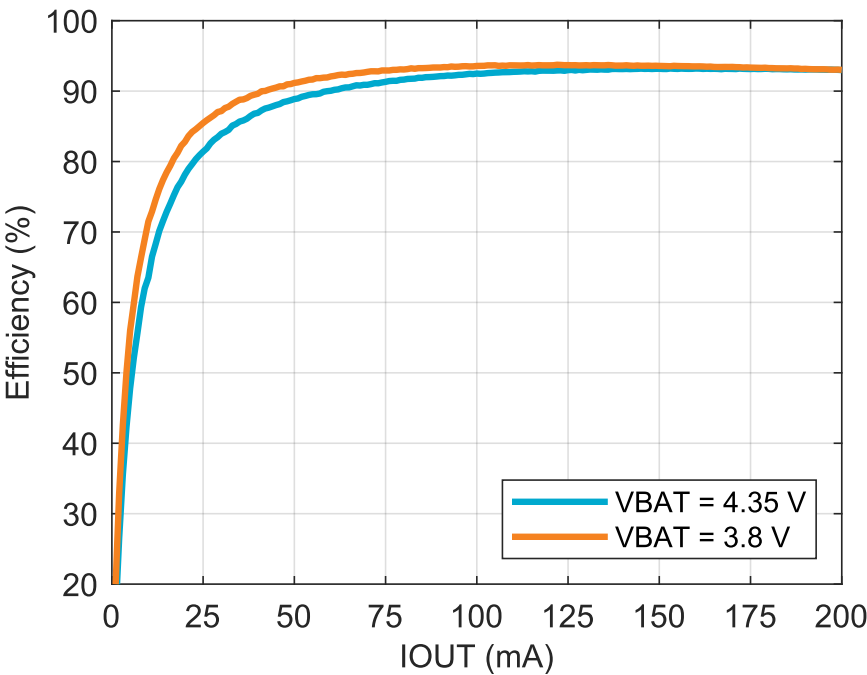


Figure 30: VOUT = 3.3 V: PWM efficiency

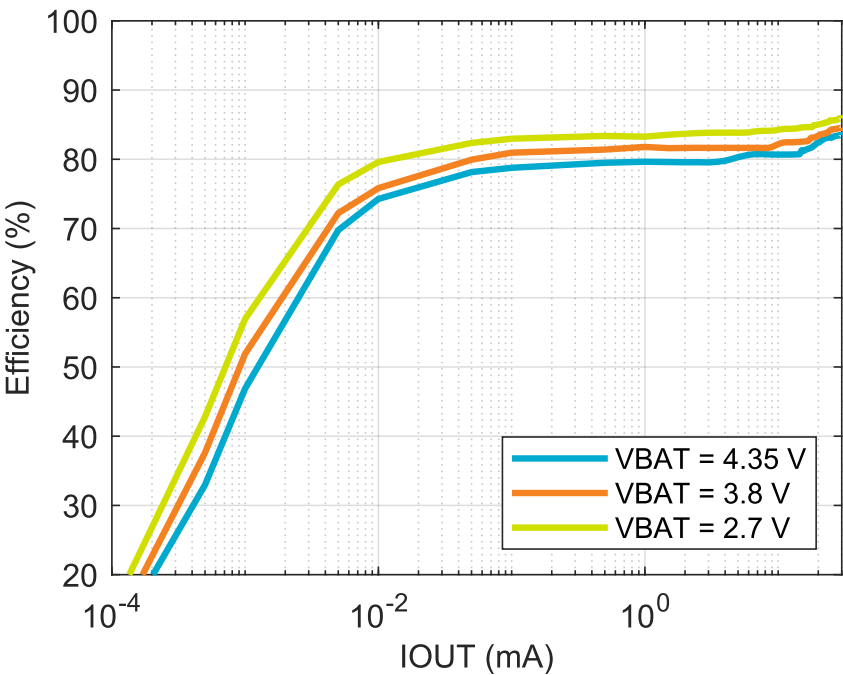


Figure 31: VOUT = 1.0 V: Hysteretic efficiency

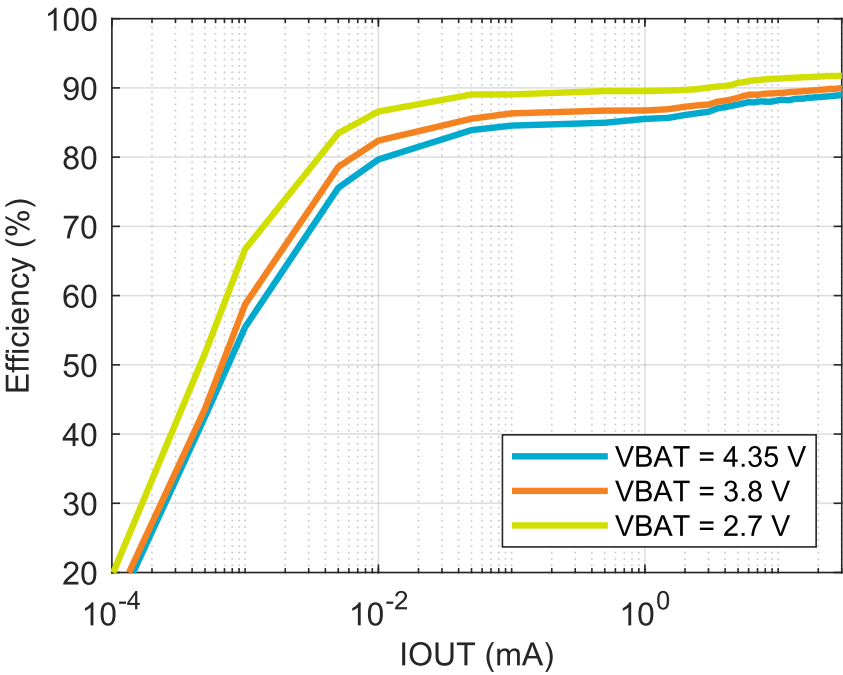


Figure 32: $V_{OUT} = 1.8$ V: Hysteretic efficiency

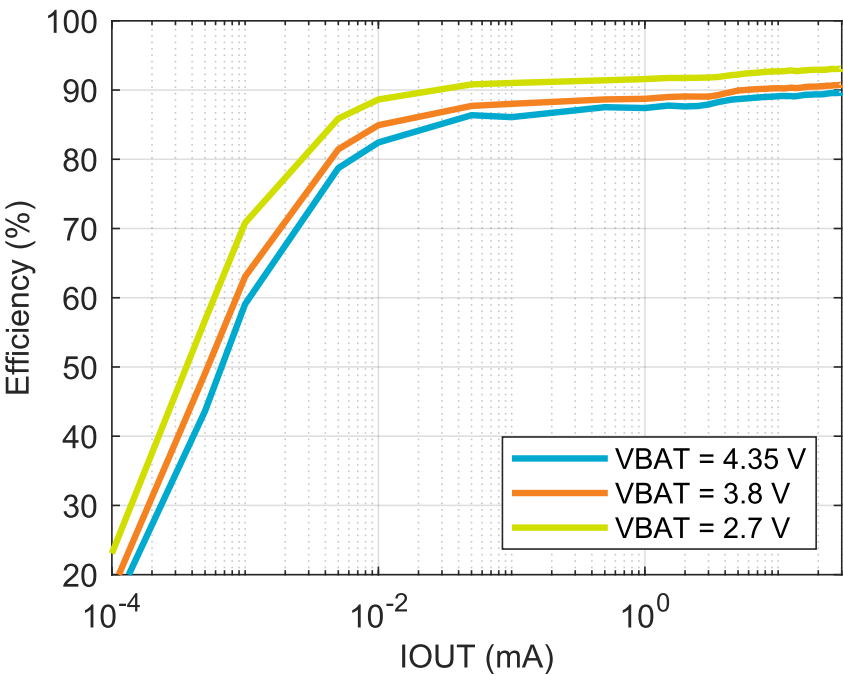


Figure 33: $V_{OUT} = 2.1$ V: Hysteretic efficiency

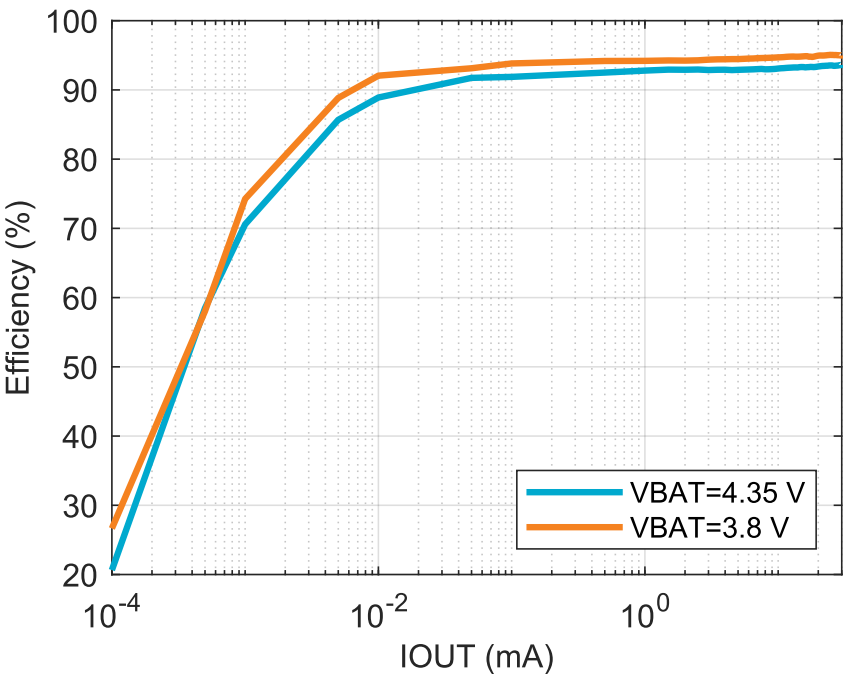


Figure 34: VOUT = 3.3 V: Hysteretic efficiency

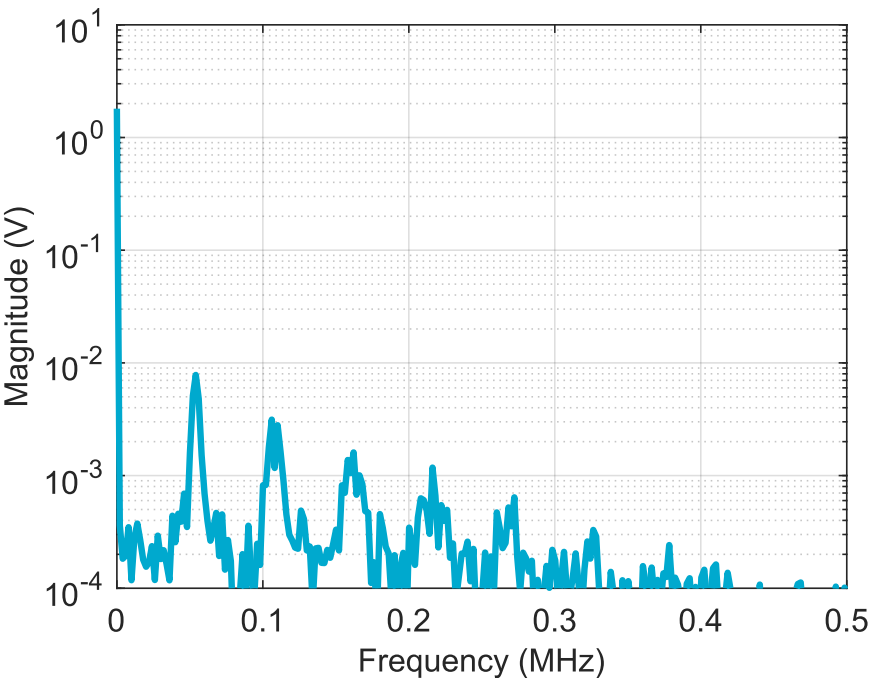


Figure 35: VOUT = 1.8 V: FFT 10 mA: PFM

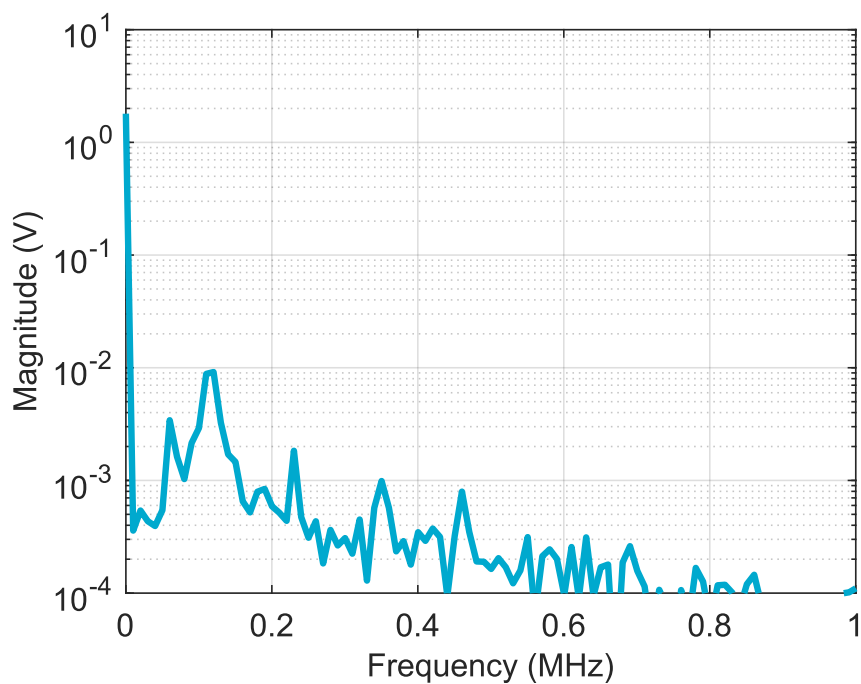


Figure 36: $V_{OUT} = 1.8\text{ V}$: FFT 50 mA: PFM

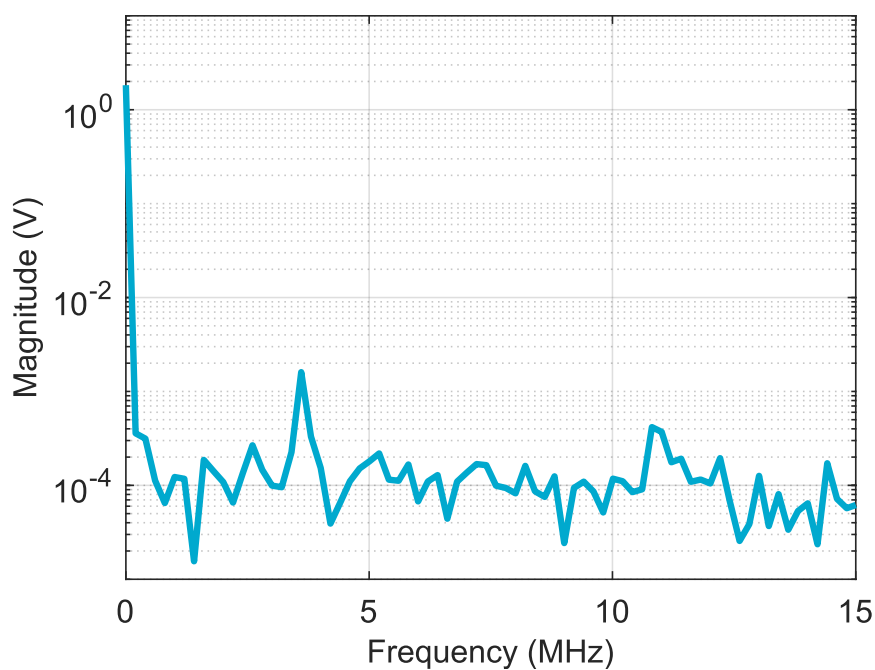


Figure 37: $V_{OUT} = 1.8\text{ V}$: FFT 100 mA: PWM: clock = 3.6 MHz

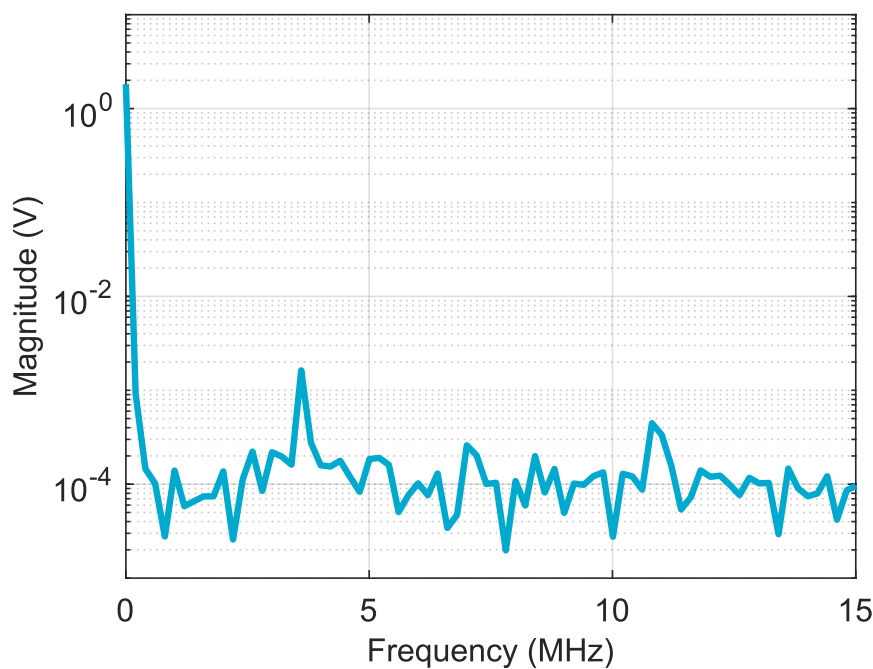


Figure 38: $V_{OUT} = 1.8\text{ V}$; FFT 200 mA; PWM: clock = 3.6 MHz

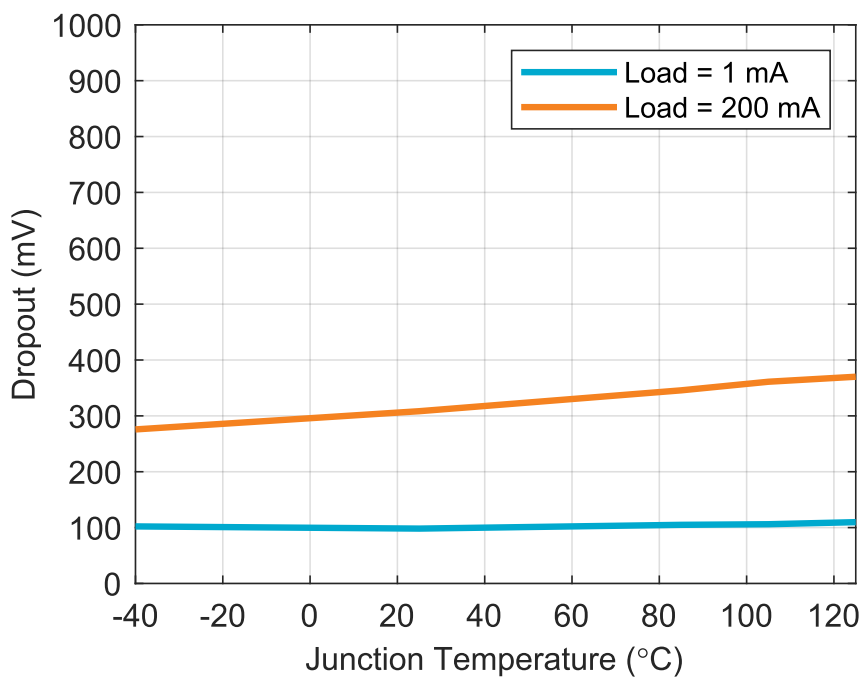


Figure 39: BUCK dropout

6.3.8 Registers

Instances

Instance	Base address	Description
BUCK	0x00000400	BUCK registers
		BUCK register

Register overview

Register	Offset	Description
BUCK1ENASET	0x0	Enable BUCK1
BUCK1ENACLR	0x1	Disable BUCK1
BUCK2ENASET	0x2	Enable BUCK2
BUCK2ENACLR	0x3	Disable BUCK2
BUCK1PWMSET	0x4	Enable PWM mode for BUCK1
BUCK1PWMCLR	0x5	Disable PWM mode for BUCK1
BUCK2PWMSET	0x6	Enable PWM mode for BUCK2
BUCK2PWMCLR	0x7	Disable PWM mode for BUCK2
BUCK1NORMVOUT	0x8	Configure BUCK1 output voltage normal mode
BUCK1RETVOUT	0x9	Configure BUCK1 output voltage retention mode
BUCK2NORMVOUT	0xA	Configure BUCK2 output voltage normal mode
BUCK2RETVOUT	0xB	Configure BUCK2 output voltage retention mode
BUCKENCTRL	0xC	Select GPIOs for BUCK1 and BUCK2 enabling
BUCKVRETCTRL	0xD	Select GPIOs for controlling BUCK1 and BUCK2 retention voltage selection
BUCKPWMCTRL	0xE	Select GPIOs for PWM mode controlling BUCK1 and BUCK2
BUCKSWCTRLSEL	0xF	BUCK output voltage set by pin or register
BUCK1VOUTSTATUS	0x10	BUCK1 output voltage target
BUCK2VOUTSTATUS	0x11	BUCK2 output voltage target
BUCKCTRL0	0x15	BUCK auto mode and pull down control
BUCKSTATUS	0x34	BUCK status register

6.3.8.1 BUCK1ENASET

Address offset: 0x0

Enable BUCK1

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKBUCK1ENASET			Enable BUCK1								
			NOEFFECT	0	No effect								
			SET	1	Enable								

6.3.8.2 BUCK1ENACLR

Address offset: 0x1

Disable BUCK1

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKBUCK1ENACLR			Disable BUCK1								
			NOEFFECT	0	No effect								
			SET	1	Disable								

6.3.8.3 BUCK2ENASET

Address offset: 0x2

Enable BUCK2

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKBUCK2ENASET			Enable BUCK2													
			NOEFFECT	0	No effect													
			SET	1	Enable													

6.3.8.4 BUCK2ENACLR

Address offset: 0x3

Disable BUCK2

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKBUCK2ENACLR			Disable BUCK2													
			NOEFFECT	0	No effect													
			SET	1	Disable													

6.3.8.5 BUCK1PWMSET

Address offset: 0x4

Enable PWM mode for BUCK1

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKBUCK1PWMSET			Enable PWM mode for BUCK1													
			NOEFFECT	0	No effect													
			SET	1	Enable													

6.3.8.6 BUCK1PWMCLR

Address offset: 0x5

Disable PWM mode for BUCK1

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKBUCK1PWMCLR			Disable PWM mode for BUCK1 and return to auto mode								
			NOEFFECT	0	No effect								
			SET	1	Disable								

6.3.8.7 BUCK2PWMSET

Address offset: 0x6

Enable PWM mode for BUCK2

Bit number											7	6	5	4	3	2	1	0
ID																		A
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKBUCK2PWMSET			Enable PWM mode for BUCK2													
			NOEFFECT	0	No effect													
			SET	1	Enable													

6.3.8.8 BUCK2PWMCLR

Address offset: 0x7

Disable PWM mode for BUCK2

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKBUCK2PWMCLR			Disable PWM mode for BUCK2 and return to auto mode													
			NOEFFECT	0	No effect													
			SET	1	Disable													

6.3.8.9 BUCK1NORMVOUT

Address offset: 0x8

Configure BUCK1 output voltage normal mode

Bit number						7	6	5	4	3	2	1	0
ID									A	A	A	A	A
Reset 0x02						0	0	0	0	0	0	1	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BUCK1NORMVOUT			Configure BUCK1 output voltage normal mode								
			1V	0	1.0 V								
			1V1	1	1.1 V								
			1V2	2	1.2 V (default)								
			1V3	3	1.3 V								
			1V4	4	1.4 V								
			1V5	5	1.5 V								
			1V6	6	1.6 V								
			1V7	7	1.7 V								
			1V8	8	1.8 V								
			1V9	9	1.9 V								
			2V0	10	2.0 V								
			2V1	11	2.1 V								
			2V2	12	2.2 V								
			2V3	13	2.3 V								
			2V4	14	2.4 V								
			2V5	15	2.5 V								

Bit number						7	6	5	4	3	2	1	0
ID						A A A A A							
Reset 0x02						0	0	0	0	0	0	1	0
ID	R/W	Field	Value ID	Value	Description								
			2V6	16	2.6 V								
			2V7	17	2.7 V								
			2V8	18	2.8 V								
			2V9	19	2.9 V								
			3V0	20	3.0 V								
			3V1	21	3.1 V								
			3V2	22	3.2 V								
			3V3	23	3.3 V								
			3V30	24	3.3 V								

6.3.8.10 BUCK1RETVOUT

Address offset: 0x9

Configure BUCK1 output voltage retention mode

Bit number						7	6	5	4	3	2	1	0
ID						A A A A A							
Reset 0x02						0	0	0	0	0	0	1	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BUCK1RETVOUT			Configure BUCK1 output voltage retention mode								
			1V	0	1.0 V								
			1V1	1	1.1 V								
			1V2	2	1.2 V (default)								
			1V3	3	1.3 V								
			1V4	4	1.4 V								
			1V5	5	1.5 V								
			1V6	6	1.6 V								
			1V7	7	1.7 V								
			1V8	8	1.8 V								
			1V9	9	1.9 V								
			2V0	10	2.0 V								
			2V1	11	2.1 V								
			2V2	12	2.2 V								
			2V3	13	2.3 V								
			2V4	14	2.4 V								
			2V5	15	2.5 V								
			2V6	16	2.6 V								
			2V7	17	2.7 V								
			2V8	18	2.8 V								
			2V9	19	2.9 V								
			3V0	20	3.0 V								
			3V1	21	3.1 V								
			3V2	22	3.2 V								
			3V3	23	3.3 V								
			3V30	24	3.3 V								

6.3.8.11 BUCK2NORMVOUT

Address offset: 0xA

Configure BUCK2 output voltage normal mode

Bit number						7	6	5	4	3	2	1	0
ID						A A A A A							
Reset 0x08						0	0	0	0	1	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BUCK2NORMVOUT			Configure BUCK2 output voltage normal mode								
			1V	0	1.0 V								
			1V1	1	1.1 V								
			1V2	2	1.2 V								
			1V3	3	1.3 V								
			1V4	4	1.4 V								
			1V5	5	1.5 V								
			1V6	6	1.6 V								
			1V7	7	1.7 V								
			1V8	8	1.8 V (default)								
			1V9	9	1.9 V								
			2V0	10	2.0 V								
			2V1	11	2.1 V								
			2V2	12	2.2 V								
			2V3	13	2.3 V								
			2V4	14	2.4 V								
			2V5	15	2.5 V								
			2V6	16	2.6 V								
			2V7	17	2.7 V								
			2V8	18	2.8 V								
			2V9	19	2.9 V								
			3V0	20	3.0 V								
			3V1	21	3.1 V								
			3V2	22	3.2 V								
			3V3	23	3.3 V								
			3V30	24	3.3 V								

6.3.8.12 BUCK2RETVOUT

Address offset: 0xB

Configure BUCK2 output voltage retention mode

Bit number						7	6	5	4	3	2	1	0
ID						A A A A A							
Reset 0x08						0	0	0	0	1	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BUCK2RETVOUT			Configure BUCK2 output voltage retention mode								
			1V	0	1.0 V								
			1V1	1	1.1 V								
			1V2	2	1.2 V								
			1V3	3	1.3 V								
			1V4	4	1.4 V								
			1V5	5	1.5 V								
			1V6	6	1.6 V								
			1V7	7	1.7 V								
			1V8	8	1.8 V (default)								
			1V9	9	1.9 V								

Bit number						7	6	5	4	3	2	1	0
ID						A A A A A							
Reset 0x08						0	0	0	0	1	0	0	0
ID	R/W	Field	Value ID	Value	Description								
			2V0	10	2.0 V								
			2V1	11	2.1 V								
			2V2	12	2.2 V								
			2V3	13	2.3 V								
			2V4	14	2.4 V								
			2V5	15	2.5 V								
			2V6	16	2.6 V								
			2V7	17	2.7 V								
			2V8	18	2.8 V								
			2V9	19	2.9 V								
			3V0	20	3.0 V								
			3V1	21	3.1 V								
			3V2	22	3.2 V								
			3V3	23	3.3 V								
			3V30	24	3.3 V								

6.3.8.13 BUCKENCTRL

Address offset: 0xC

Select GPIOs for BUCK1 and BUCK2 enabling

Bit number						7	6	5	4	3	2	1	0
ID						D	C	B	B	B	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BUCK1ENGPISEL	Select which GPIO controls BUCK1 enable										
			NOTUSED	0	Not used								
			GPIO0	1	GPIO 0 selected								
			GPIO1	2	GPIO 1 selected								
			GPIO2	3	GPIO 2 selected								
			GPIO3	4	GPIO 3 selected								
			GPIO4	5	GPIO 4 selected								
			NOTUSED1	6	No GPIO selected								
			NOTUSED2	7	No GPIO selected								
B	RW	BUCK2ENGPISEL	Select which GPIO controls BUCK2 enable										
			NOTUSED1	0	Not used								
			GPIO0	1	GPIO 0 selected								
			GPIO1	2	GPIO 1 selected								
			GPIO2	3	GPIO 2 selected								
			GPIO3	4	GPIO 3 selected								
			GPIO4	5	GPIO 4 selected								
			NOTUSED3	6	No GPIO selected								
			NOTUSED4	7	No GPIO selected								
C	RW	BUCK1ENGPIINV	Invert the sense of the selected GPIO for BUCK1										
			NORMAL	0	Not inverted								
			INVERTED	1	Inverted								
D	RW	BUCK2ENGPIINV	Invert the sense of the selected GPIO for BUCK2										
			NORMAL	0	Not inverted								
			INVERTED	1	Inverted								

6.3.8.14 BUCKVRETCTRL

Address offset: 0xD

Select GPIOs for controlling BUCK1 and BUCK2 retention voltage selection

Bit number						7	6	5	4	3	2	1	0
ID						D	C	B	B	A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BUCK1VRETGPISSEL			Select which GPIO controls BUCK1 retention voltage								
			NOTUSED	0	Not used								
			GPIO0	1	GPIO 0 selected								
			GPIO1	2	GPIO 1 selected								
			GPIO2	3	GPIO 2 selected								
			GPIO3	4	GPIO 3 selected								
			GPIO4	5	GPIO 4 selected								
			NOTUSED2	6	Not used								
			NOTUSED1	7	Not used								
B	RW	BUCK2VRETGPISSEL			Select which GPIO controls BUCK2 retention voltage								
			NOTUSED	0	Not used								
			GPIO0	1	GPIO 0 selected								
			GPIO1	2	GPIO 1 selected								
			GPIO2	3	GPIO 2 selected								
			GPIO3	4	GPIO 3 selected								
			GPIO4	5	GPIO 4 selected								
			NOTUSED2	6	Not used								
			NOTUSED1	7	Not used								
C	RW	BUCK1VRETGPININV			Invert the sense of the selected GPIO for BUCK1								
			NORMAL	0	Not inverted								
			INVERTED	1	Inverted								
D	RW	BUCK2VRETGPININV			Invert the sense of the selected GPIO for BUCK2								
			NORMAL	0	Not inverted								
			INVERTED	1	Inverted								

6.3.8.15 BUCKPWMCTRL

Address offset: 0xE

Select GPIOs for PWM mode controlling BUCK1 and BUCK2

Bit number						7	6	5	4	3	2	1	0
ID						D	C	B	B	A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BUCK1PWMGPISSEL											
			NOTUSED1	0	Not used								
			GPIO0	1	GPIO 0 selected								
			GPIO1	2	GPIO 1 selected								
			GPIO2	3	GPIO 2 selected								
			GPIO3	4	GPIO 3 selected								
			GPIO4	5	GPIO 4 selected								
			NOTUSED	6	Not used								
		NOTUSED2	7	Not used									
B	RW	BUCK2PWMGPISSEL			Select which GPIO controls BUCK2 force PWM								

Bit number						7	6	5	4	3	2	1	0
ID						D	C	B	B	A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
			NOTUSED1	0	Not used								
			GPIO0	1	GPIO 0 selected								
			GPIO1	2	GPIO 1 selected								
			GPIO2	3	GPIO 2 selected								
			GPIO3	4	GPIO 3 selected								
			GPIO4	5	GPIO 4 selected								
			NOTUSED	6	Not used								
			NOTUSED2	7	Not used								
C	RW	BUCK1PWMGPIINV			Invert the sense of the selected GPIO for BUCK1								
			NORMAL	0	Not inverted								
			INVERTED	1	Inverted								
D	RW	BUCK2PWMGPIINV			Invert the sense of the selected GPIO for BUCK2								
			NORMAL	0	Not inverted								
			INVERTED	1	Inverted								

6.3.8.16 BUCKSWCTRLSEL

Address offset: 0xF

BUCK output voltage set by pin or register

Bit number											7	6	5	4	3	2	1	0		
ID																			B	A
Reset 0x00											0	0	0	0	0	0	0	0	0	
ID	R/W	Field	Value ID	Value	Description															
A	RW	BUCK1SWCTRLSEL	Output voltage control for BUCK1																	
			VSETANDSWCTRL	0	VSET pin controls output voltage															
			SWCTRL	1	Register controls output voltage															
B	RW	BUCK2SWCTRLSEL	Output voltage control for BUCK2																	
			VSETANDSWCTRL	0	VSET pin controls output voltage															
			SWCTRL	1	Register controls output voltage															

6.3.8.17 BUCK1VOUTSTATUS

Address offset: 0x10

BUCK1 output voltage target

Bit number										7	6	5	4	3	2	1	0
ID										A A A A A A							
Reset 0x00										0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID		Value		Description										
A	R	BUCK1VOUTSTATUS				Selected VOUT target											

6.3.8.18 BUCK2VOUTSTATUS

Address offset: 0x11

BUCK2 output voltage target

Bit number						7	6	5	4	3	2	1	0
ID						A A A A A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	BUCK2VOUTSTATUS			Selected VOUT target								

6.3.8.19 BUCKCTRL0

Address offset: 0x15

BUCK auto mode and pull down control

Bit number						7	6	5	4	3	2	1	0
ID						D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BUCK1AUTOCTRLSEL			BUCK1 control								
			AUTO	0	Select auto switching between PFM and PWM								
			PFM	1	Select PFM mode only								
B	RW	BUCK2AUTOCTRLSEL			BUCK2 control								
			AUTO	0	Select auto switching between PFM and PWM								
			PFM	1	Select PFM mode only								
C	RW	BUCK1ENPULLDOWN			BUCK1 pull down								
			LOW	0	BUCK1 pull down disabled								
			HIGH	1	BUCK1 pull down enabled								
D	RW	BUCK2ENPULLDOWN			BUCK2 pull down								
			LOW	0	BUCK2 pull down disabled								
			HIGH	1	BUCK2 pull down enabled								

6.3.8.20 BUCKSTATUS

Address offset: 0x34

BUCK status register

Bit number						7	6	5	4	3	2	1	0
ID						F	E	D	D	C	B	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	BUCK1MODE			BUCK1 mode								
			AUTOMODE	0	Auto mode								
			PFMMODE	1	PFM mode								
			PWMODE	2	Force PWM mode								
			RFU	3	Reserved (PWM mode)								
B	R	BUCK1PWRGOOD			BUCK1 status								
			BUCKDISABLED	0	BUCK powered off								
			BUCKPOWERED	1	BUCK powered on								
C	R	BUCK1PWMOK			BUCK1 PWM status								
			PWMODEDISABLED0		PWM mode disabled								
			PWMODEENABLED1		PWM mode enabled								
D	R	BUCK2MODE			BUCK2 mode								
			AUTOMODE	0	Auto mode								
			PFMMODE	1	PFM mode								
			PWMODE	2	Force PWM mode								
			RFU	3	Reserved (PWM mode)								

Bit number						7	6	5	4	3	2	1	0
ID						F	E	D	D	C	B	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
E	R	BUCK2PWRGOOD			BUCK2 status								
			BUCKDISABLED	0	BUCK powered off								
			BUCKPOWERED	1	BUCK powered on								
F	R	BUCK2PWMOK			BUCK2 PWM status								
			PWMMODEDISABLED	0	PWM mode disabled								
			PWMMODEENABLED	1	PWM mode enabled								

6.4 LOADSW/LDO — Load switches/LDO regulators

Two load switches are available for use as switches or LDO regulators (LDO). They have dedicated input pins where voltage cannot exceed V_{SYS} . The input voltage can be equal to V_{OUT1} , V_{OUT2} , or any voltage up to the V_{SYS} voltage.

The mode is selected using registers [LDSW\[n\]LDOSEL](#).

The load switches (and LDOs) are OFF by default and can be controlled through control registers or GPIO pins using the following bits.

- Control register bits for each load switch or LDO:
 - [TASK.LDSW\[n\]SET](#)
 - [TASK.LDSW\[n\]CLR](#)
- GPIO[n], once configured by host software:
 - [LDSW\[n\]GPISEL](#)

When a GPIO is used to control LOADSW/LDO, it uses edges. When the GPIO toggles from LOW to HIGH, LOADSW/LDO turns ON (switch conducting). When the GPIO toggles from HIGH to LOW, LOADSW/LDO turns OFF.

Each load switch or LDO can be assigned to a separate GPIO, or a single GPIO can control both switches or LDOs.

A pull-down resistor R_{LSPD} on the **LSOUT [n]** pin is enabled in a register bit for LOADSW/LDO. See register [LDSW.LDSWCONFIG](#).

Load switch mode

Soft start is enabled by default. This ensures that the current to **LSOUT [n]** is limited for t_{SS} . Once soft start is complete, there will be no current limiting provided by the load switch. Current limiting is only available from the power source connected to **LSIN [n]**.

The soft start current limit for a load switch can be set in register [LDSW.LDSWCONFIG](#).

LDO mode

The load switches can be independently configured as LDOs. The LDO output voltage is configurable in registers [LDSW\[n\]VOUTSEL](#).

The LDO can be supplied from V_{OUT1} , V_{OUT2} or V_{SYS} , but must comply with V_{INLDO} .

6.4.1 Electrical specification

Symbol	Description	Min.	Typ.	Max.	Unit
$R_{DS_{ON_{LS}}}$	Switch on-resistance $LS_{IN} = 3.3\text{ V}$		200		m Ω
I_{LS}	Current $LS_{OUT} \geq 1.2\text{ V}$			100	mA
t_{SS}	Soft start time Soft start current limit = 25 mA, 10 μ F, 0 V to 5 V		1.8		ms
R_{LSPD}	Pull-down resistor (active discharge) at LS_{OUT}		2		k Ω
$V_{IN_{LS}}$	Input voltage range	1.0		V_{SYS}	V
$I_{Q_{LS}}$	Additional quiescent current from VBAT when load switch is enabled ($V_{BAT} = 3.8\text{ V}$, $V_{IN_{LS}} = V_{SYS}$, $I_{LOAD} = 0$, VBUS disconnected, both BUCKs running in Hysteretic mode)		60		nA

Table 22: LOADSW electrical specification

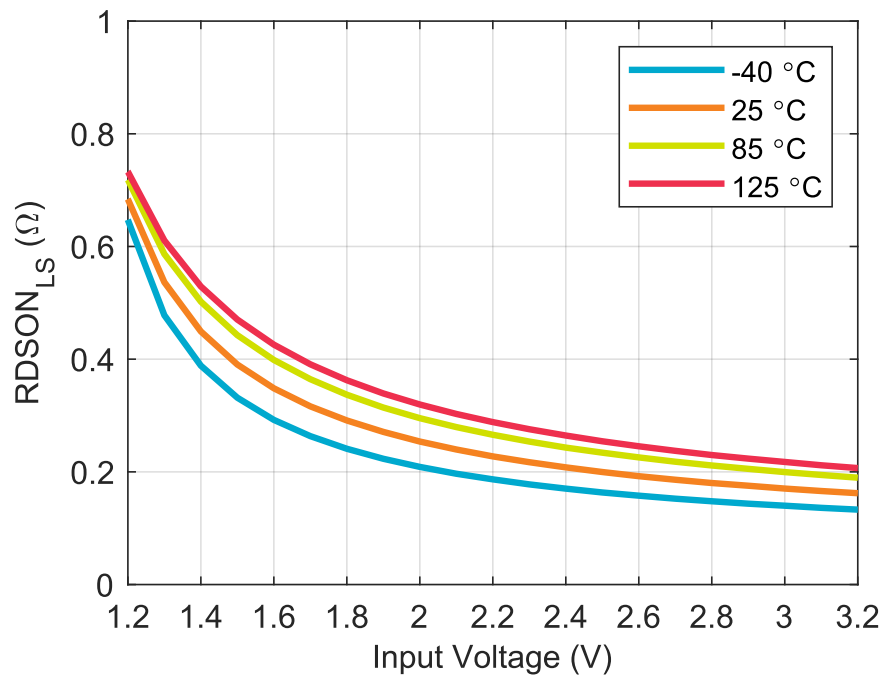
Symbol	Description	Min.	Typ.	Max.	Unit
I_{OUT_LDO}	Output current $V_{OUT} > 1.2\text{ V}$			50	mA
I_{OUT_LDO}	Output current $V_{OUT} < 1.2\text{ V}$			10	mA
V_{IN_LDO}	Input voltage range	2.6		V_{SYS}	V
V_{OUT_LDO}	Minimum setting output voltage		1.0		V
V_{OUT_LDO}	Maximum setting output voltage		3.3		V
$V_{OUT_LDO_STEP}$	Output voltage step size		100		mV
V_{OUT_ACC}	DC output voltage accuracy ($V_{IN_LDO} = 3.8\text{ V}$ to 5.5 V , $I_{LOAD} = 1\text{ mA}$)		± 3		%
$LINE_{REG_LDO}$	DC line regulation ($V_{OUT_LDO} = 1.0\text{ V}$, $V_{IN_LDO} = 3\text{ V}$ to 5.5 V , $I_{LOAD} = 1\text{ mA}$)		2		mV/V
	DC line regulation ($V_{OUT_LDO} = 1.8\text{ V}$, $V_{IN_LDO} = 3\text{ V}$ to 5.5 V , $I_{LOAD} = 1\text{ mA}$)		4		mV/V
	DC line regulation ($V_{OUT_LDO} = 3.3\text{ V}$, $V_{IN_LDO} = 3.7\text{ V}$ to 5.5 V , $I_{LOAD} = 1\text{ mA}$)		11		mV/V
$LOAD_{REG_LDO}$	DC load regulation ($V_{OUT_LDO} = 1\text{ V}$ to 1.2 V , $V_{IN_LDO} = 3.8\text{ V}$, $I_{LOAD} = 50\text{ }\mu\text{A}$ to 10 mA)		0.4		mV/mA
	DC load regulation ($V_{OUT_LDO} = 1.8\text{ V}$ to 3.3 V , $V_{IN_LDO} = 3.8\text{ V}$, $I_{LOAD} = 50\text{ }\mu\text{A}$ to 50 mA)		0.25		mV/mA
PSRR	Power supply rejection ratio ($V_{OUT_LDO} = 3\text{ V}$, $V_{IN_LDO} = 3.8\text{ V}$, $I_{LOAD} = 20\text{ mA}$, $f = 1\text{ kHz}$)		50		dB
V_{RIPPLE}	Output voltage ripple in ULP mode (VBUS disconnected, both BUCKs running in Hysteretic mode, $V_{OUT_LDO} = 1.8\text{ V}$, $V_{IN_LDO} = 3.8\text{ V}$, $I_{LOAD} = 1\text{ mA}$)		17		mV _{p-p}
	Output voltage ripple in ULP mode (VBUS disconnected, both BUCKs running in Hysteretic mode, $V_{OUT_LDO} = 3.3\text{ V}$, $V_{IN_LDO} = 3.8\text{ V}$, $I_{LOAD} = 1\text{ mA}$)		35		mV _{p-p}
$LDO_{DROPOUT}$	Dropout voltage $V_{OUT_LDO} = 3.3\text{ V}$, $I_{LOAD} = 50\text{ mA}$		25		mV
I_{Q_LDO}	Additional quiescent current from VBAT when LDO is enabled (VBAT = 3.8 V , $V_{IN_LDO} = V_{SYS}$, $I_{LOAD} = 0$, VBUS disconnected, both BUCKs running in Hysteretic mode)		500		nA
	Additional quiescent current from VBAT when LDO is enabled (VBAT = 3.8 V , V_{IN_LDO}		10		μA

Symbol	Description	Min.	Typ.	Max.	Unit
	= V _{SYS} , I _{LOAD} = 0, V _{BUS} disconnected, both BUCKs running in PWM mode)				

Table 23: LDO electrical specification

6.4.2 Electrical characteristics

The following graphs show typical electrical characteristics for LOADSW.

Figure 40: LOADSW $R_{DS_{ON}}$ vs. junction temperature

The following graphs show electrical characteristics for LDO.

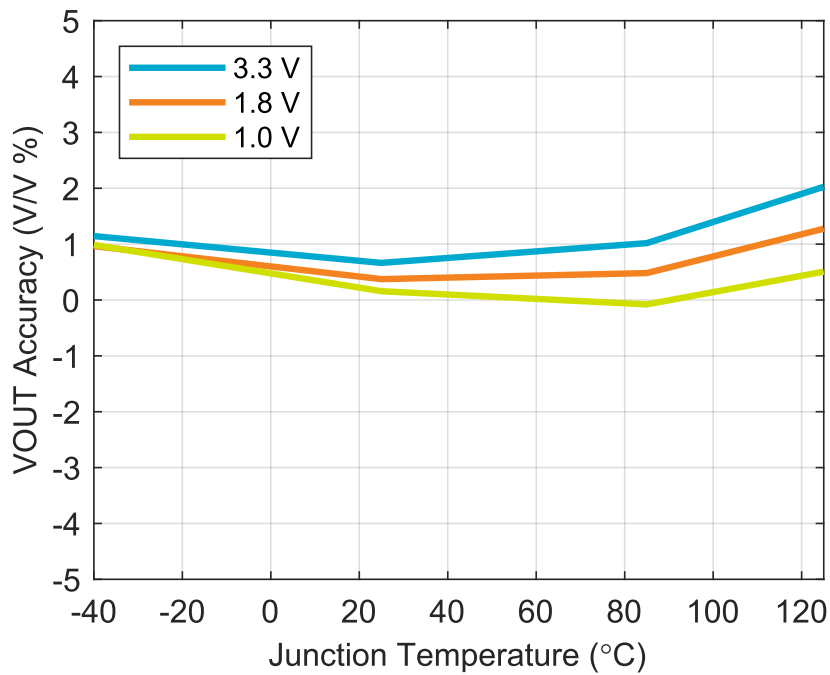


Figure 41: LDO voltage accuracy vs. junction temperature (VBUS = 5.5 V)

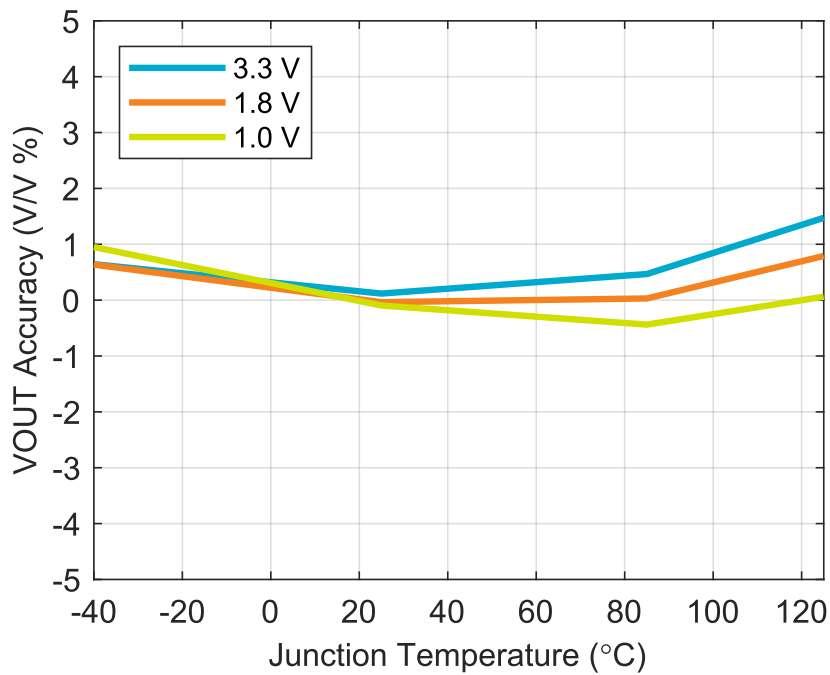


Figure 42: LDO voltage accuracy vs. junction temperature (VBAT = 3.8 V)

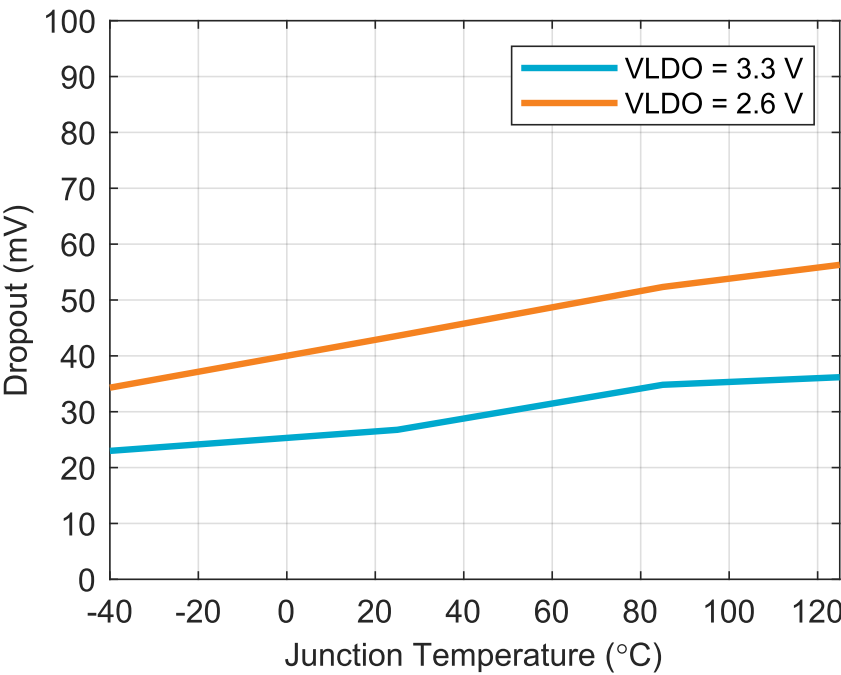


Figure 43: LDO dropout vs. junction temperature

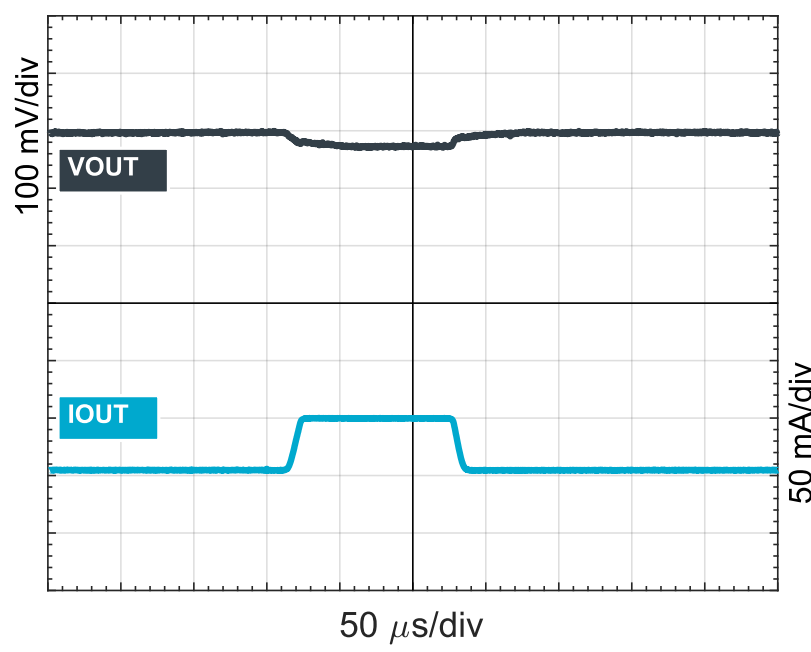


Figure 44: VINLDO = 3.8 V: VOUTLDO = 1.8 V: LDO load transient

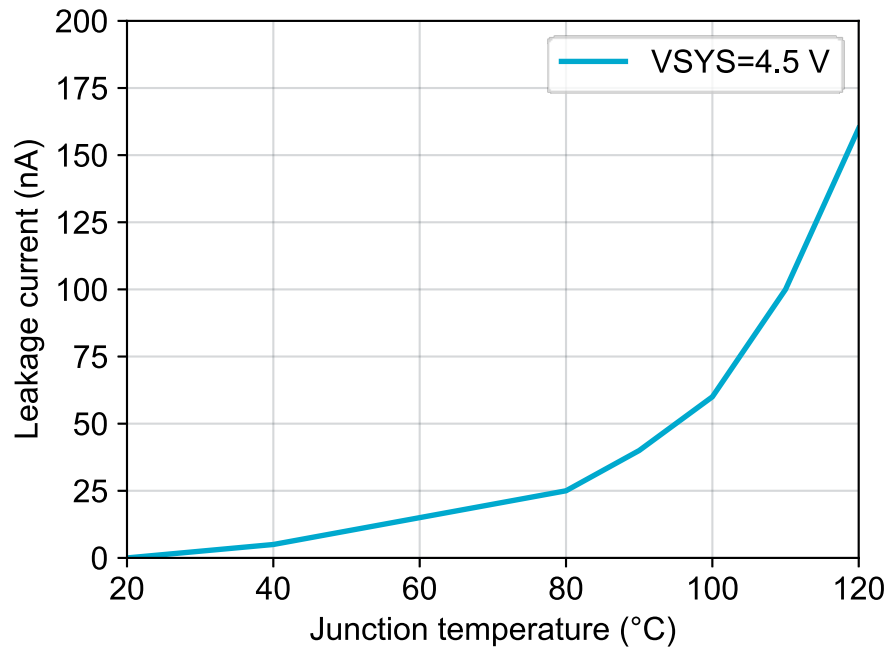


Figure 45: $V_{OUTLDO} = 1.8\text{ V}$ or 3.0 V : Leakage current vs. junction temperature (no load)

6.4.3 Registers

Instances

Instance	Base address	Description
LDSW	0x00000800	LOADSW registers LDSW register map

Register overview

Register	Offset	Description
TASKLDSW1SET	0x0	Enable LDSW1
TASKLDSW1CLR	0x1	Disable LDSW1
TASKLDSW2SET	0x2	Enable LDSW2
TASKLDSW2CLR	0x3	Disable LDSW2
LDSWSTATUS	0x4	Load switch and LDO status
LDSW1GPISEL	0x5	LDSW1 GPIO control select
LDSW2GPISEL	0x6	LDSW2 GPIO control select
LDSWCONFIG	0x7	Load switch or LDO configuration
LDSW1LDOSEL	0x8	Select LDSW1 mode
LDSW2LDOSEL	0x9	Select LDSW2 mode
LDSW1VOUTSEL	0xC	LDO1 programmable output voltage
LDSW2VOUTSEL	0xD	LDO2 programmable output voltage

6.4.3.1 TASKLDSW1SET

Address offset: 0x0

Enable LDSW1

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKLDSW1SET			Set LDSW1 enable								
			NOEFFECT	0	No effect								
			SET	1	Set enable								

6.4.3.2 TASKLDSW1CLR

Address offset: 0x1

Disable LDSW1

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKLDSW1CLR			Clear LDSW1 enable								
			NOEFFECT	0	No effect								
			CLR	1	Clear enable								

6.4.3.3 TASKLDSW2SET

Address offset: 0x2

Enable LDSW2

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKLDSW2SET			Set LDSW2 enable								
			NOEFFECT	0	No effect								
			SET	1	Set enable								

6.4.3.4 TASKLDSW2CLR

Address offset: 0x3

Disable LDSW2

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKLDSW2CLR			Clear LDSW2 enable								
			NOEFFECT	0	No effect								
			CLR	1	Clear enable								

6.4.3.5 LDSWSTATUS

Address offset: 0x4

Load switch and LDO status

Bit number						7	6	5	4	3	2	1	0
ID						E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	LDSW1PWRUPLDSW			Load switch 1 status								
			OFF	0	OFF								
			ON	1	ON								
B	R	LDSW1PWRUPLDO			LDO1 status								
			OFF	0	OFF								
			ON	1	ON								
C	R	LDSW2PWRUPLDSW			Load switch 2 status								
			OFF	0	OFF								
			ON	1	ON								
D	R	LDSW2PWRUPLDO			LDO2 status								
			OFF	0	OFF								
			ON	1	ON								
E	R	LDSWENABLE			LDSW1 or LDSW2 in use								
			OFF	0	OFF								
			ON	1	ON								

6.4.3.6 LDSW1GPISEL

Address offset: 0x5

LDSW1 GPIO control select

Bit number						7	6	5	4	3	2	1	0
ID						B A A A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	LDSW1GPISEL			Select which GPIO controls LDSW1								
			NOTUSED1	0	No GPIO selected								
			GPIO0	1	GPIO 0 selected								
			GPIO1	2	GPIO 1 selected								
			GPIO2	3	GPIO 2 selected								
			GPIO3	4	GPIO 3 selected								
			GPIO4	5	GPIO 4 selected								
			NOTUSED2	6	No GPIO selected								
			NOTUSED3	7	No GPIO selected								
B	RW	LDSW1GPIINV			Invert the sense of the selected GPIO								
			NORMAL	0	Not inverted								
			INVERTED	1	Inverted								

6.4.3.7 LDSW2GPISEL

Address offset: 0x6

LDSW2 GPIO control select

Bit number						7	6	5	4	3	2	1	0
ID						B A A A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	LDSW2GPISEL			Select which GPIO controls LDSW2								
			NOTUSED1	0	No GPIO selected								

Bit number						7	6	5	4	3	2	1	0
ID						B				A	A	A	
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
			GPIO0	1	GPIO 0 selected								
			GPIO1	2	GPIO 1 selected								
			GPIO2	3	GPIO 2 selected								
			GPIO3	4	GPIO 3 selected								
			GPIO4	5	GPIO 4 selected								
			NOTUSED2	6	No GPIO selected								
			NOTUSED3	7	No GPIO selected								
B	RW	LDSW2GPIINV			Invert the sense of the selected GPIO								
			NORMAL	0	Not inverted								
			INVERTED	1	Inverted								

6.4.3.8 LDSWCONFIG

Address offset: 0x7

Load switch or LDO configuration

Bit number						7	6	5	4	3	2	1	0
ID						F	E	D	D	C	C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	LDSW1SOFTSTARTDISABLE			LDSW1 soft start disable								
			NOEFFECT	0	No effect								
			NOSOFTSTART	1	Soft start disabled								
B	RW	LDSW2SOFTSTARTDISABLE			LDSW2 soft start disable								
			NOEFFECT	0	No effect								
			NOSOFTSTART	1	Soft start disabled								
C	RW	LDSW1SOFTSTARTSEL			Select soft start level for LDSW1								
			25MA	0	25 mA								
			50MA	1	50 mA								
			75MA	2	75 mA								
			100MA	3	100 mA								
D	RW	LDSW2SOFTSTARTSEL			Select soft start level for LDSW2								
			25MA	0	25 mA								
			50MA	1	50 mA								
			75MA	2	75 mA								
			100MA	3	100 mA								
E	RW	LDSW1ACTIVEDISCHARGE			LDSW1 active discharge enable								
			NODISCHARGE	0	No discharge								
			ACTIVE	1	Active discharge enabled								
F	RW	LDSW2ACTIVEDISCHARGE			LDSW2 active discharge enable								
			NODISCHARGE	0	No discharge								
			ACTIVE	1	Active discharge enabled								

6.4.3.9 LDSW1LDOSEL

Address offset: 0x8

Select LDSW1 mode

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00													0 0 0 0 0 0 0 0
ID	R/W	Field	Value ID	Value	Description								
A	RW	LDSW1LDOSEL			Select Load switch or LDO								
			LDSW	0	Load switch								
			LDO	1	LDO								

6.4.3.10 LDSW2LDOSEL

Address offset: 0x9

Select LDSW2 mode

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00													0 0 0 0 0 0 0 0
ID	R/W	Field	Value ID	Value	Description								
A	RW	LDSW2LDOSEL			Select Load switch or LDO								
			LDSW	0	Load switch								
			LDO	1	LDO								

6.4.3.11 LDSW1VOUTSEL

Address offset: 0xC

LDO1 programmable output voltage

Bit number						7	6	5	4	3	2	1	0
ID													A A A A A
Reset 0x00													0 0 0 0 0 0 0 0
ID	R/W	Field	Value ID	Value	Description								
A	RW	LDSW1VOUTSEL			LDO1 programmable output voltage								
			1V	0	1.0 V								
			1V1	1	1.1 V								
			1V2	2	1.2 V								
			1V3	3	1.3 V								
			1V4	4	1.4 V								
			1V5	5	1.5 V								
			1V6	6	1.6 V								
			1V7	7	1.7 V								
			1V8	8	1.8 V								
			1V9	9	1.9 V								
			2V	10	2.0 V								
			2V1	11	2.1 V								
			2V2	12	2.2 V								
			2V3	13	2.3 V								
			2V4	14	2.4 V								
			2V5	15	2.5 V								
			2V6	16	2.6 V								
			2V7	17	2.7 V								
			2V8	18	2.8 V								
			2V9	19	2.9 V								
			3V	20	3.0 V								
			3V1	21	3.1 V								

Bit number						7	6	5	4	3	2	1	0
ID						A A A A A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
			3V2	22	3.2 V								
			3V3	23	3.3 V								

6.4.3.12 LDSW2VOUTSEL

Address offset: 0xD

LDO2 programmable output voltage

Bit number						7	6	5	4	3	2	1	0
ID						A A A A A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	LDSW2VOUTSEL			LDO2 programmable output voltage								
			1V	0	1.0 V								
			1V1	1	1.1 V								
			1V2	2	1.2 V								
			1V3	3	1.3 V								
			1V4	4	1.4 V								
			1V5	5	1.5 V								
			1V6	6	1.6 V								
			1V7	7	1.7 V								
			1V8	8	1.8 V								
			1V9	9	1.9 V								
			2V	10	2.0 V								
			2V1	11	2.1 V								
			2V2	12	2.2 V								
			2V3	13	2.3 V								
			2V4	14	2.4 V								
			2V5	15	2.5 V								
			2V6	16	2.6 V								
			2V7	17	2.7 V								
			2V8	18	2.8 V								
			2V9	19	2.9 V								
			3V	20	3.0 V								
			3V1	21	3.1 V								
			3V2	22	3.2 V								
			3V3	23	3.3 V								

6.5 LEDDRV — LED drivers

LEDDRV is made of three identical low-side LED drivers on pins **LED0**, **LED1**, and **LED2**. Pin configurations are independent of each other.

The pins can be configured in registers for the following purposes:

- Charge indication
- Charge error indication
- An RGB LED (requires all three pins)
- A general purpose, open-drain output

When a pin is used as a charging indication, the charging state machine controls LEDDRV.

Pins that are used as general purpose LED drivers have a control register containing separate bits for enabling each driver, see registers [LEDDRV0SET](#) on page 87 and [LEDDRV0CLR](#) on page 88. The host software will set or reset the control register bit, which alters the state of the LED associated with that register bit.

LEDDRV can be used as open-drain digital output. Open Drain mode is the same as the general purpose LED drivers but with the LED removed. An external pull up resistor is required for each LED pin operating in Open Drain mode.

The system can control an RGB LED component (or three separate LEDs) . The **LED0**, **LED1**, and **LED2** pins can connect to any of the RGB LED cathodes (low-side). The anodes (common or individual) must be connected to the **VSYS** pin. The R, G, or B value is activated by enabling the associated LED register. Combinations of RG, RB, GB, and RGB are possible.

6.5.1 Electrical specification

Symbol	Description	Min.	Typ.	Max.	Unit
I_{LED}	LED driver current		5		mA
V_{LEDn}	Voltage on pin LED0 , LED1 , and LED2	0.5		VSYS	V

Table 24: LEDDRV electrical specification

6.5.2 Registers

Instances

Instance	Base address	Description
LEDDRV	0x00000A00	LEDDRV registers LEDDRV register map

Register overview

Register	Offset	Description
LEDDRV0MODESEL	0x0	Select LED0 mode
LEDDRV1MODESEL	0x1	Select LED1 mode
LEDDRV2MODESEL	0x2	Select LED2 mode
LEDDRV0SET	0x3	Turn LED0 On
LEDDRV0CLR	0x4	Turn LED0 Off
LEDDRV1SET	0x5	Turn LED1 On
LEDDRV1CLR	0x6	Turn LED1 Off
LEDDRV2SET	0x7	Turn LED2 On
LEDDRV2CLR	0x8	Turn LED2 Off

6.5.2.1 LEDDRV0MODESEL

Address offset: 0x0

Select LED0 mode

Bit number						7	6	5	4	3	2	1	0
ID												A	A
Reset 0x00												0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	LEDDRV0MODESEL			Select LED0 mode								
			ERROR	0	Error condition from charger								
			CHARGING	1	Charging indicator (On during charging)								
			HOST	2	Driven from register LEDDRV0SET/CLR								
			NOTUSED	3	Not used								

6.5.2.2 LEDDRV1MODESEL

Address offset: 0x1

Select LED1 mode

Bit number						7	6	5	4	3	2	1	0
ID												A	A
Reset 0x01												0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	LEDDRV1MODESEL			Select LED1 mode								
			ERROR	0	Error condition from charger								
			CHARGING	1	Charging indicator (On during charging)								
			HOST	2	Driven from register LEDDRV1SET/CLR								
			NOTUSED	3	Not used								

6.5.2.3 LEDDRV2MODESEL

Address offset: 0x2

Select LED2 mode

Bit number						7	6	5	4	3	2	1	0
ID												A	A
Reset 0x02												0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	LEDDRV2MODESEL			Select LED2 mode								
			ERROR	0	Error condition from charger								
			CHARGING	1	Charging indicator (On during charging)								
			HOST	2	Driven from register LEDDRV2SET/CLR								
			NOTUSED	3	Not used								

6.5.2.4 LEDDRV0SET

Address offset: 0x3

Turn LED0 On

Bit number						7	6	5	4	3	2	1	0
ID												A	
Reset 0x00												0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	LEDDRV0ON			Set LED0 to be On								
			NOEFFECT	0	No effect								
			SET	1	Turns On LED0 if LEDDRV0MODESEL = HOST								

6.5.2.5 LEDDRV0CLR

Address offset: 0x4

Turn LED0 Off

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	LEDDRV0OFF			Set LED0 to be Off													
			NOEFFECT	0	No effect													
			CLR	1	Turns Off LED0 if LEDDRVMODESEL = HOST													

6.5.2.6 LEDDRV1SET

Address offset: 0x5

Turn LED1 On

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	LEDDRV1ON			Set LED1 to be On													
			NOEFFECT	0	No effect													
			SET	1	Turns On LED1 if LEDDRVMODESEL = HOST													

6.5.2.7 LEDDRV1CLR

Address offset: 0x6

Turn LED1 Off

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	LEDDRV1OFF			Set LED1 to be Off													
			NOEFFECT	0	No effect													
			CLR	1	Turns Off LED1 if LEDDRVMODESEL = HOST													

6.5.2.8 LEDDRV2SET

Address offset: 0x7

Turn LED2 On

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	LEDDRV2ON			Set LED2 to be On													
			NOEFFECT	0	No effect													
			SET	1	Turns On LED2 if LEDDRVMODESEL = HOST													

Turn LED2 Off

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	LEDDRV2OFF			Set LED2 to be Off								
			NOEFFECT	0	No effect								
			CLR	1	Turns Off LED2 if LEDDRVMODESEL = HOST								

6.6 GPIO — General purpose input/output

By default, the general purpose input/output pins, **GPIO[n]**, are set as input with weak pull-down. GPIO is supplied by the **VDDIO** pin.

The number of GPIOs varies with product variant and package. See [Pin assignments](#) on page 154 for more information about the number of supported GPIOs.

GPIO has the following configurable features:

- General purpose input
- Control input
- Output
- BUCK control
- LOADSW control

Note: Events may occur when GPIO configuration is changed on the fly.

Pull-down is prioritized if both pull-up and pull-down are activated on a **GPIO** pin at the same time.

The following figure shows BUCK control.

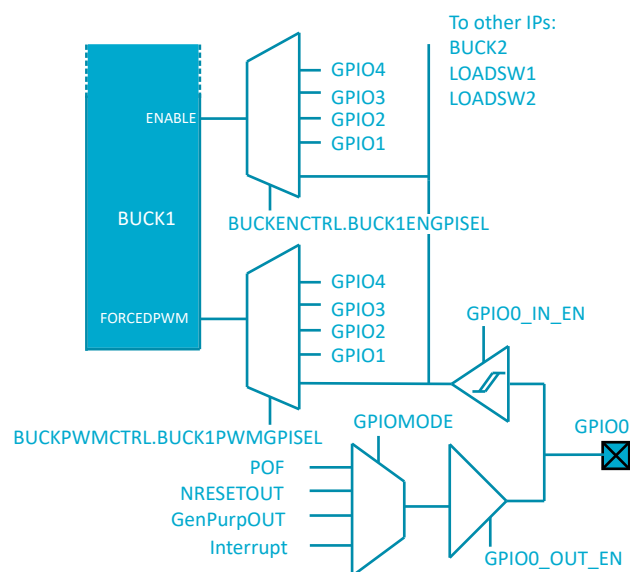


Figure 46: GPIO concept

Pins **LED0**, **LED1**, and **LED2** can be used as open-drain outputs, see [LEDDRV — LED drivers](#) on page 85.

6.6.1 Pin configuration

The GPIO peripheral implements up to 5 pins, **GPIO[0 . . . 4]**. Each of these pins can be individually configured in the **GPIONODE[n]** registers.

General purpose input

GPIO can be used as a general purpose input to monitor the input logic level. Debounce is set in register **GPIO.DEBOUNCE[n]**. Set **GPI.INPUT** to use **GPIO[n]** without setting an event.

It can also be used as an input to trigger an event. Set bit **GPI.RISING.EVENT** to generate an event on the rising edge. To generate an event on a falling edge, set bit **GPI.FALLING.EVENT**. The events are visible in the register **EVENTSGPIOSET** on page 146.

To override GPIO input states, set bit **GPI.LOGIC[n]**.

Control input

For a pin to function as a control input, write 0 in bit **GPI.INPUT**. Debounce is set in register **GPIO.DEBOUNCE[n]**. The following components can be controlled through GPIO once enabled in the corresponding register.

- **LOADSW** – Registers [LDSW1GPISSEL](#) on page 82 or [LDSW2GPISSEL](#) on page 82
- **BUCK** – Register [BUCKENCTRL](#) on page 70
- **BUCK forced PWM mode** – Register [BUCKPWMCTRL](#) on page 71
- **BUCK VOUT[n]** voltage level selection for active and retention modes – Register [BUCKVRETCTRL](#) on page 71
- **Second reset button** – **GPIO0** only, see [Two-button reset](#) on page 125

Output

The GPIO outputs can be configured as logic outputs or open drain outputs in register **GPIO.OPEN.DRAIN[n]**.

When setting a GPIO as output, the host software disables any pull-up or pull-down on that GPIO. After a reset, the default is for pull-down to be enabled.

GPIO can be used as a general purpose output by setting bit **GPO.LOGIC[n]**.

GPIO can be used as an interrupt by setting one or more from the following registers:

- [INTENEVENTSADCSET](#) on page 130
- [INTEN.EVENTS.BCHARGER\[n\]SET](#)
- [INTENEVENTSSHPLDSET](#) on page 140
- [INTEN.EVENTS.VBUSIN\[n\]SET](#)
- [INTENEVENTSGPIOSET](#) on page 148

GPIO can indicate a watchdog event when the watchdog expires. Select bit **GPO.RESET** to enable watchdog events.

An imminent power failure warning can be set by selecting bit **GPO.PLW**.

Drive strength can be selected in register [GPIDRIVE\[0\]](#) on page 94. Weak pull-up and pull-down resistors are available in the following registers:

- [GPIO.PDEN\[n\]](#)
- [GPIO.PUEN\[n\]](#)

6.6.2 Electrical specification

Symbol	Description	Min.	Typ.	Max.	Unit
V_{IH}	Input high voltage	$0.7 \times V_{DDIO}$		V_{DDIO}	V
V_{IL}	Input low voltage	$AVSS$		$0.3 \times V_{DDIO}$	V
PU_{GPIO}	Weak pull-up resistor		500		k Ω
PD_{GPIO}	Weak pull-down resistor		500		k Ω
DB_{GPIO}	Input debounce time (DEBOUNCE1=1)		20		ms

Table 25: GPIO electrical specification

6.6.3 Registers

Instances

Instance	Base address	Description
GPIOs	0x00000600	GPIO Registers GPIOs register map

Register overview

Register	Offset	Description
GPIONODE[0]	0x0	GPIO mode configuration
GPIONODE[1]	0x1	GPIO mode configuration
GPIONODE[2]	0x2	GPIO mode configuration
GPIONODE[3]	0x3	GPIO mode configuration
GPIONODE[4]	0x4	GPIO mode configuration
GPIDRIVE[0]	0x5	GPIO drive strength configuration
GPIDRIVE[1]	0x6	GPIO drive strength configuration
GPIDRIVE[2]	0x7	GPIO drive strength configuration
GPIDRIVE[3]	0x8	GPIO drive strength configuration
GPIDRIVE[4]	0x9	GPIO drive strength configuration
GPIOPUEN[0]	0xA	GPIO pull up enable configuration
GPIOPUEN[1]	0xB	GPIO pull up enable configuration
GPIOPUEN[2]	0xC	GPIO pull up enable configuration
GPIOPUEN[3]	0xD	GPIO pull up enable configuration
GPIOPUEN[4]	0xE	GPIO pull up enable configuration
GIOPDEN[0]	0xF	GPIO pull down enable configuration
GIOPDEN[1]	0x10	GPIO pull down enable configuration
GIOPDEN[2]	0x11	GPIO pull down enable configuration
GIOPDEN[3]	0x12	GPIO pull down enable configuration
GIOPDEN[4]	0x13	GPIO pull down enable configuration
GPIOPENDRAIN[0]	0x14	GPIO open drain configuration
GPIOPENDRAIN[1]	0x15	GPIO open drain configuration
GPIOPENDRAIN[2]	0x16	GPIO open drain configuration
GPIOPENDRAIN[3]	0x17	GPIO open drain configuration
GPIOPENDRAIN[4]	0x18	GPIO open drain configuration

Register	Offset	Description
GPIODEBOUNCE[0]	0x19	GPIO debounce configuration
GPIODEBOUNCE[1]	0x1A	GPIO debounce configuration
GPIODEBOUNCE[2]	0x1B	GPIO debounce configuration
GPIODEBOUNCE[3]	0x1C	GPIO debounce configuration
GPIODEBOUNCE[4]	0x1D	GPIO debounce configuration
GPIOSTATUS	0x1E	GPIO status

6.6.3.1 GPIOMODE[0]

Address offset: 0x0

GPIO mode configuration

Bit number						7	6	5	4	3	2	1	0
ID										A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOMODE			GPIO mode selection								
			GPIINPUT	0	Input								
			GPILOGIC1	1	Input, internally driven high								
			GPILOGIC0	2	Input, internally driven low								
			GPIEVENTRISE	3	Input, event from rising edge								
			GPIEVENTFALL	4	Input, event from falling edge								
			GPOIRQ	5	Interrupt output								
			GPORESET	6	Reset output								
			GPOPLW	7	Power failure warning output								
			GPOLOGIC1	8	Output, pin driven high								
			GPOLOGIC0	9	Output, pin driven low								

6.6.3.2 GPIOMODE[1]

Address offset: 0x1

GPIO mode configuration

Bit number						7	6	5	4	3	2	1	0
ID										A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOMODE			GPIO mode selection								
			GPIINPUT	0	Input								
			GPILOGIC1	1	Input, internally driven high								
			GPILOGIC0	2	Input, internally driven low								
			GPIEVENTRISE	3	Input, event from rising edge								
			GPIEVENTFALL	4	Input, event from falling edge								
			GPOIRQ	5	Interrupt output								
			GPORESET	6	Reset output								
			GPOPLW	7	Power failure warning output								
			GPOLOGIC1	8	Output, pin driven high								
			GPOLOGIC0	9	Output, pin driven low								

6.6.3.3 GPIOMODE[2]

Address offset: 0x2

GPIO mode configuration

Bit number						7	6	5	4	3	2	1	0
ID						A				A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOMODE			GPIO mode selection								
			GPIINPUT	0	Input								
			GPILOGIC1	1	Input, internally driven high								
			GPILOGIC0	2	Input, internally driven low								
			GPIEVENTRISE	3	Input, event from rising edge								
			GPIEVENTFALL	4	Input, event from falling edge								
			GPOIRQ	5	Interrupt output								
			GPORESET	6	Reset output								
			GPOPLW	7	Power failure warning output								
			GPOLOGIC1	8	Output, pin driven high								
			GPOLOGIC0	9	Output, pin driven low								

6.6.3.4 GPIOMODE[3]

Address offset: 0x3

GPIO mode configuration

Bit number						7	6	5	4	3	2	1	0
ID						A				A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOMODE			GPIO mode selection								
			GPIINPUT	0	Input								
			GPILOGIC1	1	Input, internally driven high								
			GPILOGIC0	2	Input, internally driven low								
			GPIEVENTRISE	3	Input, event from rising edge								
			GPIEVENTFALL	4	Input, event from falling edge								
			GPOIRQ	5	Interrupt output								
			GPORESET	6	Reset output								
			GPOPLW	7	Power failure warning output								
			GPOLOGIC1	8	Output, pin driven high								
			GPOLOGIC0	9	Output, pin driven low								

6.6.3.5 GPIOMODE[4]

Address offset: 0x4

GPIO mode configuration

Bit number						7	6	5	4	3	2	1	0
ID						A				A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOMODE			GPIO mode selection								
			GPIINPUT	0	Input								
			GPILOGIC1	1	Input, internally driven high								
			GPILOGIC0	2	Input, internally driven low								

Bit number						7	6	5	4	3	2	1	0
ID						A				A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
			GPIEVENTRISE	3	Input, event from rising edge								
			GPIEVENTFALL	4	Input, event from falling edge								
			GPOIRQ	5	Interrupt output								
			GPORESET	6	Reset output								
			GPOPLW	7	Power failure warning output								
			GPOLOGIC1	8	Output, pin driven high								
			GPOLOGIC0	9	Output, pin driven low								

6.6.3.6 GPIODRIVE[0]

Address offset: 0x5

GPIO drive strength configuration

Bit number						7	6	5	4	3	2	1	0
ID										A			
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIODRIVE			GPIO drive strength								
			1MA	0	1 mA								
			6MA	1	6 mA								

6.6.3.7 GPIODRIVE[1]

Address offset: 0x6

GPIO drive strength configuration

Bit number						7	6	5	4	3	2	1	0
ID										A			
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIODRIVE			GPIO drive strength								
			1MA	0	1 mA								
			6MA	1	6 mA								

6.6.3.8 GPIODRIVE[2]

Address offset: 0x7

GPIO drive strength configuration

Bit number						7	6	5	4	3	2	1	0
ID										A			
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIODRIVE			GPIO drive strength								
			1MA	0	1 mA								
			6MA	1	6 mA								

6.6.3.9 GPIODRIVE[3]

Address offset: 0x8

GPIO drive strength configuration

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIODRIVE			GPIO drive strength								
			1MA	0	1 mA								
			6MA	1	6 mA								

6.6.3.10 GPIODRIVE[4]

Address offset: 0x9

GPIO drive strength configuration

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIODRIVE			GPIO drive strength								
			1MA	0	1 mA								
			6MA	1	6 mA								

6.6.3.11 GPIOPUEN[0]

Address offset: 0xA

GPIO pull up enable configuration

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOPUEN			GPIO pull up enable								
			PULLUP0	0	Pull up disable								
			PULLUP1	1	Pull up enable								

6.6.3.12 GPIOPUEN[1]

Address offset: 0xB

GPIO pull up enable configuration

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOPUEN			GPIO pull up enable								
			PULLUP0	0	Pull up disable								
			PULLUP1	1	Pull up enable								

6.6.3.13 GPIOPUEN[2]

Address offset: 0xC

GPIO pull up enable configuration

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOPUEN			GPIO pull up enable								
			PULLUP0	0	Pull up disable								
			PULLUP1	1	Pull up enable								

6.6.3.14 GPIOPUEN[3]

Address offset: 0xD

GPIO pull up enable configuration

Bit number										7	6	5	4	3	2	1	0
ID										A							
Reset 0x00										0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description												
A	RW	GPIOPUEN			GPIO pull up enable												
			PULLUP0	0	Pull up disable												
			PULLUP1	1	Pull up enable												

6.6.3.15 GPIOPUEN[4]

Address offset: 0xE

GPIO pull up enable configuration

Bit number										7	6	5	4	3	2	1	0
ID										A							
Reset 0x00										0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description												
A	RW	GPIOPUEN			GPIO pull up enable												
			PULLUP0	0	Pull up disable												
			PULLUP1	1	Pull up enable												

6.6.3.16 GPIOPDEN[0]

Address offset: 0xF

GPIO pull down enable configuration

Bit number										7	6	5	4	3	2	1	0
ID										A							
Reset 0x01										0	0	0	0	0	0	0	1
ID	R/W	Field	Value ID	Value	Description												
A	RW	GPIOPDEN			GPIO pull down enable												
			PULLDOWN0	0	Pull down disable												
			PULLDOWN1	1	Pull down enable												

6.6.3.17 GPIOPDEN[1]

Address offset: 0x10

GPIO pull down enable configuration

Bit number											7	6	5	4	3	2	1	0
ID																		A
Reset 0x01											0	0	0	0	0	0	0	1
ID	R/W	Field	Value ID	Value	Description													
A	RW	GPIOPDEN			GPIO pull down enable													
			PULLDOWN0	0	Pull down disable													
			PULLDOWN1	1	Pull down enable													

6.6.3.18 GPIOPDEN[2]

Address offset: 0x11

GPIO pull down enable configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x01											0	0	0	0	0	0	0	1
ID	R/W	Field	Value ID	Value	Description													
A	RW	GPIOPDEN			GPIO pull down enable													
			PULLDOWN0	0	Pull down disable													
			PULLDOWN1	1	Pull down enable													

6.6.3.19 GPIOPDEN[3]

Address offset: 0x12

GPIO pull down enable configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x01											0	0	0	0	0	0	0	1
ID	R/W	Field	Value ID	Value	Description													
A	RW	GPIOPDEN			GPIO pull down enable													
			PULLDOWN0	0	Pull down disable													
			PULLDOWN1	1	Pull down enable													

6.6.3.20 GPIOPDEN[4]

Address offset: 0x13

GPIO pull down enable configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x01											0	0	0	0	0	0	0	1
ID	R/W	Field	Value ID	Value	Description													
A	RW	GPIOPDEN			GPIO pull down enable													
			PULLDOWN0	0	Pull down disable													
			PULLDOWN1	1	Pull down enable													

6.6.3.21 GPIOOPENDRAIN[0]

Address offset: 0x14

GPIO open drain configuration

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOOPENDRAIN			GPIO open drain								
			OPENDRAIN0	0	Open drain disable								
			OPENDRAIN1	1	Open drain enable								

6.6.3.22 GPIOOPENDRAIN[1]

Address offset: 0x15

GPIO open drain configuration

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOOPENDRAIN			GPIO open drain								
			OPENDRAIN0	0	Open drain disable								
			OPENDRAIN1	1	Open drain enable								

6.6.3.23 GPIOOPENDRAIN[2]

Address offset: 0x16

GPIO open drain configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	RW	GPIOOPENDRAIN			GPIO open drain													
			OPENDRAIN0	0	Open drain disable													
			OPENDRAIN1	1	Open drain enable													

6.6.3.24 GPIOOPENDRAIN[3]

Address offset: 0x17

GPIO open drain configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	RW	GPIOOPENDRAIN			GPIO open drain													
			OPENDRAIN0	0	Open drain disable													
			OPENDRAIN1	1	Open drain enable													

6.6.3.25 GPIOOPENDRAIN[4]

Address offset: 0x18

GPIO open drain configuration

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	GPIOOPENDRAIN			GPIO open drain								
			OPENDRAIN0	0	Open drain disable								
			OPENDRAIN1	1	Open drain enable								

6.6.3.26 GPIODEBOUNCE[0]

Address offset: 0x19

GPIO debounce configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	RW	GPIODEBOUNCE			GPIO debounce													
			DEBOUNCE0	0	Debounce disable													
			DEBOUNCE1	1	Debounce enable													

6.6.3.27 GPIODEBOUNCE[1]

Address offset: 0x1A

GPIO debounce configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	RW	GPIODEBOUNCE			GPIO debounce													
			DEBOUNCE0	0	Debounce disable													
			DEBOUNCE1	1	Debounce enable													

6.6.3.28 GPIODEBOUNCE[2]

Address offset: 0x1B

GPIO debounce configuration

Bit number										7	6	5	4	3	2	1	0
ID										A							
Reset 0x00										0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description												
A	RW	GPIODEBOUNCE			GPIO debounce												
			DEBOUNCE0	0	Debounce disable												
			DEBOUNCE1	1	Debounce enable												

6.6.3.29 GPIODEBOUNCE[3]

Address offset: 0x1C

GPIO debounce configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	RW	GPIODEBOUNCE			GPIO debounce													
			DEBOUNCE0	0	Debounce disable													
			DEBOUNCE1	1	Debounce enable													

6.6.3.30 GPIODEBOUNCE[4]

Address offset: 0x1D

GPIO debounce configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	RW	GPIODEBOUNCE			GPIO debounce													
			DEBOUNCE0	0	Debounce disable													
			DEBOUNCE1	1	Debounce enable													

6.6.3.31 GPIOSTATUS

Address offset: 0x1E

GPIO status

Bit number						7	6	5	4	3	2	1	0
ID						E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	GPIO0STATUS			GPIO0 status								
			LOW	0	Low								
			HIGH	1	High								
B	R	GPIO1STATUS			GPIO1 status								
			LOW	0	Low								
			HIGH	1	High								
C	R	GPIO2STATUS			GPIO2 status								
			LOW	0	Low								
			HIGH	1	High								
D	R	GPIO3STATUS			GPIO3 status								
			LOW	0	Low								
			HIGH	1	High								
E	R	GPIO4STATUS			GPIO4 status								
			LOW	0	Low								
			HIGH	1	High								

7 System features

7.1 System Monitor

The chip includes a 10-bit ADC which is used for measuring internal parameters. It can be used in the following measurement modes:

- Single-shot
- Automatic
- Timed

Measurement request priority

When multiple measurement requests happen at the same time, the priority is as follows:

1. VBAT
2. Battery temperature, T_{BAT}
3. Battery current, I_{BAT}
4. Die temperature, T_{DIE}
5. VSYS
6. VBUS

If a measurement has been requested but the measurement has not started, a higher priority can be requested.

When a low priority measurement is requested and the system has started the measurement, a higher priority can be requested. The system will complete the lower priority measurement before the higher priority measurement.

7.1.1 Single-shot measurements

Single-shot measurements are triggered by a task specific for each measurement.

Value	Task
Battery temperature	TASKNTCMEASURE on page 106
Battery voltage, Single-shot mode and Burst mode	TASKVBATMEASURE on page 106 ADCCONFIG on page 108
VSYS voltage	TASKVSYSMEASURE on page 107
Battery current	ADCIBATMEASEN on page 112 (occurs after VBAT measurement)
VBUS voltage	TASKVBUS7MEASURE on page 107
Die temperature	TASKTEMPMEASURE on page 106

Table 26: Tasks for single-shot measurements

A VBAT measurement triggered in Burst mode performs four consecutive measurements, with each result available separately. Conversions are run back-to-back and complete in t_{CONV} .

Note: To repeat a measurement, it must be requested once the previous request is complete. Repeat measurement requests are lost when made while the previous conversion is still ongoing. Alternate measurements can be requested, which are queued. See [System Monitor](#) on page 101 for more information.

7.1.2 Automatic measurements

Automatic measurements for battery voltage are enabled in register [ADCCONFIG](#) on page 108. The default interval is 1024 ms.

7.1.2.1 Automatic measurements during charging

Battery temperature and die temperature are measured automatically at regular intervals when the battery is charging. The host software can read this value and returns the latest measurement.

The measurement intervals are as follows:

- Battery temperature – once every 64, 128, or 1024 ms.
- Die temperature – once every 4 ms, see [Charger thermal regulation](#) on page 30.

Note: To enable automatic thermistor and die temperature monitoring, set register [TASKAUTOTIMUPDATE](#) on page 109. This should also be set after changing the automated period.

7.1.3 Timed measurements

Timed measurements for battery voltage in Single-shot mode and Burst mode are initiated in register [ADCDELTIMCONF](#) on page 109. See [Monitor battery state of charge](#) on page 104 for more information.

7.1.4 Measurement results

Results from the ADC are stored in registers according to the following table. Some registers hold alternate results when that feature is requested. Host software must concatenate the LSB to the MSB of the result register for full accuracy.

Value/alternate result	Register
VBAT	ADCVBATRESULTMSB on page 109
Battery temperature	ADCNTCRESULTMSB on page 110
Die temperature	ADCTEMPRESULTMSB on page 110
VSYS Single-shot mode	ADCVSYSRESULTMSB on page 110
LSBs for Single-shot mode VSYS, Die temperature, NTC thermistor, and VBAT	ADCGP0RESULTLSBS on page 110
Burst mode VBAT0	ADCVBAT0RESULTMSB on page 110
Burst mode VBAT1	ADCVBAT1RESULTMSB on page 111
Burst mode VBAT2 Battery current IBAT	ADCVBAT2RESULTMSB on page 111
Burst mode VBAT3 Single-shot mode VBUS	ADCVBAT3RESULTMSB on page 111
LSBs for Burst mode VBAT0, VBAT1, VBAT2, VBAT3, IBAT and VBUS	ADCGP1RESULTLSBS on page 111

Table 27: ADC measurements

The following equations can be used to read the results.

VBAT

The equation for VBAT is given by the following:

$$V_{BAT} = \frac{V_{BATADC}}{1023} VFS_{VBAT}$$

Here, V_{BATADC} is the ADC value from the VBAT register and VFS_{VBAT} is the full scale voltage for measuring VBAT.

VBUS

The equation for VBUS is given by the following:

$$V_{BUS} = \frac{V_{BUSADC}}{1023} VFS_{VBUS}$$

Here, V_{BUSADC} is the ADC value from the VBUS register and VFS_{VBUS} is the full scale voltage for measuring VBUS.

VSYS

Equation for VSYS is given by the following:

$$V_{SYS} = \frac{V_{SYSADC}}{1023} VFS_{VSYS}$$

Here, V_{SYSADC} is the ADC value from the VSYS register and VFS_{VSYS} is the full scale voltage for measuring VSYS.

Battery temperature (Kelvin)

The battery temperature T_{BAT} (in Kelvin) is given by the following equation:

$$T_{BAT} = \frac{1}{\frac{1}{T_0} - \frac{1}{\beta} \cdot \ln\left(\frac{1023}{T_{BATADC}} - 1\right)}$$

Here, $T_0 = 298.15$ K, T_{BATADC} is the ADC value from the battery temperature register `ADCNTCRESULTMSB` and β is the NTC beta parameter.

Die temperature in °C

The die temperature, T_D (in °C), is given by the following equation:

$$T_D = 394.67 - 0.7926 \cdot K_{DIETEMP}$$

Here, $K_{DIETEMP}$ is the ADC value for the die temperature.

7.1.5 Events and interrupts

An event register and interrupt are available for each measurement and are issued once the measurement has been completed.

See registers `EVENTSADCSET` on page 128, `EVENTSADCCLR` on page 129, `INTENEVENTSADCSET` on page 130, and `INTENEVENTSADCCLR` on page 131.

7.1.6 Battery temperature measurement

Before using a battery temperature measurement, the appropriate NTC thermistor must be configured. See [Monitor battery temperature](#) on page 29 for information about suitable thermistors and how to configure.

7.1.7 Monitor battery state of charge

The host runs the fuel gauge algorithm and periodically requests measurements from the ADC. These measurements update the algorithm parameters and allow the state of charge to be determined.

The algorithm must be provided with the battery model parameters for accurate fuel gauge readings. The battery model parameters can be created from the nPM PowerUP application.

Once the battery is modeled over the operating temperature range, the fuel gauge algorithm is optimized to operate over the full range of battery voltages, temperatures, and application currents.

7.1.8 Battery current measurement

Host software can request a IBAT measurement by setting bit `IBAT.MEAS.ENABLE` to 1 in register `ADCIBATMEASEN` on page 112. This allows consecutive VBAT and IBAT measurements. When both measurements are available in the ADC registers, the `ADCIBATRDY` event is generated. See register `ADCIBATMEASSTATUS` on page 109 for more information about the IBAT measurement.

Measurements are invalid and a new measurement is needed when bit `IBAT.MEASE.INVALID` is set.

Direction of current flow is shown in bit `BCHARGER.MODE`.

A value of 01 means the battery is discharging. During a discharge, the full scale current (in Amps) is the battery discharge current limit (as configured in registers `BCHGSETDISCHARGEMSB` and `BCHGSETDISCHARGELSB`) multiplied by 1.12.

A value of 10 means the system is supplied by VBUS.

A value of 11 means the battery is charging. When charging, the full scale current (in Amps) is the charge current setting (as configured in registers `BCHGSETMSB` and `BCHGSETLSB`) multiplied by 1.25.

7.1.9 Electrical specification

Symbol	Description	Min.	Typ.	Max.	Unit
$V_{FS_{VBAT}}$	Full scale voltage for VBAT measurement		5.0		V
$V_{BATACCUR}$	Accuracy of the VBAT measurement (3 V < VBAT < 4.5 V and -10°C < T _J < +125°C)	-1		+1	%
$V_{FS_{VBUS}}$	Full scale voltage for VBUS measurement		7.5		V
$V_{BUSACCUR}$	Accuracy of the VBUS measurement		±1.5		%
$V_{FS_{V_{SYS}}}$	Full scale voltage for V _{SYS} measurement		6.375		V
$V_{SYSACCUR}$	Accuracy of the V _{SYS} measurement		±1.5		%
C_{BATNTC}	Capacitance in parallel with the thermistor	0		100	pF
$V_{FS_{TEMP}}$	Full scale for battery and die temperature measurements		1.5		V
t_{CONV}	Conversion time		250		μs
DNL	Differential non-linearity		< 0.5		LSB

Table 28: System Monitor electrical specification

7.1.10 Registers

Instances

Instance	Base address	Description
ADC	0x00000500	SAADC registers ADC register map

Register overview

Register	Offset	Description
TASKVBATMEASURE	0x0	Battery voltage measurement
TASKNTCMEASURE	0x1	Battery temperature measurement
TASKTEMPMEASURE	0x2	Die temperature measurement
TASKVSYSTEMMEASURE	0x3	V _{SYS} measurement
TASKIBATMEASURE	0x6	Battery current measurement
TASKVBUSMEASURE	0x7	VBUS measurement
TASKDELAYEDVBATMEASURE	0x8	Delayed battery voltage measurement
ADCCONFIG	0x9	ADC configuration
ADCNTCRSEL	0xA	Select battery NTC thermistor
ADCAUTOTIMCONF	0xB	Auto measurement intervals
TASKAUTOTIMUPDATE	0xC	Strobe for AUTOTIMCONF
ADCDELTIMCONF	0xD	Configure delay for battery voltage measurement

Register	Offset	Description
ADCIBATMEASSTATUS	0x10	Battery current measurement status
ADCVBATRESULTMSB	0x11	Battery voltage measurement result MSB
ADCNTCRESULTMSB	0x12	Battery temperature measurement result MSB
ADCTEMPRESULTMSB	0x13	Die temperature measurement result MSB
ADCVSYSRESULTMSB	0x14	VSYS measurement result MSB
ADCGP0RESULTLSBS	0x15	Result LSBs (VBAT, battery temperature, die temperature and VSYS)
ADCVBAT0RESULTMSB	0x16	VBAT0 burst measurement result MSB
ADCVBAT1RESULTMSB	0x17	VBAT1 burst measurement result MSB
ADCVBAT2RESULTMSB	0x18	VBAT2 burst measurement result MSB
ADCVBAT3RESULTMSB	0x19	VBAT3 burst or VBUS measurement result MSB
ADCGP1RESULTLSBS	0x1A	VBAT burst measurement result LSBs (VBAT0, VBAT1, VBAT2 and VBAT3)
ADCIBATMEASEN	0x24	Enable automatic battery current measurement

7.1.10.1 TASKVBATMEASURE

Address offset: 0x0

Battery voltage measurement

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKVBATMEASURE			Start VBAT measurement								
			NOEFFECT	0	No effect								
			TRIGGER	1	Trigger task								

7.1.10.2 TASKNTCMEASURE

Address offset: 0x1

Battery temperature measurement

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKNTCMEASURE			Start battery NTC thermistor measurement								
			NOEFFECT	0	No effect								
			TRIGGER	1	Trigger task								

7.1.10.3 TASKTEMPMEASURE

Address offset: 0x2

Die temperature measurement

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKTEMPMEASURE			Start die temperature measurement								
			NOEFFECT	0	No effect								
			TRIGGER	1	Trigger task								

7.1.10.4 TASKVSYSMESURE

Address offset: 0x3

VSYS measurement

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKVSYSMEASURE			Start VSYS measurement													
			NOEFFECT	0	No effect													
			TRIGGER	1	Trigger task													

7.1.10.5 TASKIBATMEASURE

Address offset: 0x6

Battery current measurement

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKIBATMEASURE			Start IBAT measurement								
			NOEFFECT	0	No effect								
			TRIGGER	1	Trigger task								

7.1.10.6 TASKVBUS7MEASURE

Address offset: 0x7

VBUS measurement

Bit number										7	6	5	4	3	2	1	0
ID										A							
Reset 0x00										0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description												
A	W	TASKVBUS7MEASURE			Start VBUS measurement												
			NOEFFECT	0	No effect												
			TRIGGER	1	Trigger task												

7.1.10.7 TASKDELAYEDVBATMEASURE

Address offset: 0x8

Delayed battery voltage measurement

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKDLVDVATMEASURE			Start delayed VBAT measurement													
			NOEFFECT	0	No effect													
			TRIGGER	1	Trigger task													

7.1.10.8 ADCCONFIG

Address offset: 0x9

ADC configuration

Bit number						7	6	5	4	3	2	1	0
ID												B	A
Reset 0x00												0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	VBATAUTOENABLE			Enable VBAT Auto measurement every 1 s								
			NOAUTO	0	No automatic measurements								
			AUTOENABLE	1	Trigger measurement every 1 s								
B	RW	VBATBURSTENABLE			Enable VBAT burst mode VBAT0, VBAT1, VBAT2, VBAT3								
			SINGLEMODE	0	Trigger a single measurement								
			BURSTMODE	1	Trigger 4 consecutive measurements								

7.1.10.9 ADCNTCRSEL

Address offset: 0xA

Select battery NTC thermistor

Bit number						7	6	5	4	3	2	1	0
ID												A	A
Reset 0x01												0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	ADCNTCRSEL			Select thermistor value								
			Hi_Z	0	No thermistor								
			10K	1	NTC 10 kOhm								
			47K	2	NTC 47 kOhm								
			100K	3	NTC 100 kOhm								

7.1.10.10 ADCAUTOTIMCONF

Address offset: 0xB

Auto measurement intervals

Bit number						7	6	5	4	3	2	1	0
ID												B	B
Reset 0x03												0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	NTCAUTOTIM			Battery thermistor measurement interval when charging								
			4MS	0	4 ms								
			64MS	1	64 ms								
			128MS	2	128 ms								
			1024MS	3	1024 ms								
B	RW	TEMPAUTOTIM			Die temperature measurement interval when charging								
			4MS	0	4 ms								
			8MS	1	8 ms								
			16MS	2	16 ms								
			32MS	3	32 ms								

7.1.10.11 TASKAUTOTIMUPDATE

Address offset: 0xC

Strobe for AUTOTIMCONF

Bit number	7 6 5 4 3 2 1 0						
ID							A
Reset 0x00	0 0 0 0 0 0 0 0						
ID	R/W	Field	Value ID	Value	Description		
A	W	TASKAUTOTIMUPDATE			Update NTCAUTOTIM and TEMPAUTOTIM measurement intervals		
			NOEFFECT	0	No effect		
			UPDATEAUTOTIM	1	Update		

7.1.10.12 ADCDELTIMCONF

Address offset: 0xD

Configure delay for battery voltage measurement

Bit number	7 6 5 4 3 2 1 0						
ID	A A A A A A A A						
Reset 0x00	0 0 0 0 0 0 0 0						
ID	R/W	Field	Value ID	Value	Description		
A	RW	VBATDELTIM			Delay from 4 ms (value 0) to 514 ms (value 255) in 2 ms steps		

7.1.10.13 ADCIBATMEASSTATUS

Address offset: 0x10

Battery current measurement status

Bit number	7 6 5 4 3 2 1 0						
ID	C B B A A						
Reset 0x00	0 0 0 0 0 0 0 0						
ID	R/W	Field	Value ID	Value	Description		
A	R	BCHARGERICHARGE			Scaling factor for charging current (00=10%, 01=50%, 11=100%)		
B	R	BCHARGERMODE			Battery charger mode (01=discharging, 10=idle, no battery current, 11=charging)		
C	R	IBATMEASEINVALID			IBAT measurement invalid		

7.1.10.14 ADCVBATRESULTMSB

Address offset: 0x11

Battery voltage measurement result MSB

Bit number	7 6 5 4 3 2 1 0						
ID	A A A A A A A A						
Reset 0x00	0 0 0 0 0 0 0 0						
ID	R/W	Field	Value ID	Value	Description		
A	R	VBATRESULTMSB			Battery voltage measurement result MSB		

7.1.10.15 ADCNTCRESULTMSB

Address offset: 0x12

Battery temperature measurement result MSB

Bit number	7 6 5 4 3 2 1 0						
ID	A A A A A A A						
Reset 0x00	0 0 0 0 0 0 0						
ID	R/W	Field	Value ID	Value	Description		
A	R	NTCRESULTMSB			Battery temperature measurement result MSB		

7.1.10.16 ADCTEMPRESULTMSB

Address offset: 0x13

Die temperature measurement result MSB

Bit number	7 6 5 4 3 2 1 0						
ID	A A A A A A A						
Reset 0x00	0 0 0 0 0 0 0						
ID	R/W	Field	Value ID	Value	Description		
A	R	TEMPRESULTMSB			Die temperature measurement result MSB		

7.1.10.17 ADCVSYRESULTMSB

Address offset: 0x14

VSYS measurement result MSB

Bit number	7 6 5 4 3 2 1 0						
ID	A A A A A A A						
Reset 0x00	0 0 0 0 0 0 0						
ID	R/W	Field	Value ID	Value	Description		
A	R	VSYRESULTMSB			VSYS measurement result MSB		

7.1.10.18 ADCGPORESULTLSBS

Address offset: 0x15

Result LSBs (VBAT, battery temperature, die temperature and VSYS)

Bit number	7 6 5 4 3 2 1 0						
ID	D D C C B B A A						
Reset 0x00	0 0 0 0 0 0 0						
ID	R/W	Field	Value ID	Value	Description		
A	R	VBATRESULTLSB			Battery voltage measurement result LSBs		
B	R	NTCRESULTLSB			Battery temperature measurement result LSBs		
C	R	TEMPRESULTLSB			Die temperature measurement result LSBs		
D	R	VSYRESULTLSB			VSYS measurement result LSBs		

7.1.10.19 ADCVBATORESULTMSB

Address offset: 0x16

VBAT0 burst measurement result MSB

Bit number						7	6	5	4	3	2	1	0
ID						A	A	A	A	A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	VBAT0RESULTMSB			VBAT0 burst measurement result MSB								

7.1.10.20 ADCVBAT1RESULTMSB

Address offset: 0x17

VBAT1 burst measurement result MSB

Bit number						7	6	5	4	3	2	1	0
ID						A	A	A	A	A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	VBAT1RESULTMSB			VBAT1 burst measurement result MSB								

7.1.10.21 ADCVBAT2RESULTMSB

Address offset: 0x18

VBAT2 burst measurement result MSB

Bit number						7	6	5	4	3	2	1	0
ID						A	A	A	A	A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	VBAT2RESULTMSB			ADC VBAT2 burst measurement result MSB								

7.1.10.22 ADCVBAT3RESULTMSB

Address offset: 0x19

VBAT3 burst or VBUS measurement result MSB

Bit number						7	6	5	4	3	2	1	0
ID						A	A	A	A	A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	VBAT3RESULTMSB			If TASKVBATMEASURE is triggered in burst mode, this register will contain ADC VBAT3 burst measurement result MSB. If TASKVBUS7MEASURE is triggered, this register will contain VBUS measurement result MSB.								

7.1.10.23 ADCGP1RESULTLSBS

Address offset: 0x1A

VBAT burst measurement result LSBs (VBAT0, VBAT1, VBAT2 and VBAT3)

Bit number						7	6	5	4	3	2	1	0
ID						D	D	C	C	B	B	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	VBAT0RESULTLSB			Burst VBAT0 measurement result LSBs								
B	R	VBAT1RESULTLSB			Burst VBAT1 measurement result LSBs								
C	R	VBAT2RESULTLSB			Burst VBAT2 measurement result LSBs								
D	R	VBAT3RESULTLSB			Burst VBAT3 measurement result LSBs								

7.1.10.24 ADCIBATMEASEN

Address offset: 0x24

Enable automatic battery current measurement

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	IBATMEASENABLE			Enable IBAT measurement after VBAT measurement								

7.2 POF — Power-fail comparator

The power-fail comparator (POF) provides the host with an early warning of an impending power supply failure.

POF is generated from an always active comparator monitoring the voltage on the **VSYS** pin. It can be configured through [POFCONFIG](#) on page 114 to give a warning through a GPIO to the host.

If voltage on the **VSYS** pin drops below $VSYS_{POF}$, but voltage remains above the respective BOR threshold on the **VBAT** or **VBUS** pins, the **VSYS** pin is disabled after $t_{POFWAIT}$ and registers are reset after t_{PWRDN} . If $VSYS > VSYS_{POF}$, the chip powers up after t_{PWRDN} . See [Power fail warning](#) on page 113.

Note: Before setting $VSYS_{POF}$, voltage on the **VSYS** pin must be higher than the selected threshold or it triggers a POF event and resets the device. The POF threshold is also reset to the default setting. $VSYS_{POF}$ must be set to a higher voltage than the battery undervoltage protection level to avoid triggering the protection circuit. When $VSYS > VSYS_{POF}$, BUCK may start up again depending on **VSET[n]** pin configuration.

If configured, a power failure warning is issued in the following cases:

- VBUS is removed while the battery is empty or not connected ($VBAT < VSYS_{POF}$)
- VBUS rises above $VBUS_{OVP}$ while the battery is empty or not connected ($VBAT < VSYS_{POF}$)
- The battery is removed when VBUS is not connected
- The battery discharges until $VBAT < VSYS_{POF}$ and VBUS is not connected
- Battery voltage drops momentarily below $VSYS_{POF}$ and VBUS is not connected

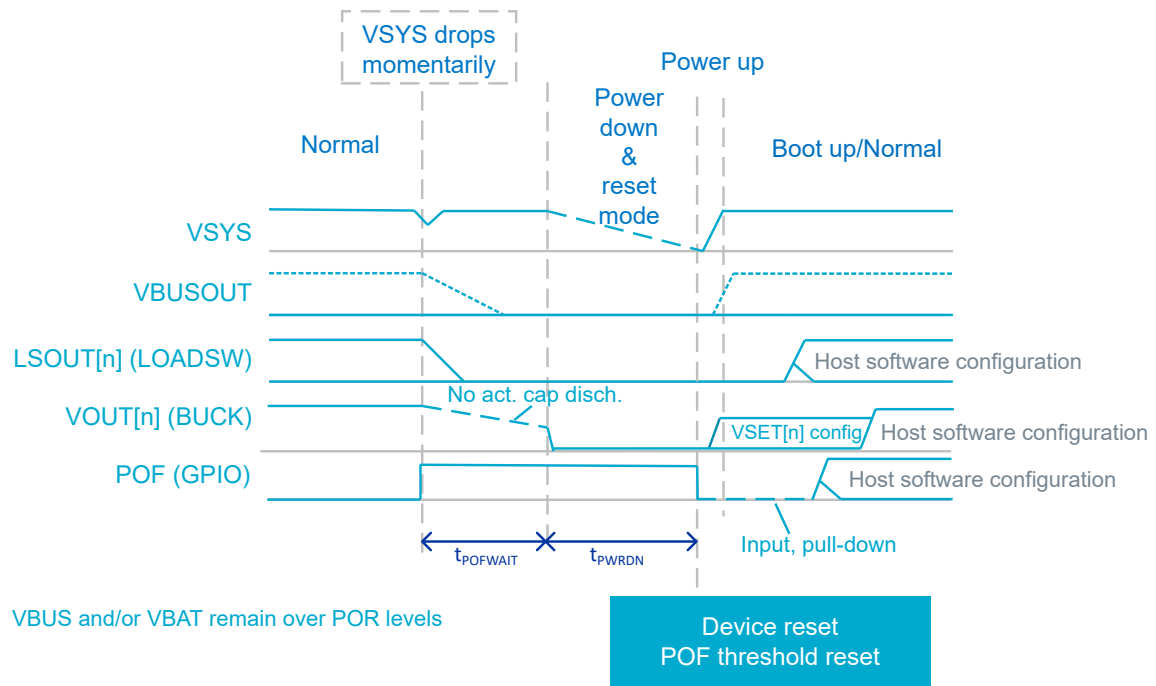


Figure 47: Power fail warning

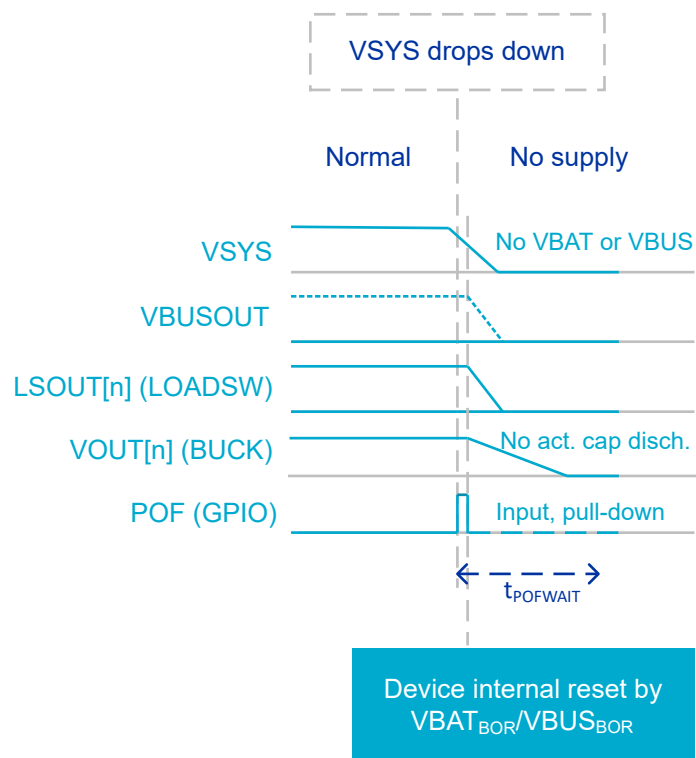


Figure 48: Power removal

To use the POF warning feature, set **POFWARNPOLARITY** and **POFENA** to 1 in register **POFCONFIG** on page 114. GPIO settings are located in **GPIO — General purpose input/output** on page 89.

7.2.1 Electrical specification

Symbol	Description	Min.	Typ.	Max.	Unit
POF	VSYS _{POF} rising threshold, default Always 100 mV (typ.) above the falling threshold		2.8		V
VSYS _{POF}	Minimum setting VSYS _{POF} falling threshold		2.5		V
VSYS _{POF}	Default setting VSYS _{POF} falling threshold		2.7		V
VSYS _{POF}	Maximum setting VSYS _{POF} falling threshold		3.4		V
t _{POF}	Reaction time (from crossing the threshold to edge on the warning signal)		1		ms
t _{PWRDN}	Time in power-down mode		100		ms
t _{POFWAIT}	Delay before enabling the active output capacitor discharge and disconnecting VBAT and VBUS from VSYS		30		ms

Table 29: POF electrical specification

7.2.2 Registers

Instances

Instance	Base address	Description
POF	0x00000900	POF registers POF register map

Register overview

Register	Offset	Description
POFCONFIG	0x0	Configuration for power failure

7.2.2.1 POFCONFIG

Address offset: 0x0

Configuration for power failure

Bit number						7	6	5	4	3	2	1	0
ID						C C C C B A							
Reset 0x00						0 0 0 0 0 0 0 0							
ID	R/W	Field	Value ID	Value	Description								
A	RW	POFENA			Enable power failure warning								
			OFF	0	Warning disabled								
			ENABLED	1	Warning enabled								
B	RW	POFWARNPOLARITY			Power failure warning polarity								
			LOACTIVE	0	Active low								
			HIACTIVE	1	Active high								
C	RW	POFVSYSTHRESHSEL			Power failure threshold select								
			2V8	0	2.8 V								

Bit number					7	6	5	4	3	2	1	0
ID					C C C C B A							
Reset 0x00					0 0 0 0 0 0 0 0							
ID	R/W	Field	Value ID	Value	Description							
			2V6	1	2.6 V							
			2V7	2	2.7 V							
			2V9	3	2.9 V							
			3V0	4	3.0 V							
			3V1	5	3.1 V							
			3V2	6	3.2 V							
			3V3	7	3.3 V							
			3V4	8	3.4 V							
			3V5	9	3.5 V							
			unused10	10	2.8 V							
			unused11	11	2.8 V							
			unused12	12	2.8 V							
			unused13	13	2.8 V							
			unused14	14	2.8 V							
			unused15	15	2.8 V							

7.3 TIMER — Timer/monitor

TIMER can be used in the following ways, depending on configuration.

- Boot monitor
- Watchdog timer
- Wake-up timer
- General purpose timer

TIMER is a 24-bit timer running at the frequency of the timer clock, f_{TIMER} , and has a prescaler.

TIMER only runs one configuration at a time because it is shared for all functions. The wake-up timer wakes the system at a programmable interval when the device is in Hibernate mode. Do not use the watchdog timer or general purpose timer when the system is in Ship or Hibernate mode.

TIMER is controlled by register [TIMERCONFIG](#) on page 119. The start value is configured with [TIMERHIBYTE](#) on page 120, [TIMERMIDBYTE](#) on page 120, and [TIMERLOBYTE](#) on page 121. The settings are applied with [TIMERTARGETSTROBE](#) on page 119. TIMER is started with [TIMERSET](#) on page 118 and is stopped with [TIMERCLR](#) on page 119.

Example settings are shown in the following table.

f_{TIMER}	TIMERHIBYTE	TIMERMIDBYTE	TIMERLOBYTE	Time
2 ms	0	0	250	0.5 s
16 ms	0	0	250	4 s
16 ms	0	1	0	4.096 s
16 ms	1	0	0	1048.576 s
16 ms	255	255	255	74.5 h

Table 30: Example timer register settings

7.3.1 Boot monitor

After a power-on reset, the default timer is boot monitor and this is disabled. When enabled, it allows an automatic power cycle if the host does not set bit `TASK.TIMER.DIS` within t_{BOOT} .

Host software can enable the boot monitor with bit `BOOT.TIMER.EN`. It can disable the boot monitor to prevent interference with firmware updates. When enabled, the boot monitor remains enabled even if the chip is reset, except for a power-on reset. Removing both `VBAT` and `VBUS`, or clearing the `BOOT.TIMER.EN` bit, deactivates the timer during the next power-up.

7.3.2 Watchdog timer

Watchdog timer expiration can be configured by host software to generate an `NRESETOUT` through a GPIO or a power cycle.

Power cycle means internally disconnecting `VSYS` from `VBAT` and `VBUS`. `BUCK` and `LOADSW` are actively pulled low for 100 ms. The device is reset and `BUCK` is re-enabled. Active pull-downs are present at pin `VOUT1`, `VOUT2`, `LSOUT1`, and `LS2OUT2` during t_{PWRDN} .

The watchdog timer can issue a pre-warning interrupt, $t_{PREWARN}$, before expiration. The reset pulse, which is active-low, through the `NRESETOUT` GPIO lasts for t_{RESET} . Watchdog can be configured in register `WATCHDOGKICK` on page 119.

The pre-warning interrupt is generated one cycle of the selected prescaler, either 2 ms or 16 ms, before expiry of the watchdog occurs.

The following figure shows a watchdog reset where the nPM1300 device is not reset internally.

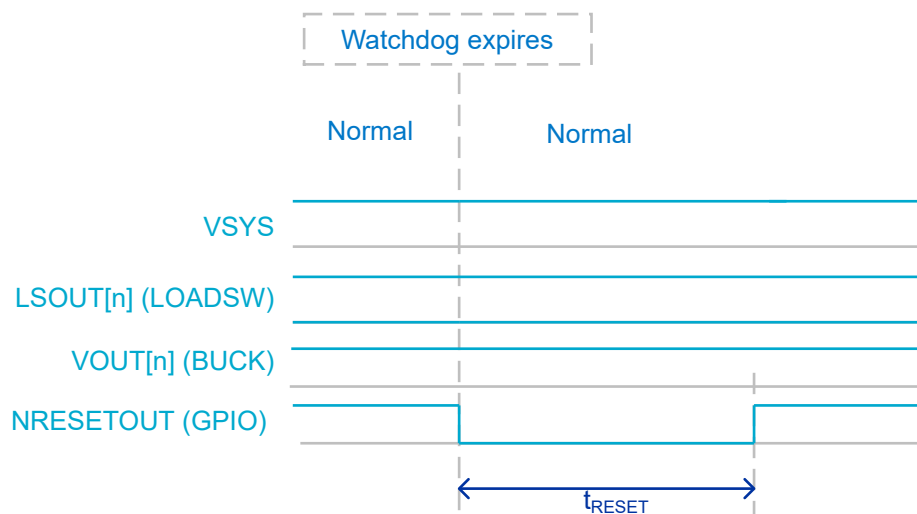


Figure 49: Watchdog reset

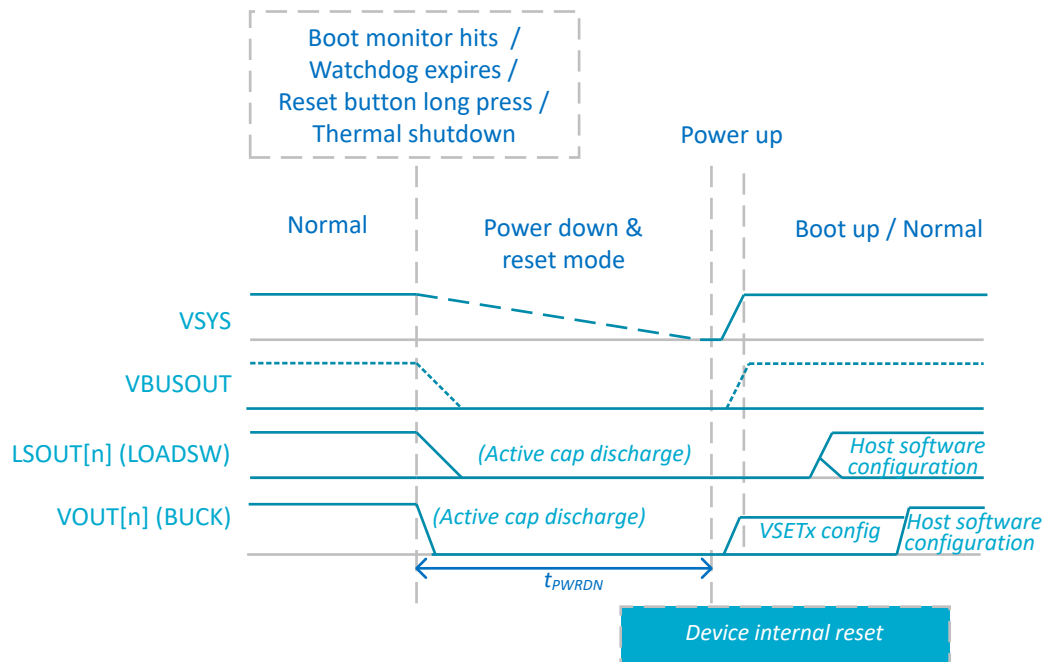


Figure 50: Power cycle

Note: For the thermal shutdown case, t_{PWRDN} will be longer as it waits for the die temperature to cool down below $TSD - TSD_{HYST}$.

7.3.3 Wake-up timer

The wake-up timer wakes the system from Hibernate mode.

Host software configures the timer before the device enters Hibernate mode, see [Ship and Hibernate modes](#) on page 121.

7.3.4 General purpose timer

The general purpose timer interrupts the host after a timeout with the WATCHDOG.WARNING event.

Prescaler is configured in register [TIMERCONFIG](#) on page 119 with the default set to 16 ms.

When the prescaler is configured to 16 ms in [TIMERCONFIG](#) on page 119 and [TIMERHIBYTE](#) on page 120 is 5, [TIMERMIDBYTE](#) on page 120 is 2 and [TIMERLOBYTE](#) on page 121 is 1, then the general purpose timer will wake after 5251 seconds.

7.3.5 Electrical specification

Both prescaler settings 16 ms and (2 ms) are included. Values in parenthesis are for the 2 ms prescaler.

Symbol	Description	Min.	Typ.	Max.	Unit
f_{TIMER}	Frequency of timer clock		64 (512)		Hz
t_{PREWARN}	Time between watchdog timer interrupt and reset/power cycle		16 (2)		ms
$t_{\text{PER_MIN}}$	Minimum time period		16 (2)		ms
$t_{\text{PER_MAX}}$	Maximum time period		3 (9)		days (hours)
t_{BOOT}	Amount of time before a power cycle is performed when no traffic is observed on TWI and BOOT.TIMER.EN is set		10		s
t_{PWRDN}	Length of power cycle		100		ms
t_{RESET}	Length of reset pulse		100		ms
f_{ACCUR}	Accuracy of timer clock		± 3		%

Table 31: TIMER electrical specification

7.3.6 Registers

Instances

Instance	Base address	Description
TIMER	0x00000700	TIMER registers TIMER register map

Register overview

Register	Offset	Description
TIMERSET	0x0	Start timer
TIMERCLR	0x1	Stop timer
TIMERTARGETSTROBE	0x3	Strobe for timer target
WATCHDOGKICK	0x4	Watchdog kick
TIMERCONFIG	0x5	Timer mode selection
TIMERSTATUS	0x6	Timers status
TIMERHIBYTE	0x8	Timer most significant byte
TIMERMIDBYTE	0x9	Timer middle byte
TIMERLOBYTE	0xA	Timer least significant byte

7.3.6.1 TIMERSET

Address offset: 0x0

Start timer

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKTIMEREN			Start timer													
			NOEFFECT	0	No effect													
			SET	1	Timer start request													

7.3.6.2 TIMERCLR

Address offset: 0x1

Stop timer

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKTIMERDIS			Stop timer													
			NOEFFECT	0	No effect													
			SET	1	Timer stop request													

7.3.6.3 TIMERTARGETSTROBE

Address offset: 0x3

Strobe for timer target

Bit number										7	6	5	4	3	2	1	0
ID										A							
Reset 0x00										0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description												
A	W	TASKTIMERTARGETSTROBE			Timer target strobe												
			NOEFFECT	0	No effect												
			SET	1	Load timer target (24-bit timer value)												

7.3.6.4 WATCHDOGKICK

Address offset: 0x4

Watchdog kick

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKWATCHDOGKICK			Watchdog kick													
		NOEFFECT	0	No effect														
		Kick	1	Kick watchdog														

7.3.6.5 TIMERCONFIG

Address offset: 0x5

Timer mode selection

Bit number						7	6	5	4	3	2	1	0
ID						B				A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	TIMERMODESEL			Select watchdog and timer modes								
			BOOTMONITOR	0	Boot monitor								
			WATCHDOGWARNING	1	Watchdog warning								
			WATCHDOGRESET	2	Watchdog reset								
			GENPURPOSETIMER	3	General purpose timer								
B	RW	TIMERPRESCALER	WAKEUPTIMER	4	Wakeup timer								
					Select between 16 ms and 2 ms timer prescaler								
			SLOW	0	16 ms prescaler								
			FAST	1	2 ms prescaler								

7.3.6.6 TIMERSTATUS

Address offset: 0x6

Timers status

Bit number						7	6	5	4	3	2	1	0
ID						B				A			
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	BOOTMONITORACTIVE			Boot monitor active								
			INACTIVE	0	Boot monitor not running								
			ACTIVE	1	Boot monitor running								
B	R	SLOWDOMAINCONFIGURED			Timer ready after TIMERTARGETSTROBE								
			NOTCONFIG	0	Not configured								
			CONFIG	1	Timer configured								

7.3.6.7 TIMERHIBYTE

Address offset: 0x8

Timer most significant byte

Bit number						7	6	5	4	3	2	1	0
ID						A	A	A	A	A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	TIMERHIBYTE			Timer most significant byte								

7.3.6.8 TIMERMIDBYTE

Address offset: 0x9

Timer middle byte

Bit number						7	6	5	4	3	2	1	0
ID						A	A	A	A	A	A	A	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	TIMERMIDBYTE			Timer middle byte								

7.3.6.9 TIMERLOBYTE

Address offset: 0xA

Timer least significant byte

Bit number	7	6	5	4	3	2	1	0
ID	A	A	A	A	A	A	A	A
Reset 0x00	0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description			
A	RW	TIMERLOBYTE			Timer least significant byte			

7.4 Ship and Hibernate modes

Ship and Hibernate modes isolate the battery from the system and minimize the quiescent current.

Hibernate mode is identical to Ship mode with the exception that, in Hibernate mode, the timer is running and functions as an additional wake-up source.

The device enters Ship mode through register [TASKENTERSHIPMODE](#) on page 123. Register [SHPHLDCONFIG](#) on page 123 configures the **SHPHLD** button press time, and register [TASKSHPHLD CFGSTROBE](#) on page 123 applies the configured value. When VBUS is not present, the device enters Ship mode immediately. The host software must wait until [EVENTSVBUSIN0SET](#) on page 141 to ensure VBUS is disconnected and discharged before writing to the register.

The device enters Hibernate mode through register [TASKENTERHIBERNATE](#) on page 122. The host software must wait until [EVENTSVBUSIN0SET](#) on page 141 to ensure VBUS is disconnected and discharged before writing to the register. To apply the timer value, registers [TIMERHIBYTE](#) on page 120, registers [TIMERMIDBYTE](#) on page 120, and [TIMERLOBYTE](#) on page 121 must be configured before register [TIMERTARGETSTROBE](#) on page 119. In Hibernate mode, the quiescent current is higher compared to Ship mode because the low-power timer is running.

Exiting Hibernate mode using a button press must be configured in register [SHPHLDCONFIG](#) on page 123 and [TASKSHPHLD CFGSTROBE](#) on page 123.

When entering Ship mode, BUCK can be configured to discharge by enabling their pull downs, see [BUCKCTRL0](#) on page 73.

Note: [SHPHLDCONFIG](#) on page 123 and [TASKSHPHLD CFGSTROBE](#) on page 123 must be set before entering either Ship or Hibernate modes.

The following are alternative ways to exit Ship and Hibernate modes.

- Pulling pin SHPHLD low for a minimum period of $t_{\text{shipToActive}}$ (see [SHPHLDCONFIG](#) on page 123). A push button to GND is required.
- Applying a voltage on VBUS > VBUS_{POR} .
- Exiting automatically through the Wake-up timer (only from Hibernate mode).

7.4.1 Electrical specification

Symbol	Description	Min.	Typ.	Max.	Unit
$t_{\text{shipToActive}}$	Duration SHPHLD pin must be held low to exit Ship or Hibernate mode		16 32 64 96 (default) 304 608 1008 3008		ms
V_{IL}	Input low voltage for SHPHLD pin	AVSS		0.4	V
V_{IH}	Input high voltage for SHPHLD pin	1.2		VBAT	V
t_{RESETBUT}	Amount of time for a button press to cause a power cycle		10		s
R_{SHPHLD}	Pull-up resistor on SHPHLD pin		50		k Ω

Table 32: Ship mode electrical specification

7.4.2 Registers

Instances

Instance	Base address	Description
SHIP	0x00000B00	SHIP registers SHPHLD register map

Register overview

Register	Offset	Description
TASKENTERHIBERNATE	0x0	Enter Hibernate
TASKSHPHLD CFGSTROBE	0x1	Load SHPHLD configuration
TASKENTERSHIPMODE	0x2	Enter Ship mode
TASKRESETCFG	0x3	Reset configuration
SHPHLD CONFIG	0x4	Configuration for SHPHLD
SHPHLD STATUS	0x5	Status of the SHPHLD pin
LPRESET CONFIG	0x6	Long press reset configuration

7.4.2.1 TASKENTERHIBERNATE

Address offset: 0x0

Enter Hibernate

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKENTERHIBERNATE			Enter Hibernate (Ship mode with Wake-up timer)													
			NOEFFECT	0	No effect													
			TRIGGER	1	Trigger task													

7.4.2.2 TASKSHPHLDCFGSTROBE

Address offset: 0x1

Load SHPHLD configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKSHPHLDCONFIGSTROBE			Load the SHPHLD configuration													
			NOEFFECT	0	No effect													
			TRIGGER	1	Strobe config													

7.4.2.3 TASKENTERSHIPMODE

Address offset: 0x2

Enter Ship mode

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKENTERSHIPMODE			Enter Ship mode (without Wake-up timer)													
		NOEFFECT			0	No effect												
		TRIGGER			1	Trigger task												

7.4.2.4 TASKRESETCFG

Address offset: 0x3

Reset configuration

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	W	TASKSHPHLDRSTCONFIG			Reset configuration settings													
			NOEFFECT	0	No effect													
			TRIGGER	1	Reset													

7.4.2.5 SHPHLDCONFIG

Address offset: 0x4

Configuration for SHPHLD

Bit number											7	6	5	4	3	2	1	0
ID														A			A	A
Reset 0x03											0	0	0	0	0	0	1	1
ID	R/W	Field	Value ID	Value	Description													
A	RW	SHPHLDTIM			Configuration for SHPHLD debounce													
			16ms	0	16 ms													
			32ms	1	32 ms													
			64ms	2	64 ms													
			96ms	3	96 ms (default)													
			304ms	4	304 ms													
			608ms	5	608 ms													
			1008ms	6	1008 ms													
			3008ms	7	3008 ms													

7.4.2.6 SHPHLDSTATUS

Address offset: 0x5

Status of the SHPHLD pin

Bit number											7	6	5	4	3	2	1	0
ID											A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	R	SHPHLDPINSTATUS			SHPHLD pin status													
			LOW	0	Low													
			HIGH	1	High													

7.4.2.7 LPRESETCONFIG

Address offset: 0x6

Long press reset configuration

Bit number											7	6	5	4	3	2	1	0
ID											B A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
A	RW	LONGTIMRESETDIS			Long press reset disable													
			ENABLED	0	Long press reset enabled													
			DISABLED	1	Long press reset disabled													
B	RW	LONGTIMTWOBUTTONSEL			Select one (default) or two buttons to perform long press reset													
			SHPHLD	0	One button (SHPHLD)													
			SHPHLDGPIO0	1	Two buttons (SHPHLD and GPIO0)													

7.5 RESET — Reset control

The **SHPHLD** pin is a reset control, in addition to being used for exiting Ship and Hibernate mode.

The **SHPHLD** pin has an internal pull-up resistor R_{SHPHLD} to VBAT or VBUS depending on which has the highest voltage. The functionality of the pin is determined by the device mode.

Normal operation

If configured, a short logic-low pulse on **SHPHLD** sends an interrupt to the host. Host software reads the pin state in register [SHPHLDSTATUS](#) on page 124.

A long logic-low ($> t_{\text{RESETBUT}}$) on **SHPHLD** causes a power cycle and resets the whole system. This feature is enabled by default after power-up, but can be disabled by the host software. See register [LPRESETCONFIG](#) on page 124 for more information.

Ship and Hibernate modes

When a logic-low occurs for longer than $t_{\text{ShipToActive}}$, the device wakes up from Ship or Hibernate mode, performs an internal reset, and transitions to normal operation.

Two-button reset

A two-button reset is implemented by connecting one button to the **SHPHLD** pin and another button to **GPIO0**. This feature is enabled by setting [LPRESETCONFIG](#) on page 124, and then [TASKSHPHLD CFGSTROBE](#) on page 123 to apply the configured value. In addition, **GPIO0** needs to be configured as input using [GPIOMODE\[0\]](#) on page 92 and the internal pull-down needs to be disabled in [GPIOPDEN\[0\]](#) on page 96. Internal pull-up can be enabled in [GPIOPUEN\[0\]](#) on page 95 in case there is no external pull-up resistor. Pressing and holding both buttons for longer than t_{RESETBUT} starts a power cycle.

Host software reset

Host software can reset the device by writing the TASKSWRESET bit in register [TASKSWRESET](#) on page 128. As a consequence, a power cycle is performed. A reset is not possible in Ship or Hibernate mode.

Scratch registers, reason for reset

Only POR and TASKCLRERRLOG can initialize the context registers found at [SCRATCH\[n\]](#). The cause of the first reset is reported in register [RSTCAUSE](#) on page 150.

7.6 TWI — I²C compatible two-wire interface

TWI is a two-wire interface that controls and monitors the device state through registers.

Main Features

- I²C compatible up to 400 kHz
- TWI clock supports 100 kHz to 1 MHz

A GPIO pin can be set as an interrupt pin, see [GPIO — General purpose input/output](#) on page 89.

Interface supply

TWI is supplied by VDDIO. It is recommended to connect **VDDIO** to a BUCK output, **VOUT1**, or **VOUT2**. VDDIO must be present in all operating modes of the chip, except in Ship and Hibernate modes.

Addressing

The 7-bit slave address is 110 1011.

The registers have 16-bit addressing and 8-bit data. The upper address byte is the register instance base address (bank address). The lower byte is the offset within an instance (bank).

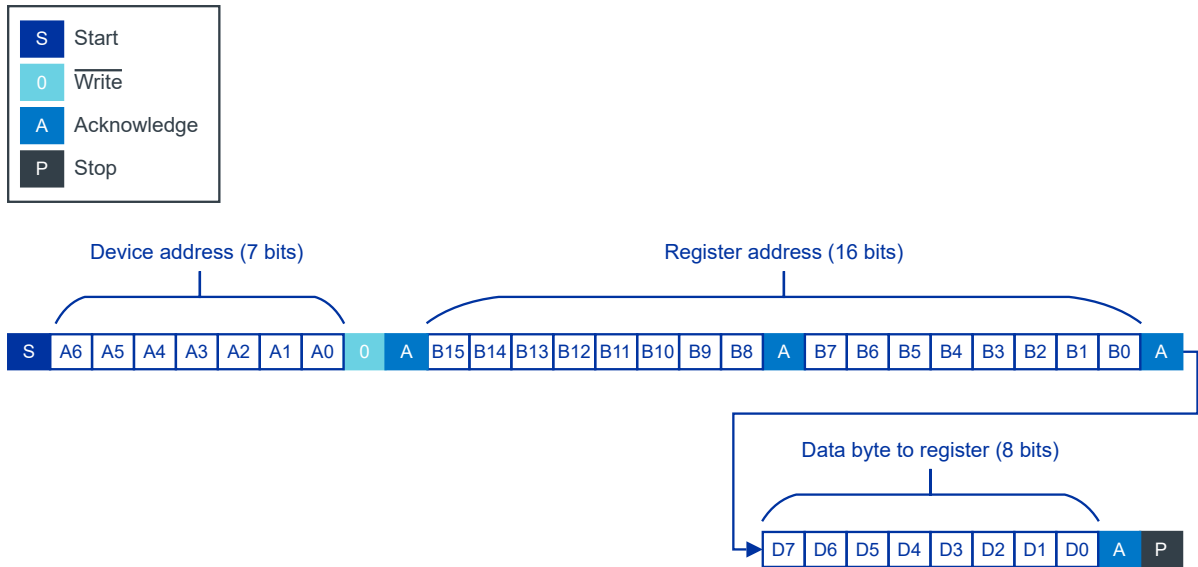


Figure 51: TWI write example

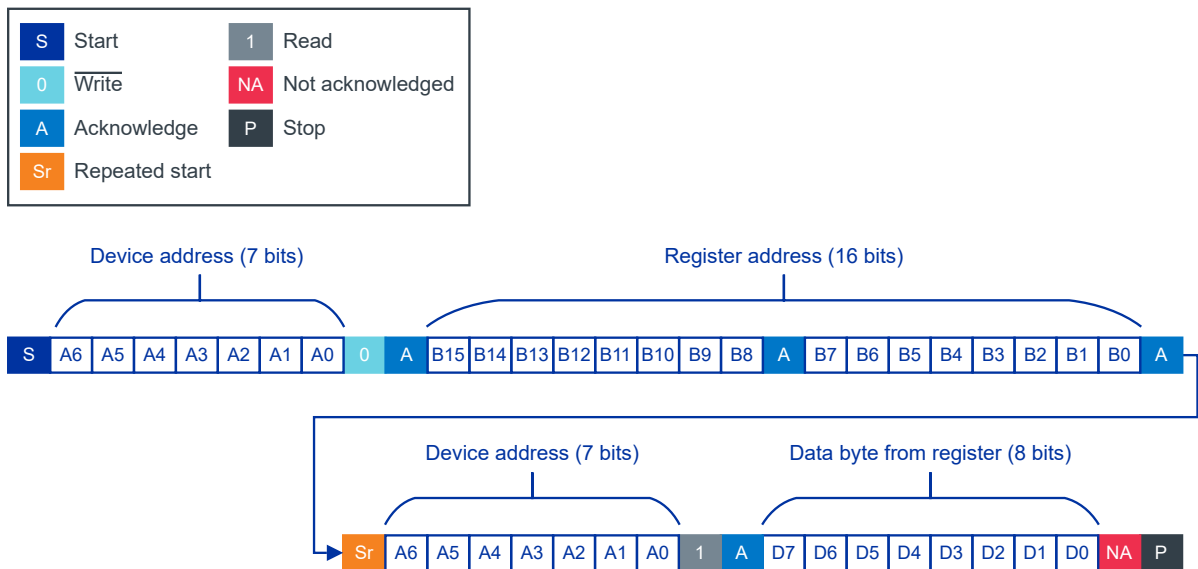


Figure 52: TWI read example

7.6.1 TWI timing diagram

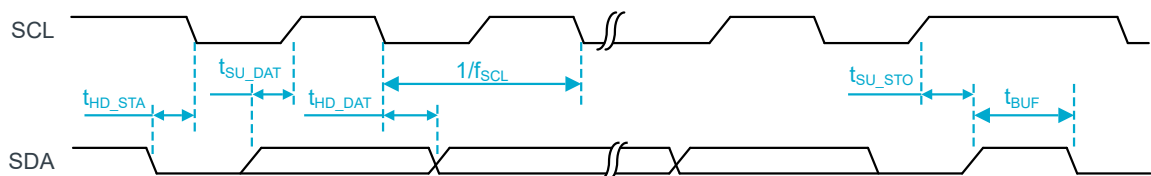


Figure 53: TWI timing diagram

7.6.2 Electrical specification

Symbol	Description	Min.	Typ.	Max.	Units
f_{SCL}	Bit rate for TWI	100		1000	kbps
t_{SU_DAT}	Data setup time before positive edge on SCL, all modes	50			ns
t_{HD_DAT}	Data hold time after negative edge on SCL, all modes	0			ns
t_{HD_STA}	Hold time from for START condition (SDA low to SCL low), 100 kbps	260			ns
t_{SU_STO}	Setup time from SCL high to STOP condition, 100 kbps	260			ns
t_{BUF}	Bus free time between STOP and START conditions		500		ns

Table 33: TWI electrical specification

7.7 Event and interrupt registers

This section details the event and interrupt related registers.

7.7.1 Registers

Instances

Instance	Base address	Description
MAIN	0x00000000	MAIN registers
		MAIN Register map

Register overview

Register	Offset	Description
TASKSWRESET	0x1	Task force a full reboot power-cycle
EVENTSADCSET	0x2	ADC Event Set
EVENTSADCCLR	0x3	ADC Event Clear
INTENEVENTSADCSET	0x4	ADC Interrupt Enable Set
INTENEVENTSADCCLR	0x5	ADC Interrupt Enable Clear
EVENTSBCHARGER0SET	0x6	Battery temperature region and die temperature Event Set
EVENTSBCHARGER0CLR	0x7	Battery temperature region and die temperature Event Clear
INTENEVENTSBCHARGER0SET	0x8	Battery temperature region and die temperature Interrupt Enable Set
INTENEVENTSBCHARGER0CLR	0x9	Battery temperature region and die temperature Interrupt Enable Clear
EVENTSBCHARGER1SET	0xA	Charger Event Set
EVENTSBCHARGER1CLR	0xB	Charger Event Clear
INTENEVENTSBCHARGER1SET	0xC	Charger Interrupt Enable Set
INTENEVENTSBCHARGER1CLR	0xD	Charger Interrupt Enable Clear
EVENTSBCHARGER2SET	0xE	Battery Event Set
EVENTSBCHARGER2CLR	0xF	Battery Event Clear
INTENEVENTSBCHARGER2SET	0x10	Battery Interrupt Enable Set
INTENEVENTSBCHARGER2CLR	0x11	Battery Interrupt Enable Clear

Register	Offset	Description
EVENTSSHPLDSET	0x12	SHPHLD pin and watchdog Event Set
EVENTSSHPLDCLR	0x13	SHPHLD pin and watchdog Event Clear
INTENEVENTSSHPLDSET	0x14	SHPHLD pin and watchdog Interrupt Enable Set
INTENEVENTSSHPLDCLR	0x15	SHPHLD pin and watchdog Interrupt Enable Clear
EVENTSVBUSIN0SET	0x16	VBUS Event Set
EVENTSVBUSIN0CLR	0x17	VBUS Event Clear
INTENEVENTSVBUSIN0SET	0x18	VBUS Interrupt Enable Set
INTENEVENTSVBUSIN0CLR	0x19	VBUS Interrupt Enable Clear
EVENTSVBUSIN1SET	0x1A	Thermal and charger detection Event Set
EVENTSVBUSIN1CLR	0x1B	Thermal and charger detection Event Clear
INTENEVENTSVBUSIN1SET	0x1C	Thermal and charger detection Interrupt Enable Set
INTENEVENTSVBUSIN1CLR	0x1D	Thermal and charger detection Interrupt Enable Clear
EVENTSGPIOSET	0x22	GPIO Event Set
EVENTSGPIOCLR	0x23	GPIO Event Clear
INTENEVENTSGPIOSET	0x24	GPIO Interrupt Enable Set
INTENEVENTSGPIOCLR	0x25	GPIO Interrupt Enable Clear

7.7.1.1 TASKSWRESET

Address offset: 0x1

Task force a full reboot power-cycle

Bit number						7	6	5	4	3	2	1	0
ID						A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKSWRESET			Turn off all supplies and apply internal reset								
			NOEFFECT	0	No effect								
			TRIGGER	1	Trigger task								

7.7.1.2 EVENTSADCSET

Address offset: 0x2

ADC Event Set

Bit number						7	6	5	4	3	2	1	0
ID						H	G	F	E	D	C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTADCVBATRDY			VBAT measurement finished. Writing 1 sets the event (for debugging).								
		W1S											
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTADCNTCRDY			Battery NTC measurement finished. Writing 1 sets the event (for debugging).								
		W1S											
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTADCTEMPRDY			Internal die temperature measurement finished. Writing 1 sets the event (for debugging).								
		W1S											
			LOW	0	Low								
			HIGH	1	High								

Bit number						7	6	5	4	3	2	1	0
ID						H	G	F	E	D	C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
D	RW	EVENTADCVSYSRDY W1S			VSYS voltage measurement measurement finished. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTADCSET1RDY W1S			VSET1 pin measurement finished. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTADCSET2RDY W1S			VSET2 pin measurement finished. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
G	RW	EVENTADCIBATRDY W1S			IBAT measurement finished. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
H	RW	EVENTADCVBUS7V0RDY W1S			VBUS measurement finished. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.3 EVENTSADCCLR

Address offset: 0x3

ADC Event Clear

Bit number						7	6	5	4	3	2	1	0
ID						H	G	F	E	D	C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTADCVBATRDY W1C			VBAT measurement finished. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTADCNTCRDY W1C			Battery NTC measurement finished. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTADCTEMPRDY W1C			Internal die temperature measurement finished. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTADCVSYSRDY W1C			VSYS voltage measurement measurement finished. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTADCSET1RDY W1C			VSET1 pin measurement finished. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								

Bit number						7	6	5	4	3	2	1	0
ID						H	G	F	E	D	C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
F	RW	EVENTADCSET2RDY W1C	HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
G	RW	EVENTADCIBATRDY W1C			IBAT measurement finished. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
H	RW	EVENTADCVBUS7V0RDY W1C			VBUS measurement finished. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.4 INTENEVENTSADCSET

Address offset: 0x4

ADC Interrupt Enable Set

Bit number						7	6	5	4	3	2	1	0
ID						H	G	F	E	D	C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTADCVBATRDY W1S			Writing 1 enables interrupts from EVENTADCVBATRDY								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTADCNTCRDY W1S			Writing 1 enables interrupts from EVENTADCNTCRDY								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTADCTEMPRDY W1S			Writing 1 enables interrupts from EVENTADCTEMPRDY								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTADCVSYSRDY W1S			Writing 1 enables interrupts from EVENTADCVSYSRDY								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTADCVSET1RDY W1S			Writing 1 enables interrupts from EVENTADCVSET1RDY								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTADCVSET2RDY W1S			Writing 1 enables interrupts from EVENTADCVSET2RDY								
			LOW	0	Low								
			HIGH	1	High								
G	RW	EVENTADCIBATRDY W1S			Writing 1 enables interrupts from EVENTADCIBATRDY								

Bit number						7	6	5	4	3	2	1	0
ID						H	G	F	E	D	C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
H	RW	EVENTADCVBUS7V0RDY W1S	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.5 INTENEVENTSADCCLR

Address offset: 0x5

ADC Interrupt Enable Clear

Bit number						7	6	5	4	3	2	1	0
ID						H	G	F	E	D	C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTADCVBATRDY W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTADCNTCRDY W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTADCTEMPRDY W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTADCVSYSRDY W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTADCSET1RDY W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTADCSET2RDY W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
G	RW	EVENTADCIBATRDY W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
H	RW	EVENTADCVBUS7V0RDY W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.6 EVENTSBCHARGER0SET

Address offset: 0x6

Battery temperature region and die temperature Event Set

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTNTCCOLD W1S			Battery temperature in cold region. Writing 1 sets the event (for debugging).								
			LOW	0		Low							
			HIGH	1		High							
B	RW	EVENTNTCCOOL W1S			Battery temperature in cool region. Writing 1 sets the event (for debugging).								
			LOW	0		Low							
			HIGH	1		High							
C	RW	EVENTNTCWARM W1S			Battery temperature in warm region. Writing 1 sets the event (for debugging).								
			LOW	0		Low							
			HIGH	1		High							
D	RW	EVENTNTCHOT W1S			Battery temperature in hot region. Writing 1 sets the event (for debugging).								
			LOW	0		Low							
			HIGH	1		High							
E	RW	EVENTDIETEMPHIGH W1S			Die temperature is over TCHGSTOP. Charging stops. Writing 1 sets the event (for debugging).								
			LOW	0		Low							
			HIGH	1		High							
F	RW	EVENTDIETEMPRESUME W1S			Die temperature is under TCHGRESUME. Charging resumes. Writing 1 sets the event (for debugging).								
			LOW	0		Low							
			HIGH	1		High							

7.7.1.7 EVENTSBCHARGER0CLR

Address offset: 0x7

Battery temperature region and die temperature Event Clear

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTNTCCOLD W1C			Battery temperature in cold region. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0		Low							
			HIGH	1		High							
B	RW	EVENTNTCCOOL W1C			Battery temperature in cool region. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0		Low							
			HIGH	1		High							
C	RW	EVENTNTCWARM W1C			Battery temperature in warm region. Writing 1 clears the event (e.g. to acknowledge an interrupt).								

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
D	RW	EVENTNTCHOT W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTDIETEMPHIGH W1C				Battery temperature in hot region. Writing 1 clears the event (e.g. to acknowledge an interrupt).							
			LOW	0	Low								
			HIGH	1	High								
						Die temperature is over TCHGSTOP. Charging stops. Writing 1 clears the event (e.g. to acknowledge an interrupt).							
F	RW	EVENTDIETEMPRESUME W1C				Die temperature is under TCHGRESUME. Charging resumes. Writing 1 clears the event (e.g. to acknowledge an interrupt).							
			LOW	0	Low								
			HIGH	1	High								

7.7.1.8 INTENEVENTSBCHARGER0SET

Address offset: 0x8

Battery temperature region and die temperature Interrupt Enable Set

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTNTCCOLD W1S				Writing 1 enables interrupts from EVENTNTCCOLD							
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTNTCCOOL W1S				Writing 1 enables interrupts from EVENTNTCCOOL							
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTNTCWARM W1S				Writing 1 enables interrupts from EVENTNTCWARM							
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTNTCHOT W1S				Writing 1 enables interrupts from EVENTNTCHOT							
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTDIETEMPHIGH W1S				Writing 1 enables interrupts from EVENTDIETEMPHIGH							
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTDIETEMPRESUME W1S				Writing 1 enables interrupts from EVENTDIETEMPRESUME							
			LOW	0	Low								
			HIGH	1	High								

7.7.1.9 INTENEVENTSBCHARGER0CLR

Address offset: 0x9

Battery temperature region and die temperature Interrupt Enable Clear

Bit number						7	6	5	4	3	2	1	0
ID						F	E	D	C	B	A		
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTNTCCOLD W1C			Writing 1 disables interrupts from EVENTNTCCOLD								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTNTCCOOL W1C			Writing 1 disables interrupts from EVENTNTCCOOL								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTNTCWARM W1C			Writing 1 disables interrupts from EVENTNTCWARM								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTNTCHOT W1C			Writing 1 disables interrupts from EVENTNTCHOT								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTDIETEMPHIGH W1C			Writing 1 disables interrupts from EVENTDIETEMPHIGH								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTDIETEMPRESUME W1C			Writing 1 disables interrupts from EVENTDIETEMPRESUME								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.10 EVENTSCHARGER1SET

Address offset: 0xA

Charger Event Set

Bit number						7	6	5	4	3	2	1	0
ID						F	E	D	C	B	A		
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTSUPPLEMENT W1S			Supplement mode activated. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTCHGTRICKLE W1S			Tickle charge started. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTCHGCC W1S			Constant current charging started. Writing 1 sets the event (for debugging).								

Bit number						7	6	5	4	3	2	1	0
ID							F	E	D	C	B	A	
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTCHGCV				Constant voltage charging started. Writing 1 sets the event (for debugging).							
		W1S	LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTCHGCOMPLETED				Charging completed (battery full). Writing 1 sets the event (for debugging).							
		W1S	LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTCHGERROR				Charging error. Writing 1 sets the event (for debugging).							
		W1S	LOW	0	Low								
			HIGH	1	High								

7.7.1.11 EVENTSBCCHARGER1CLR

Address offset: 0xB

Charger Event Clear

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTSUPPLEMENT			Supplement mode activated. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
				LOW	0	Low							
				HIGH	1	High							
B	RW	EVENTCHGTRICKLE			Tickle charge started. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
				LOW	0	Low							
				HIGH	1	High							
C	RW	EVENTCHGCC			Constant current charging started. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
				LOW	0	Low							
				HIGH	1	High							
D	RW	EVENTCHGCV			Constant voltage charging started. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
				LOW	0	Low							
				HIGH	1	High							
E	RW	EVENTCHGCOMPLETED			Charging completed (battery full). Writing 1 clears the event (e.g. to acknowledge an interrupt).								
				LOW	0	Low							
				HIGH	1	High							
F	RW	EVENTCHGERROR			Charging error. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
				LOW	0	Low							
				HIGH	1	High							

7.7.1.12 INTENEVENTSBCHARGER1SET

Address offset: 0xC

Charger Interrupt Enable Set

Bit number						7	6	5	4	3	2	1	0
ID						F	E	D	C	B	A		
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTSUPPLEMENT W1S			Writing 1 enables interrupts from EVENTSUPPLEMENT								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTCHGTRICKLE W1S			Writing 1 enables interrupts from EVENTCHGTRICKLE								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTCHGCC W1S			Writing 1 enables interrupts from EVENTCHGCC								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTCHGCV W1S			Writing 1 enables interrupts from EVENTCHGCV								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTCHGCOMPLETED W1S			Writing 1 enables interrupts from EVENTCHGCOMPLETED								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTCHGERROR W1S			Writing 1 enables interrupts from EVENTCHGERROR								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.13 INTENEVENTSBCHARGER1CLR

Address offset: 0xD

Charger Interrupt Enable Clear

Bit number						7	6	5	4	3	2	1	0
ID						F	E	D	C	B	A		
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTSUPPLEMENT W1C			Writing 1 disables interrupts from EVENTSUPPLEMENT								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTCHGTRICKLE W1C			Writing 1 disables interrupts from EVENTCHGTRICKLE								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTCHGCC W1C			Writing 1 disables interrupts from EVENTCHGCC								
			LOW	0	Low								
			HIGH	1	High								

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
D	RW	EVENTCHGCV W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTCHGCOMPLETED W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTCHGERROR W1C	LOW	0	Low								
			HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.14 EVENTSBCHARGER2SET

Address offset: 0xE

Battery Event Set

Bit number						7	6	5	4	3	2	1	0
ID						C B A							
Reset 0x00						0	0	0	0	0	0	0	0
D	R/W	Field	Value ID	Value	Description								
A	RW	EVENTBATDETECTED	W1S		Reserved (Battery detected). Writing 1 sets the event (for debugging).								
				LOW	0	Low							
				HIGH	1	High							
B	RW	EVENTBATLOST	W1S		Reserved (Battery lost). Writing 1 sets the event (for debugging).								
				LOW	0	Low							
				HIGH	1	High							
C	RW	EVENTBATRECHARGE	W1S		Battery re-charge needed. Writing 1 sets the event (for debugging).								
				LOW	0	Low							
				HIGH	1	High							

7.7.1.15 EVENTSBCHARGER2CLR

Address offset: 0xF

Battery Event Clear

Bit number										7	6	5	4	3	2	1	0	
ID										C B A								
Reset 0x00										0	0	0	0	0	0	0	0	
ID	R/W	Field	Value ID	Value	Description													
A	RW	EVENTBATDETECTED W1C			Reserved (Battery detected). Writing 1 clears the event (e.g. to acknowledge an interrupt).													
			LOW	0														Low
			HIGH	1														High
B	RW	EVENTBATLOST W1C			Reserved (Battery lost). Writing 1 clears the event (e.g. to acknowledge an interrupt).													
			LOW	0														Low
			HIGH	1														High
C	RW	EVENTBATRECHARGE W1C			Battery re-charge needed. Writing 1 clears the event (e.g. to acknowledge an interrupt).													
			LOW	0														Low
			HIGH	1														High

7.7.1.16 INTENEVENTSBCHARGER2SET

Address offset: 0x10

Battery Interrupt Enable Set

Bit number						7	6	5	4	3	2	1	0
ID											C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTBATDETECTED W1S			Reserved (Writing 1 enables interrupts from EVENTBATDETECTED)								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTBATLOST W1S			Reserved (Writing 1 enables interrupts from EVENTBATLOST)								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTBATRECHARGE W1S			Writing 1 enables interrupts from EVENTBATRECHARGE								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.17 INTENEVENTSBCHARGER2CLR

Address offset: 0x11

Battery Interrupt Enable Clear

Bit number										7	6	5	4	3	2	1	0
ID										C B A							
Reset 0x00										0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description												
A	RW	EVENTBATDETECTED			Reserved (Writing 1 disables interrupts from EVENTBATDETECTED)												
			W1C														
				LOW	0	Low											
				HIGH	1	High											
B	RW	EVENTBATLOST			Reserved (Writing 1 disables interrupts from EVENTBATLOST)												
			W1C														
				LOW	0	Low											
				HIGH	1	High											
C	RW	EVENTBATRECHARGE			Writing 1 disables interrupts from EVENTBATRECHARGE												
			W1C														
				LOW	0	Low											
				HIGH	1	High											

7.7.1.18 EVENTSSPHLDSET

Address offset: 0x12

SHPHLD pin and watchdog Event Set

Bit number										7	6	5	4	3	2	1	0
ID										D C B A							
Reset 0x00										0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description												
A	RW	EVENTSHPHLDBTNPRESS			SHPHLD button is pressed. Writing 1 sets the event (for debugging).												
			W1S														
			LOW	0	Low												
			HIGH	1	High												
B	RW	EVENTSHPHLDBTNRELEASE			SHPHLD button is released. Writing 1 sets the event (for debugging).												
			W1S														
			LOW	0	Low												
			HIGH	1	High												
C	RW	EVENTSHPHLDEXIT			SHPHLD button held to exit Ship mode. Writing 1 sets the event (for debugging).												
			W1S														
			LOW	0	Low												
			HIGH	1	High												
D	RW	EVENTWATCHDOGWARN			Watchdog timeout warning. Writing 1 sets the event (for debugging).												
			W1S														
			LOW	0	Low												
			HIGH	1	High												

7.7.1.19 EVENTSSPHLDCLR

Address offset: 0x13

SHPHLD pin and watchdog Event Clear

Bit number						7	6	5	4	3	2	1	0
ID						D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTSHPHLDBTNPRESS W1C			SHPHLD button is pressed. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTSHPHLDBTNRELEASE W1C			SHPHLD button is released. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTSHPHLDEXIT W1C			SHPHLD button held to exit Ship mode. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTWATCHDOGWARN W1C			Watchdog timeout warning. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.20 INTENEVENTSSHPHLDSET

Address offset: 0x14

SHPHLD pin and watchdog Interrupt Enable Set

Bit number						7	6	5	4	3	2	1	0
ID						D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTSHPHLDBTNPRESS W1S			Writing 1 enables interrupts from EVENTSHPHLDBTNPRESS								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTSHPHLDBTNRELEASE W1S			Writing 1 enables interrupts from EVENTSHPHLDBTNRELEASE								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTSHPHLDEXIT W1S			Writing 1 enables interrupts from EVENTSHPHLDEXIT								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTWATCHDOGWARN W1S			Writing 1 enables interrupts from EVENTWATCHDOGWARN								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.21 INTENEVENTSSHPHLDCLR

Address offset: 0x15

SHPHLD pin and watchdog Interrupt Enable Clear

Bit number						7	6	5	4	3	2	1	0				
ID											D	C	B	A			
Reset 0x00											0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description												
A	RW	EVENTSHPHLDBTNPRESS W1C			Writing 1 disables interrupts from EVENTSHPHLDBTNPRESS												
			LOW	0	Low												
			HIGH	1	High												
B	RW	EVENTSHPHLDBTNRELEASE W1C			Writing 1 disables interrupts from EVENTSHPHLDBTNRELEASE												
			LOW	0	Low												
			HIGH	1	High												
C	RW	EVENTSHPHLDEXIT W1C			Writing 1 disables interrupts from EVENTSHPHLDEXIT												
			LOW	0	Low												
			HIGH	1	High												
D	RW	EVENTWATCHDOGWARN W1C			Writing 1 disables interrupts from EVENTWATCHDOGWARN												
			LOW	0	Low												
			HIGH	1	High												

7.7.1.22 EVENTSVBUSIN0SET

Address offset: 0x16

VBUS Event Set

Bit number						7	6	5	4	3	2	1	0
ID						F	E	D	C	B	A		
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTVBUSDETECTED W1S			VBUS detected. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTVBUSREMOVED W1S			VBUS removed. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTVBUSOVRVOLTDETECTED W1S			VBUS over voltage detected. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTVBUSOVRVOLTREMOVED W1S			VBUS over voltage removed. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTVBUSUNDVOLTDETECTED W1S			VBUS under voltage detected. Writing 1 sets the event (for debugging).								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTVBUSUNDVOLTREMOVED W1S			VBUS under voltage removed. Writing 1 sets the event (for debugging).								
			LOW	0	Low								

Bit number	7	6	5	4	3	2	1	0
ID	F E D C B A							
Reset 0x00	0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description			
			HIGH	1	High			

7.7.1.23 EVENTSVBUSIN0CLR

Address offset: 0x17

VBUS Event Clear

Bit number	7	6	5	4	3	2	1	0
ID	F E D C B A							
Reset 0x00	0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description			
A	RW	EVENTVBUSDETECTED W1C			VBUS detected. Writing 1 clears the event (e.g. to acknowledge an interrupt).			
			LOW	0				
			HIGH	1				
B	RW	EVENTVBUSREMOVED W1C			VBUS removed. Writing 1 clears the event (e.g. to acknowledge an interrupt).			
			LOW	0				
			HIGH	1				
C	RW	EVENTVBUSOVRVOLTDETECTED W1C			VBUS over voltage detected. Writing 1 clears the event (e.g. to acknowledge an interrupt).			
			LOW	0				
			HIGH	1				
D	RW	EVENTVBUSOVRVOLTREMOVED W1C			VBUS over voltage removed. Writing 1 clears the event (e.g. to acknowledge an interrupt).			
			LOW	0				
			HIGH	1				
E	RW	EVENTVBUSUNDVOLTDETECTED W1C			VBUS under voltage detected. Writing 1 clears the event (e.g. to acknowledge an interrupt).			
			LOW	0				
			HIGH	1				
F	RW	EVENTVBUSUNDVOLTREMOVED W1C			VBUS under voltage removed. Writing 1 clears the event (e.g. to acknowledge an interrupt).			
			LOW	0				
			HIGH	1				

7.7.1.24 INTENEVENTSVBUSIN0SET

Address offset: 0x18

VBUS Interrupt Enable Set

Bit number	7	6	5	4	3	2	1	0
ID	F E D C B A							
Reset 0x00	0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description			
A	RW	EVENTVBUSDETECTED W1S			Writing 1 enables interrupts from EVENTVBUSDETECTED			
			LOW	0				
			HIGH	1				

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
B	RW	EVENTVBUSREMOVED W1S			Writing 1 enables interrupts from EVENTVBUSREMOVED								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTVBUSOVRVOLTDETECTED W1S			Writing 1 enables interrupts from EVENTVBUSOVRVOLTDETECTED								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTVBUSOVRVOLTREMOVED W1S			Writing 1 enables interrupts from EVENTVBUSOVRVOLTREMOVED								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTVBUSUNDVOLTDETECTED W1S			Writing 1 enables interrupts from EVENTVBUSUNDVOLTDETECTED								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTVBUSUNDVOLTREMOVED W1S			Writing 1 enables interrupts from EVENTVBUSUNDVOLTREMOVED								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.25 INTENEVENTSVBUSIN0CLR

Address offset: 0x19

VBUS Interrupt Enable Clear

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTVBUSDETECTED W1C			Writing 1 disables interrupts from EVENTVBUSDETECTED								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTVBUSREMOVED W1C			Writing 1 disables interrupts from EVENTVBUSREMOVED								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTVBUSOVRVOLTDETECTED W1C			Writing 1 disables interrupts from EVENTVBUSOVRVOLTDETECTED								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTVBUSOVRVOLTREMOVED W1C			Writing 1 disables interrupts from EVENTVBUSOVRVOLTREMOVED								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTVBUSUNDVOLTDETECTED W1C			Writing 1 disables interrupts from EVENTVBUSUNDVOLTDETECTED								
			LOW	0	Low								

Bit number											7	6	5	4	3	2	1	0
ID											F E D C B A							
Reset 0x00											0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description													
F	RW	EVENTVBUSUNDVOLTREMOVED W1C	HIGH	1	High													
			Writing 1 disables interrupts from EVENTVBUSUNDVOLTREMOVED															
			LOW	0	Low													
			HIGH	1	High													

7.7.1.26 EVENTSVBUSIN1SET

Address offset: 0x1A

Thermal and charger detection Event Set

Bit number					7	6	5	4	3	2	1	0	
ID					F E D C B A								
Reset 0x00					0	0	0	0	0	0	0	0	
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTTHERMALWARNDETECTED W1S			Thermal warning detected. Writing 1 sets the event (for debugging).								
			LOW	0									Low
			HIGH	1									High
B	RW	EVENTTHERMALWARNREMOVED W1S			Thermal warning removed. Writing 1 sets the event (for debugging).								
			LOW	0									Low
			HIGH	1									High
C	RW	EVENTTHERMALSHUTDOWNDETECTED W1S			Thermal shutdown detected. Writing 1 sets the event (for debugging).								
			LOW	0									Low
			HIGH	1									High
D	RW	EVENTTHERMALSHUTDOWNREMOVED W1S			Thermal shutdown removed. Writing 1 sets the event (for debugging).								
			LOW	0									Low
			HIGH	1									High
E	RW	EVENTCC1STATECHANGE W1S			Voltage changed on pin CC1. Writing 1 sets the event (for debugging).								
			LOW	0									Low
			HIGH	1									High
F	RW	EVENTCC2STATECHANGE W1S			Voltage changed on pin CC2. Writing 1 sets the event (for debugging).								
			LOW	0									Low
			HIGH	1									High

7.7.1.27 EVENTSVBUSIN1CLR

Address offset: 0x1B

Thermal and charger detection Event Clear

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTTHERMALWARNDETECTED W1C			Thermal warning detected. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTTHERMALWARNREMOVED W1C			Thermal warning removed. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTTHERMALSHUTDOWNDETECTED W1C			Thermal shutdown detected. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTTHERMALSHUTDOWNREMOVED W1C			Thermal shutdown removed. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTCC1STATECHANGE W1C			Voltage changed on pin CC1. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTCC2STATECHANGE W1C			Voltage changed on pin CC2. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.28 INTENEVENTSVBUSIN1SET

Address offset: 0x1C

Thermal and charger detection Interrupt Enable Set

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTTHERMALWARNDETECTED W1S			Writing 1 enables interrupts from EVENTTHERMALWARNDETECTED								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTTHERMALWARNREMOVED W1S			Writing 1 enables interrupts from EVENTTHERMALWARNREMOVED								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTTHERMALSHUTDOWNDETECTED W1S			Writing 1 enables interrupts from EVENTTHERMALSHUTDOWNDETECTED								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTTHERMALSHUTDOWNREMOVED W1S			Writing 1 enables interrupts from EVENTTHERMALSHUTDOWNREMOVED								
			LOW	0	Low								

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
E	RW	EVENTCC1STATECHANGE W1S	HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTCC2STATECHANGE W1S	HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.29 INTENEVENTSVBUSIN1CLR

Address offset: 0x1D

Thermal and charger detection Interrupt Enable Clear

Bit number						7	6	5	4	3	2	1	0
ID						F E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTTHERMALWARNDETECTED W1C	HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
B	RW	EVENTTHERMALWARNREMOVED W1C	HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
C	RW	EVENTTHERMALSHUTDOWNDETECTED W1C	HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTTHERMALSHUTDOWNREMOVED W1C	HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTCC1STATECHANGE W1C	HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								
F	RW	EVENTCC2STATECHANGE W1C	HIGH	1	High								
			LOW	0	Low								
			HIGH	1	High								

7.7.1.30 EVENTSGPIOSSET

Address offset: 0x22

GPIO Event Set

Bit number						7	6	5	4	3	2	1	0
ID						E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTGPIOEDGEDETECT0 W1S			Edge is detected on GPIO0. GPIOMODE = 3 : rising edge GPIOMODE = 4 : falling edge. Writing 1 sets the event (for debugging).								
			LOW	0									
			HIGH	1									
B	RW	EVENTGPIOEDGEDETECT1 W1S			Edge is detected on GPIO1. GPIOMODE = 3 : rising edge GPIOMODE = 4 : falling edge. Writing 1 sets the event (for debugging).								
			LOW	0									
			HIGH	1									
C	RW	EVENTGPIOEDGEDETECT2 W1S			Edge is detected on GPIO2. GPIOMODE = 3 : rising edge GPIOMODE = 4 : falling edge. Writing 1 sets the event (for debugging).								
			LOW	0									
			HIGH	1									
D	RW	EVENTGPIOEDGEDETECT3 W1S			Edge is detected on GPIO3. GPIOMODE = 3 : rising edge GPIOMODE = 4 : falling edge. Writing 1 sets the event (for debugging).								
			LOW	0									
			HIGH	1									
E	RW	EVENTGPIOEDGEDETECT4 W1S			Edge is detected on GPIO4. GPIOMODE = 3 : rising edge GPIOMODE = 4 : falling edge. Writing 1 sets the event (for debugging).								
			LOW	0									
			HIGH	1									

7.7.1.31 EVENTSGPIOCLR

Address offset: 0x23

GPIO Event Clear

Bit number						7	6	5	4	3	2	1	0
ID						E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	RW	EVENTGPIOEDGEDETECT0 W1C			Edge is detected on GPIO0. GPIOMODE = 3 : rising edge GPIOMODE = 4 : falling edge. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0									
			HIGH	1									
B	RW	EVENTGPIOEDGEDETECT1 W1C			Edge is detected on GPIO1. GPIOMODE = 3 : rising edge GPIOMODE = 4 : falling edge. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0									
			HIGH	1									
C	RW	EVENTGPIOEDGEDETECT2 W1C			Edge is detected on GPIO2. GPIOMODE = 3 : rising edge GPIOMODE = 4 : falling edge. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0									
			HIGH	1									
D	RW	EVENTGPIOEDGEDETECT3 W1C			Edge is detected on GPIO3. GPIOMODE = 3 : rising edge GPIOMODE = 4 : falling edge. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0									
			HIGH	1									
E	RW	EVENTGPIOEDGEDETECT4 W1C			Edge is detected on GPIO4. GPIOMODE = 3 : rising edge GPIOMODE = 4 : falling edge. Writing 1 clears the event (e.g. to acknowledge an interrupt).								
			LOW	0									

Bit number	7	6	5	4	3	2	1	0
ID	E D C B A							
Reset 0x00	0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description			
			HIGH	1	High			

7.7.1.32 INTENEVENTSGPIOSET

Address offset: 0x24

GPIO Interrupt Enable Set

Bit number	7	6	5	4	3	2	1	0
ID	E D C B A							
Reset 0x00	0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description			
A	RW	EVENTGPIOEDGEDETECT0 W1S			Writing 1 enables interrupts from EVENTGPIOEDGEDETECT0			
			LOW	0	Low			
			HIGH	1	High			
B	RW	EVENTGPIOEDGEDETECT1 W1S			Writing 1 enables interrupts from EVENTGPIOEDGEDETECT1			
			LOW	0	Low			
			HIGH	1	High			
C	RW	EVENTGPIOEDGEDETECT2 W1S			Writing 1 enables interrupts from EVENTGPIOEDGEDETECT2			
			LOW	0	Low			
			HIGH	1	High			
D	RW	EVENTGPIOEDGEDETECT3 W1S			Writing 1 enables interrupts from EVENTGPIOEDGEDETECT3			
			LOW	0	Low			
			HIGH	1	High			
E	RW	EVENTGPIOEDGEDETECT4 W1S			Writing 1 enables interrupts from EVENTGPIOEDGEDETECT4			
			LOW	0	Low			
			HIGH	1	High			

7.7.1.33 INTENEVENTSGPIOCLR

Address offset: 0x25

GPIO Interrupt Enable Clear

Bit number	7	6	5	4	3	2	1	0
ID	E D C B A							
Reset 0x00	0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description			
A	RW	EVENTGPIOEDGEDETECT0 W1C			Writing 1 disables interrupts from EVENTGPIOEDGEDETECT0			
			LOW	0	Low			
			HIGH	1	High			
B	RW	EVENTGPIOEDGEDETECT1 W1C			Writing 1 disables interrupts from EVENTGPIOEDGEDETECT1			
			LOW	0	Low			
			HIGH	1	High			

Bit number						7	6	5	4	3	2	1	0
ID						E D C B A							
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
C	RW	EVENTGPIOEDGEDETECT2 W1C			Writing 1 disables interrupts from EVENTGPIOEDGEDETECT2								
			LOW	0	Low								
			HIGH	1	High								
D	RW	EVENTGPIOEDGEDETECT3 W1C			Writing 1 disables interrupts from EVENTGPIOEDGEDETECT3								
			LOW	0	Low								
			HIGH	1	High								
E	RW	EVENTGPIOEDGEDETECT4 W1C			Writing 1 disables interrupts from EVENTGPIOEDGEDETECT4								
			LOW	0	Low								
			HIGH	1	High								

7.8 Reset and error registers

This section details the error and reset related registers.

Note: During the cooling period after a TSD and if VSYS drops below VSYS_{POF}, VSYSLOW could be set instead of THERMALSHUTDOWN in register [RSTCAUSE](#) on page 150.

7.8.1 Registers

Instances

Instance	Base address	Description
ERRLOG	0x00000E00	Error Log registers ERRLOG register map

Register overview

Register	Offset	Description
TASKCLRERRLOG	0x0	Task to clear the Errlog registers
SCRATCH0	0x1	Boot monitor control and scratch register 0
SCRATCH1	0x2	Scratch register 1
RSTCAUSE	0x3	Reset reasons. Cleared with TASKCLRERRLOG
CHARGERERRREASON	0x4	Error log for charger. Cleared with TASKCLRERRLOG
CHARGERERRSENSOR	0x5	Error log for charger. Cleared with TASKCLRERRLOG

7.8.1.1 TASKCLRERRLOG

Address offset: 0x0

Task to clear the Errlog registers

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00													0
ID	R/W	Field	Value ID	Value	Description								
A	W	TASKCLRERRLOG			Clear registers RSTCAUSE, CHARGERERRREASON and CHARGERERRSENSOR								

7.8.1.2 SCRATCH0

Address offset: 0x1

Boot monitor control and scratch register 0

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00													0
ID	R/W	Field	Value ID	Value	Description								
A	RW	BOOTTIMEREN			Enable boot monitor timer, only cleared by POR								
			NOBOOTMON	0	Boot monitor disable								
			BOOTMON	1	Boot monitor enable								
B	RW	SCRATCH0			Scratch register, only cleared by POR								

7.8.1.3 SCRATCH1

Address offset: 0x2

Scratch register 1

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00													0
ID	R/W	Field	Value ID	Value	Description								
A	RW	SCRATCH1			Scratch register, only cleared by POR								

7.8.1.4 RSTCAUSE

Address offset: 0x3

Reset reasons. Cleared with TASKCLRERRLOG

Bit number						7	6	5	4	3	2	1	0
ID													A
Reset 0x00													0
ID	R/W	Field	Value ID	Value	Description								
A	R	SHIPMODEEXIT			Reset caused by Ship mode exit								
			NORST	0	No reset								
			RST	1	Reset activated by Ship mode exit								
B	R	BOOTMONITORTIMEOUT			Reset caused by boot monitor timeout								
			NORST	0	No reset								
			RST	1	Reset activated by boot monitor								
C	R	WATCHDOGTIMEOUT			Reset caused by watchdog timeout								
			NORST	0	No reset								
			RST	1	Reset activated by watchdog								
D	R	LONGPRESSTIMEOUT			Reset caused by long press reset								
			NORST	0	No reset								
			RST	1	Long press reset								

Bit number						7	6	5	4	3	2	1	0
ID						G	F	E	D	C	B	A	
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
E	R	THERMALSHUTDOWN			Reset caused by thermal shutdown (TSD)								
			NORST	0	No reset								
			RST	1	Reset activated by TSD								
F	R	VSYSLOW			Reset caused by power failure (POF)								
			NORST	0	No reset								
			RST	1	Reset activated by POF								
G	R	SWRESET			Reset caused by TASKSWRESET								
			NORST	0	No reset								
			RST	1	Reset activated by TASKSWRESET								

7.8.1.5 CHARGERERRREASON

Address offset: 0x4

Error log for charger. Cleared with TASKCLRERRLOG

Bit number						7	6	5	4	3	2	1	0
ID						G	F	E	D	C	B	A	
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	NTCSENSORERR			NTC thermistor sensor error								
B	R	VBATSENSORERR			VBAT sensor error								
C	R	VBATLOW			VBAT low error								
D	R	VTRICKLE			Vtrickle error								
E	R	MEASTIMEOUT			Measurement timeout error								
F	R	CHARGETIMEOUT			Charge timeout error								
G	R	TRICKLETIMEOUT			Trickle timeout error								

7.8.1.6 CHARGERERRSENSOR

Address offset: 0x5

Error log for charger. Cleared with TASKCLRERRLOG

Bit number						7	6	5	4	3	2	1	0
ID						H	G	F	E	D	C	B	A
Reset 0x00						0	0	0	0	0	0	0	0
ID	R/W	Field	Value ID	Value	Description								
A	R	SENSORNTCCOLD			NTC cold region active when error occurs								
B	R	SENSORNTCCOOL			NTC cool region active when error occurs								
C	R	SENSORNTCWARM			NTC warm region active when error occurs								
D	R	SENSORNTCHOT			NTC hot region active when error occurs								
E	R	SENSORVTERM			Vterm status when error occurs								
F	R	SENSORRECHARGE			Recharge status when error occurs								
G	R	SENSORVTRICKLE			Vtrickle status when error occurs								
H	R	SENSORVBATLOW			Vbatlow status when error occurs								

8 Application

The following application example uses nPM1300 and an nRF5x *Bluetooth*® Low Energy System on Chip (SoC). For other configurations, see [Reference circuitry](#) on page 159.

The example application is for a design with the following configuration and features:

- BUCK, LOADSW, and LDO are in use
- Host software controls the device through TWI, the interrupt on **GPIO1**, and RESET on the **GPIO0** pin
- Three LEDs available
- Battery pack with NTC thermistor
- Ship mode
- Low battery indication LED

8.1 Schematic

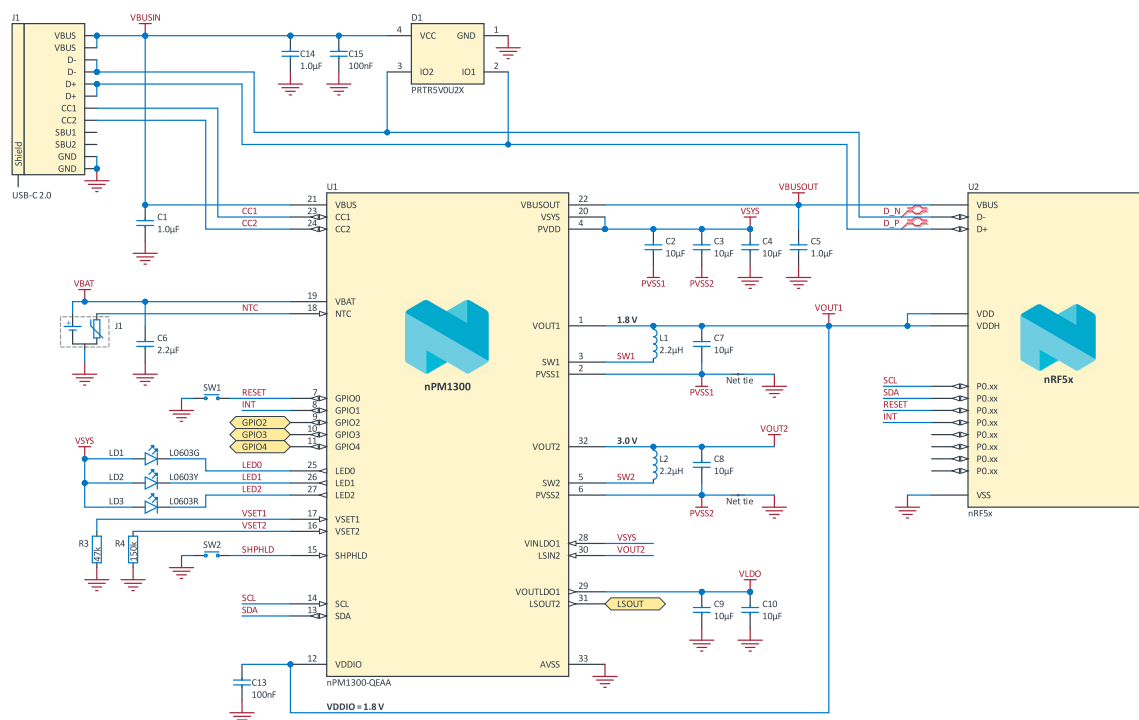


Figure 54: Application example

8.2 Supplying from BUCK

An application must not be supplied directly from **VBAT**. This can interrupt the battery charging process causing unwanted behavior from the charger. Use either **VOUT1**, **VOUT2**, or **VSYS** to supply the application.

BUCK1 starts automatically and supplies the nRF5x host SoC with 1.8 V. BUCK1 is the I/O voltage for the system. BUCK2 starts automatically with 3 V output voltage for other application features.

8.3 USB port negotiation

nRF5x can connect to a USB host.

Port negotiation is performed after nPM1300 port detection. The nRF5x device and nPM1300 are both connected to USB-C in the application example.

- The **D+** and **D-** pins are connected to nRF5x. The **CC1** and **CC2** pins are connected to nPM1300. The nRF5x SoC must wait until nPM1300 completes port detection using the USB configuration channel.
- The nRF5x device must set the correct current limit as described in [Charge current limit \(I_{CHG}\)](#) on page 28.
- **VBUS** is supplied to SYSREG on nPM1300 and **VBUSOUT** supplies the nRF5x **VBUS** input.

VBUSOUT is only for host sensing and should not be used as a source.

8.4 Charging and error states

Three LEDs can be used for charging indicators or general purpose by the application.

8.5 Termination voltage and current

The termination voltage, V_{TERM} , is configured through TWI up to 4.45 V.

Charge current is configured through TWI.

8.6 NTC thermistor configuration

The **NTC** pin connects to an external NTC thermistor. Place the NTC thermistor with thermal coupling on the battery pack.

8.7 Ship mode

Ship mode is enabled at production time through the TWI interface.

SHPHLD is connected to **SW2** and is in the circuit to exit Ship mode. If another circuit is present instead of a button, any signal that is able to pull the **SHPHLD** pin low for the required period can be connected to that net. See [Ship and Hibernate modes](#) on page 121 for more information.

9 Hardware and layout

9.1 Pin assignments

The pin assignment figures and tables describe the pinouts for the product variants of the chip.

9.1.1 QFN32 pin assignments

The pin assignment figure and table describe the assignments for this variant of the chip.

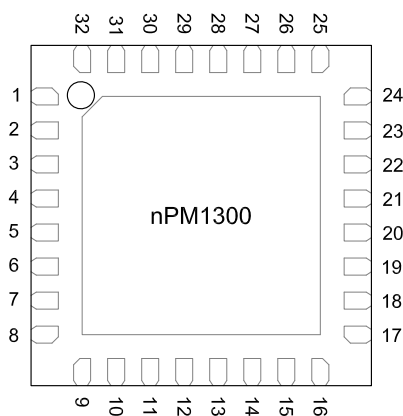


Figure 55: QFN32 pin assignments (top view)

Pin	Name	Function	Description
1	VOUT1	Power	BUCK1 output
2	PVSS1	Power	BUCK1 power ground
3	SW1	Power	BUCK1 regulator output to inductor
4	PVDD	Power	BUCK[n] power input
5	SW2	Power	BUCK2 regulator output to inductor
6	PVSS2	Power	BUCK2 power ground
7	GPIO0	Digital I/O	GPIO0
8	GPIO1	Digital I/O	GPIO1
9	GPIO2	Digital I/O	GPIO2
10	GPIO3	Digital I/O	GPIO3
11	GPIO4	Digital I/O	GPIO4
12	VDDIO	Power	Supply for TWI and GPIOs
13	SDA	Digital I/O	TWI data
14	SCL	Digital input	TWI clock
15	SHPHLD	Digital input	Ship mode hold
16	VSET2	Analog input	Voltage set for BUCK2 to resistor
17	VSET1	Analog input	Voltage set for BUCK1 to resistor
18	NTC	Analog input	Battery thermistor
19	VBAT	Power	Battery
20	VSYS	Power	System voltage output
21	VBUS	Power	Input supply
22	VBUSOUT	Analog output	VBUS output for host
23	CC1	Analog input	USB Type-C configuration channel 1
24	CC2	Analog input	USB Type-C configuration channel 2
25	LED0	Analog output	LEDDRV0 output
26	LED1	Analog output	LEDDRV1 output
27	LED2	Analog output	LEDDRV2 output
28	LSIN1/VINLDO1	Power	LOADSW1 supply or LDO1 input
29	LSOUT1/VOUTLDO1	Power	LOADSW1 or LDO1 output
30	LSIN2/VINLDO2	Power	LOADSW2 supply or LDO2 input
31	LSOUT2/VOUTLDO2	Power	LOADSW2 or LDO2 output
32	VOUT2	Power	BUCK2 output
Exposed pad	AVSS	Power	Ground

Table 34: QFN32 pin assignments

9.1.2 CSP ball assignments

The ball assignment figure and table describe the ball assignments for this variant of the chip.

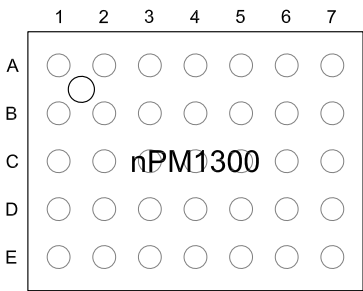


Figure 56: CSP ball assignment (top view)

Ball	Name	Function	Description
A1	LED0	Analog output	LEDDRV0 output
A2	LED1	Analog output	LEDDRV1 output
A3	LED2	Analog output	LEDDRV2 output
A4	LSOUT1/VOU TLDO1	Power	LOADSW1 or LDO1 output
A5	LSOUT2/VOU TLDO2	Power	LOADSW2 or LDO2 output
A6	AVSS	Power	Ground
A7	PVSS1	Power	BUCK1 power ground
B1, B2	VBUS	Power	Input supply
B3	CC2	Analog input	USB Type-C configuration channel 2
B4	LSIN1/VINLDO1	Power	LOADSW1 supply or LDO1 input
B5	LSIN2/VINLDO2	Power	LOADSW2 supply or LDO2 input
B6	VOUT1	Power	BUCK1 output
B7	SW1	Power	BUCK1 regulator output to inductor
C1, C2	VSYS	Power	System voltage output
C3	VBUSOUT	Analog output	VBUS output for host
C4	GPIO3	Digital I/O	GPIO3
C5	GPIO2	Digital I/O	GPIO2
C6	VOUT2	Power	BUCK2 output
C7	PVDD	Power	Power input for BUCK[n]
D1, D2	VBAT	Power	Battery
D3	NTC	Analog input	Battery thermistor
D4	SHPHLD	Digital input	Ship mode hold
D5	CC1	Analog input	USB Type-C configuration channel 1
D6	GPIO0	Digital I/O	GPIO0
D7	SW2	Power	BUCK2 regulator output to inductor
E1	VSET2	Analog input	Voltage set for BUCK2 to resistor
E2	VSET1	Analog input	Voltage set for BUCK1 to resistor
E3	SCL	Digital input	TWI clock
E4	VDDIO	Power	Supply for TWI and GPIOs
E5	SDA	Digital I/O	TWI data
E6	GPIO1	Digital I/O	GPIO1
E7	PVSS2	Power	BUCK2 power ground

Table 35: Pin descriptions

9.2 Mechanical specifications

The mechanical specifications for the packages show the dimensions.

9.2.1 QFN32 package

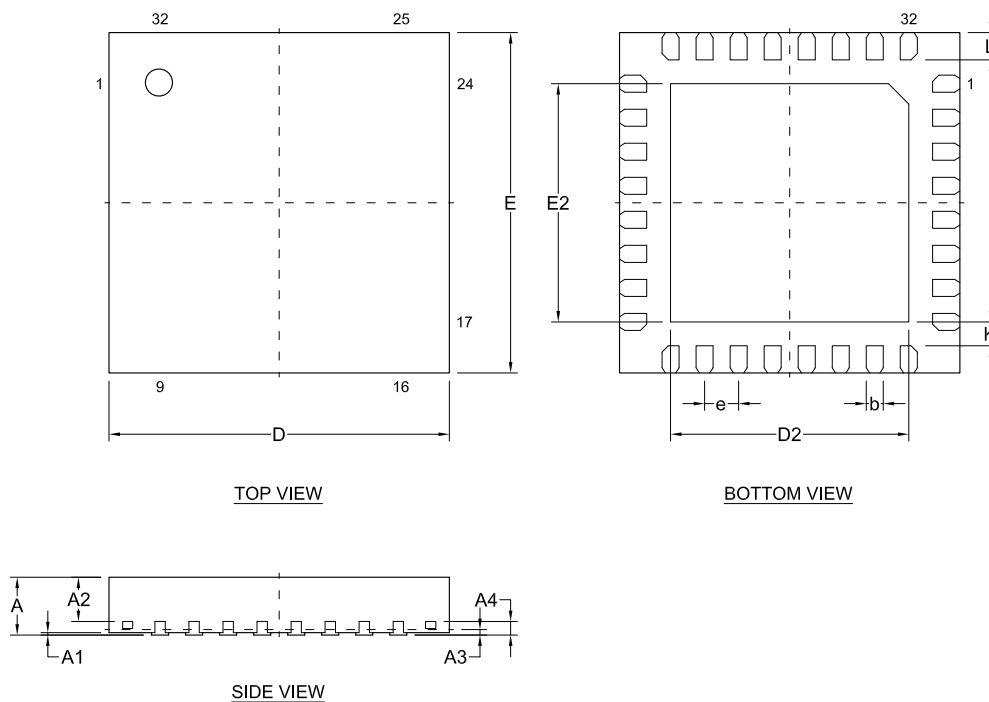


Figure 57: QFN32 5.0x5.0 mm package

	A	A1	A2	A3	A4	b	D, E	D2, E2	e	K	L
Min.	0.8	0				0.2	4.9	3.4			0.3
Nom.	0.85	0.035	0.65	0.08	0.203	0.25	5	3.5	0.5	0.35	0.4
Max.	0.9	0.05				0.3	5.1	3.6			0.45

Table 36: QFN32 dimensions in millimeters

9.2.2 CSP package

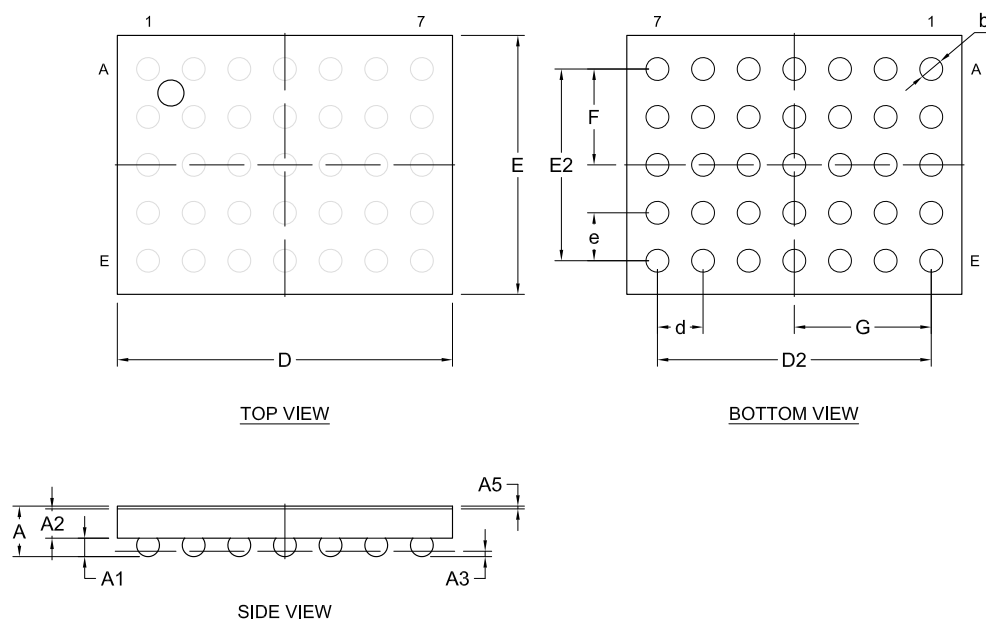


Figure 58: CSP 2.3775x3.0775 mm package

	A	A1	A2	A3	A5	D	D2	d	E	E2	e	F	G	b
Min.	0.416	0.14	0.254		0.022	3.0475			2.3475					0.195
Nom.	0.464		0.269	0.05	0.025	3.0775	2.514	0.419	2.3775	1.76	0.44	0.88	1.257	
Max.	0.512	0.2	0.284		0.028	3.1075			2.4075					0.255

Table 37: CSP dimensions in millimeters

9.3 Reference circuitry

Documentation for the different package reference circuits, including Altium Designer files, PCB layout files, and PCB production files, can be downloaded from www.nordicsemi.com.

The following reference circuits for nPM1300 show the schematics and components to support different configurations in a design.

	Configuration 1	Configuration 2	Configuration 3
Description	Full configuration	Simple configuration	Minimal configuration
BUCKs	Both configured	One configured	Not used
LOADSWs	Both configured, LDO mode	One configured, load switch mode	Not used
Ship mode exit	Configured	Configured	Not used
Charging	Available	Available	Available
Battery thermistor	Configured	Configured	Not used
LEDs	Three available	One available	Not used
GPIOs	Configured	Configured	Configured
TWI	Configured	Configured	Configured
VSET1	47 k Ω $\pm$ 1%	47 k Ω $\pm$ 1%	Not used
VSET2	150 k Ω $\pm$ 1%	Not used	Not used
VOUT1	1.8 V	1.8 V	Not used
VOUT2	3.0 V	Not used	Not used
VBUSOUT	Configured	Configured	Not used
VDDIO	Configured	Configured	Configured

Table 38: PCB application configuration

9.3.1 Configuration 1

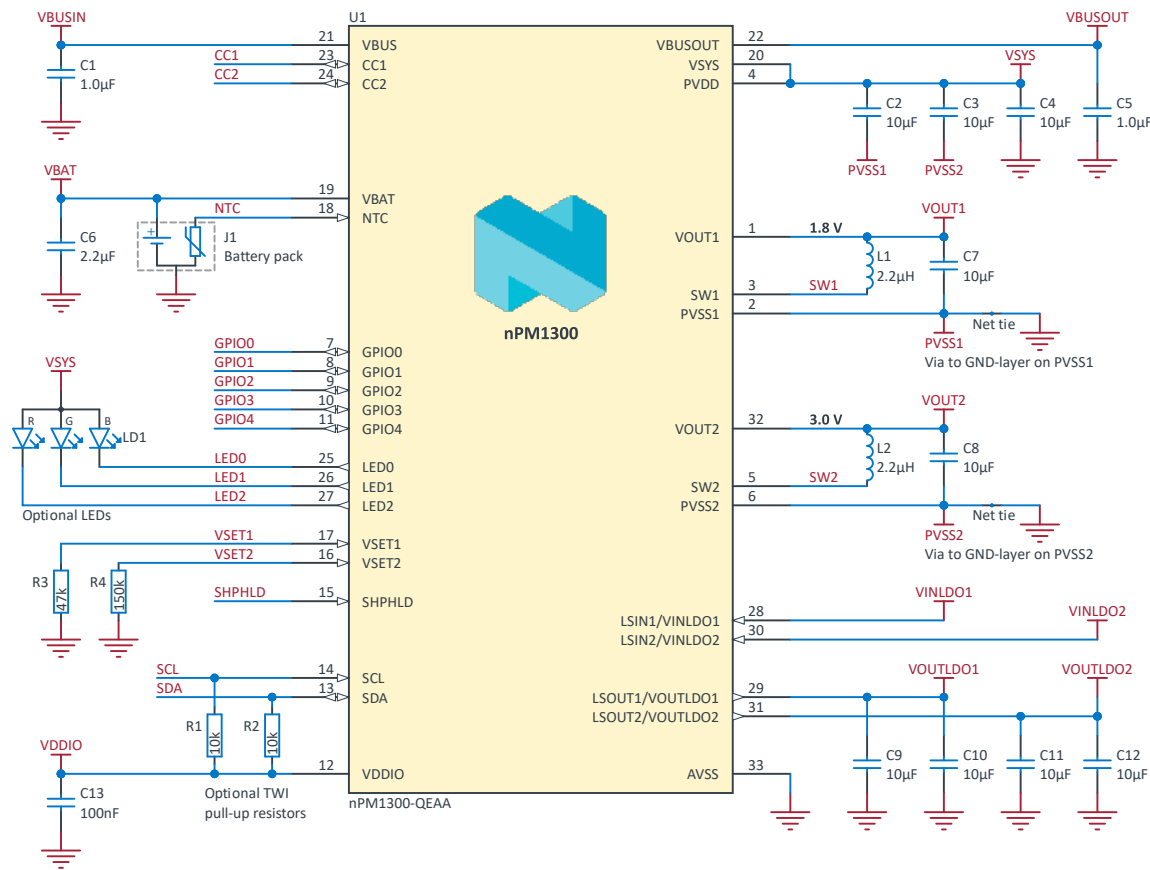


Figure 59: QFN schematic

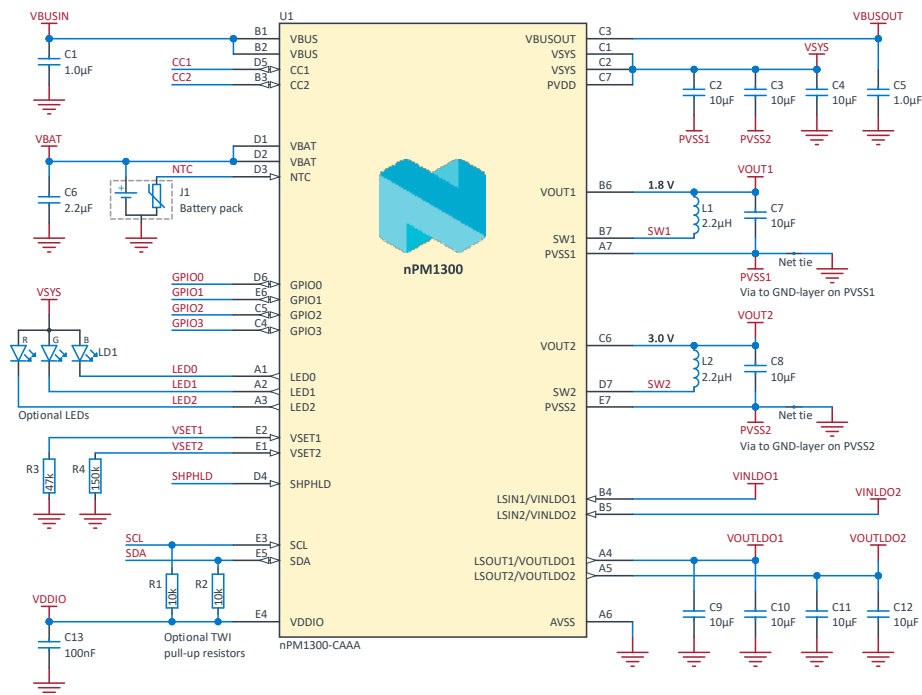


Figure 60: CSP schematic

Designator	Value	Description	Footprint
C1, C5	1.0 μ F	Capacitor, X5R, 10 V, $\pm 10\%$	0603
C2, C3, C4, C7, C8, C9, C10, C11, C12	10 μ F	Capacitor, X5R, 25 V, $\pm 20\%$	0603
C6	2.2 μ F	Capacitor, X7R, 16 V, $\pm 10\%$	0603
C13	100 nF	Capacitor, X5R, $\pm 10\%$	0201
L1, L2	2.2 μ H	Inductor, DCR < 400 m Ω , $\pm 20\%$	0806
R1, R2	Dependent on bus speed and parasitic capacitances	Optional pull-up resistors for TWI, 0.05 W, $\pm 1\%$	0201
R3, R4	See Output voltage selection on page 49	Resistors for setting the BUCK1 and BUCK2 output voltages, 0.05 W, $\pm 1\%$	0201
U1	nPM1300	nPM1300	QFN32 or CSP35

Table 39: Bill of material

9.3.2 Configuration 2

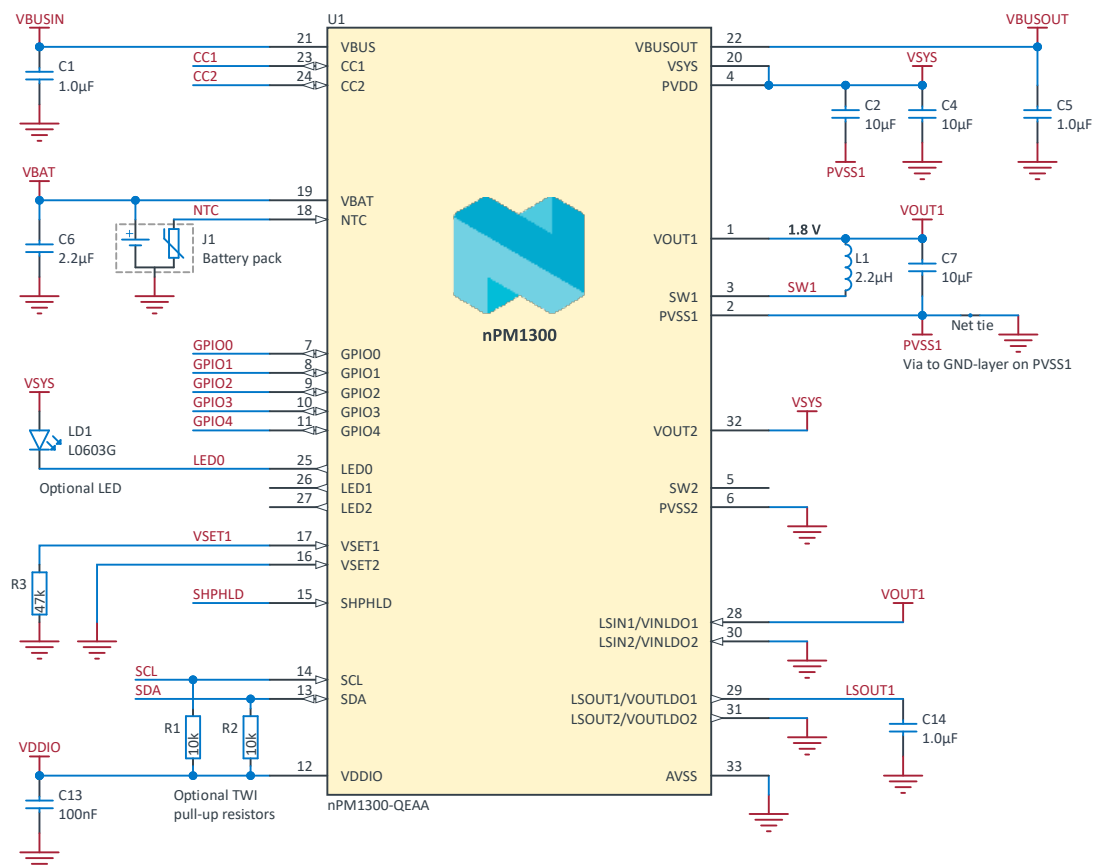


Figure 61: QFN schematic

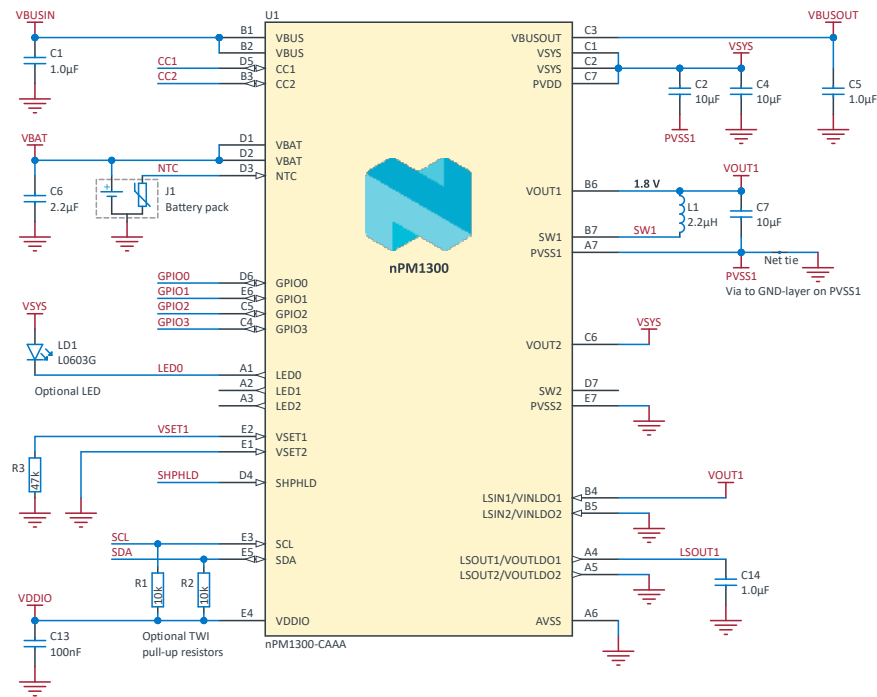


Figure 62: CSP schematic

Designator	Value	Description	Footprint
C1, C5, C14	1.0 μ F	Capacitor, X5R, 10 V, $\pm 10\%$	0603
C2, C4, C7	10 μ F	Capacitor, X5R, 25 V, $\pm 20\%$	0603
C6	2.2 μ F	Capacitor, X5R, 25 V, $\pm 10\%$	0603
C13	100 nF	Capacitor, X5R, 25 V, $\pm 10\%$	0201
L1	2.2 μ H	Inductor, DCR < 400 m Ω , $\pm 20\%$	0806
R1, R2	Dependent on bus speed and parasitic capacitances	Optional pull-up resistors for TWI, 0.05 W, $\pm 1\%$	0201
R3	See Output voltage selection on page 49	Resistors for setting the BUCK1 and BUCK2 output voltages, 0.05 W, $\pm 1\%$	0201
U1	nPM1300	nPM1300	QFN32 or CSP35

Table 40: Bill of material

9.3.3 Configuration 3

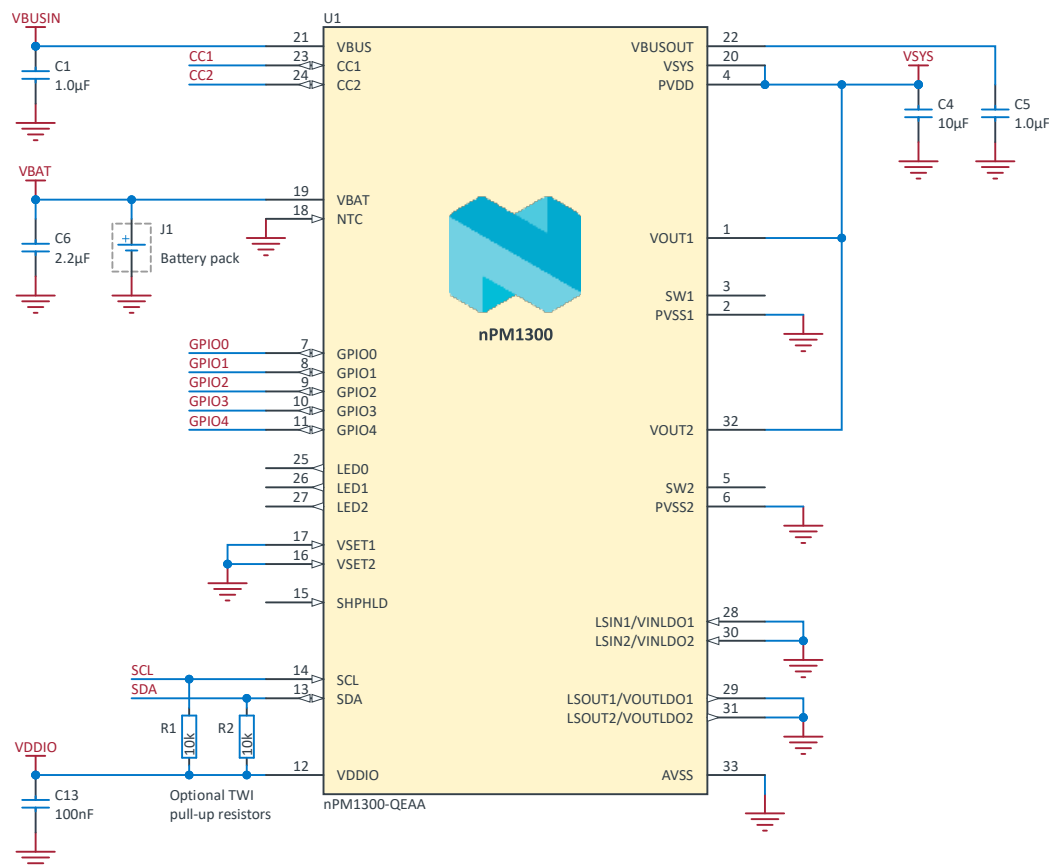


Figure 63: QFN schematic

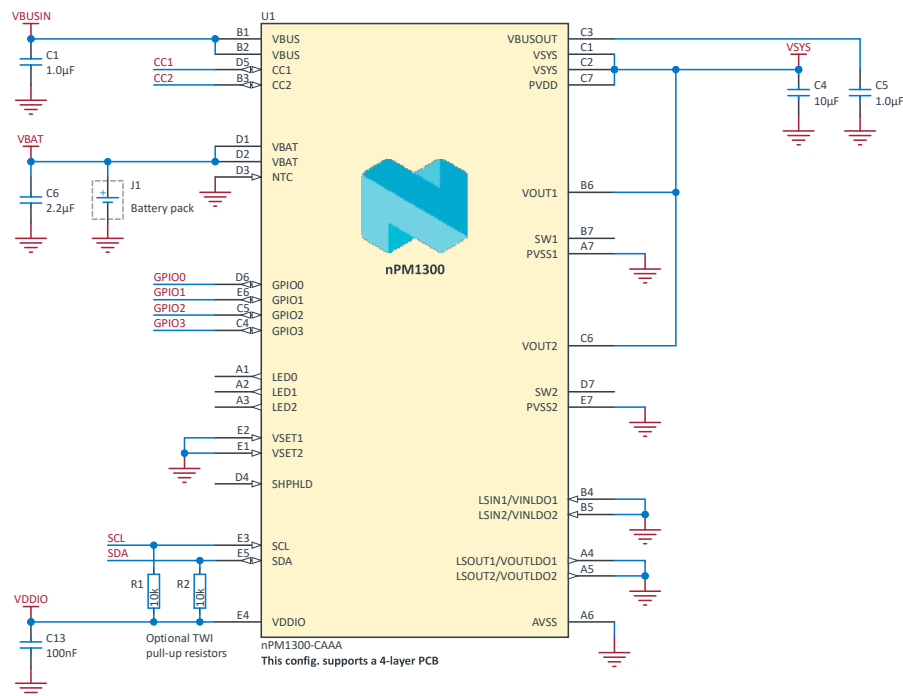


Figure 64: CSP schematic

Designator	Value	Description	Footprint
C1, C5	1.0 μ F	Capacitor, X5R, 10 V, $\pm 10\%$	0603
C4	10 μ F	Capacitor, X5R, 25 V, $\pm 20\%$	0603
C6	2.2 μ F	Capacitor, X7R, 16 V, $\pm 10\%$	0603
C13	100 nF	Capacitor, X5R, $\pm 10\%$	0201
R1, R2	Dependent on bus speed and parasitic capacitances	Optional pull-up resistors for TWI, 0.05 W, $\pm 1\%$	0201
U1	nPM1300	nPM1300	QFN32 or CSP35

Table 41: Bill of material

9.3.4 PCB guidelines

A well designed PCB is necessary to achieve good performance. A poor layout can lead to loss in performance or functionality.

To ensure functionality, it is essential to follow the schematics and layout references closely.

A PCB with a minimum of two layers, including a ground plane, is recommended for optimal performance.

The BUCK supply voltage should be decoupled with high performance capacitors as close as possible to the supply pins.

Long power supply lines on the PCB should be avoided. All device grounds, VDD connections, and VDD bypass capacitors must be connected as close as possible to the device.

9.3.5 PCB layout example

The PCB layouts for configuration 1 are shown here for QFN followed by WLCSP.

QFN PCB layout

For all available reference layouts, see the Reference Layout section on the **Downloads** tab for nPM1300 on www.nordicsemi.com.

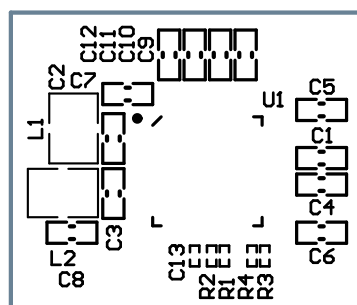


Figure 65: Top silkscreen layer QFN

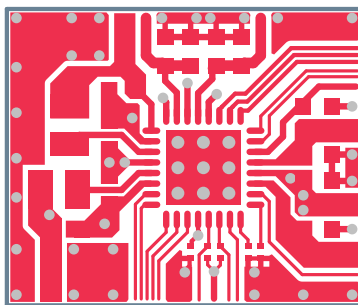


Figure 66: Top layer QFN

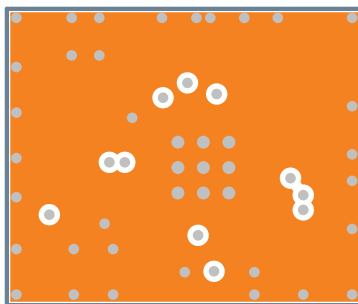


Figure 67: Mid layer 1 QFN

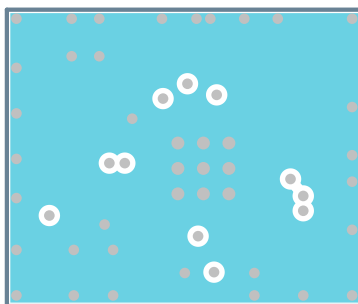


Figure 68: Mid layer 2 QFN

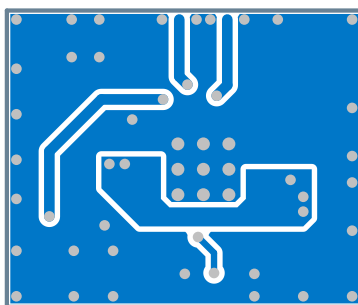


Figure 69: Bottom layer QFN

Note: No components on the bottom layer.

WLCSP PCB layout

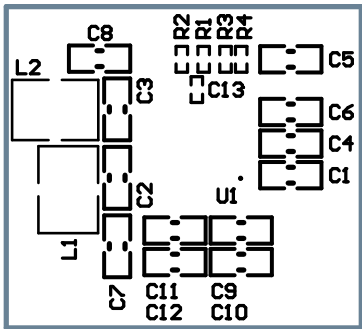


Figure 70: Top silkscreen layer WLCSP

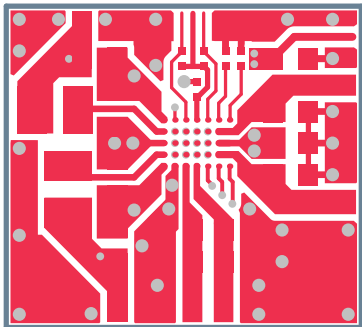


Figure 71: Top layer WLCSP

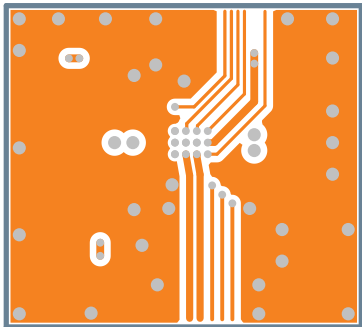


Figure 72: Mid layer 1 WLCSP

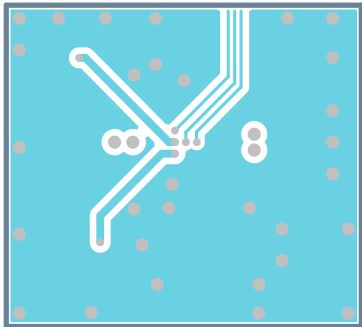


Figure 73: Mid layer 2 WLCSP

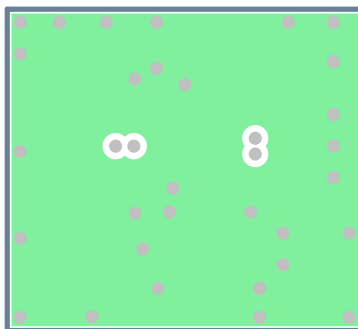


Figure 74: Mid layer 3 WLCSP

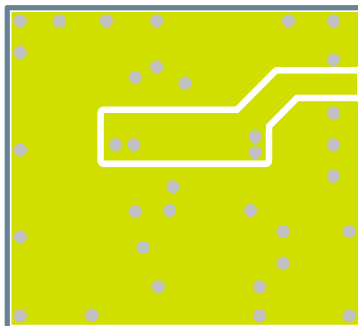


Figure 75: Mid layer 4 WLCSP

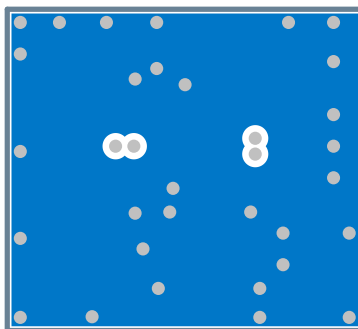


Figure 76: Bottom layer WLCSP

Note: No components are on the bottom layer.

10 Ordering information

This chapter contains information on IC marking, ordering codes, and container sizes.

10.1 IC marking

The nPM1300 PMIC package is marked as shown in the following figure.

N	P	M	1	3	0	0
<P	P>	<V	V>	<H>	<P>	
<Y	Y>	<W	W>	<L	L>	

Figure 77: IC marking

10.2 Box labels

The following figures define the box labels used for nPM1300.

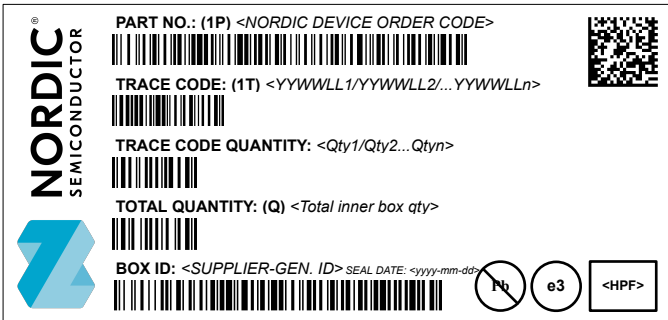


Figure 78: Inner box label

	
FROM: 	TO: 
PART NO.: (1P) <NORDIC DEVICE ORDER CODE>  <HPF>	
CUSTOMER PO NO: (K) <CUSTOMER P.O. NO>  Pb	
SALES ORDER NO: (14K) <Nordic sales order+Sales order line no.+ Delivery line no.> 	
SHIPMENT ID: (2K) <Nordic's shipment ID> 	
QUANTITY: (Q) <Total Quantity> 	
COUNTRY OF ORIGIN: (4L) <COO 2-char. Code> 	CARTON NO. x/n
DELIVERY NO: (9K) <Shipper's shipment no.> 	GROSS WEIGHT KGS 

Figure 79: Outer box label

10.3 Order code

The following tables define nPM1300 order codes and definitions.

n	P	M	1	3	0	0	-	<P	P>	<V	V>	-	<C	C>
---	---	---	---	---	---	---	---	----	----	----	----	---	----	----

Figure 80: Order code

Abbreviation	Definition and implemented codes
nPM13	nPM13 series product
00	Part code
<PP>	Package variant code
<VV>	Function variant code
<H><P><F>	Build code H - Hardware version code P - Production configuration code (production site, etc.) F - Firmware version code (only visible on shipping container label)
<YY><WW><LL>	Tracking code YY - Year code WW - Assembly week number LL - Wafer lot code
<CC>	Container code
eX	2 nd level Interconnect Symbol where value of X is based on J-STD-609

Table 42: Abbreviations

10.4 Code ranges and values

The following tables define nPM1300 code ranges and values.

<PP>	Package	Size (mm)	Pin/Ball count	Pitch (mm)
CA	CSP	3.1x2.4	35	0.419 0.440
QE	QFN	5.0x5.0	32	0.5

Table 43: Package variant codes

<VV>	Flash (kB)	RAM (kB)
AA	n/a	n/a

Table 44: Function variant codes

<H>	Description
[A . . Z]	Hardware version/revision identifier (incremental)

Table 45: Hardware version codes

<P>	Description
[0 . . 9]	Production device identifier (incremental)
[A . . Z]	Engineering device identifier (incremental)

Table 46: Production configuration codes

<F>	Description
[A . . N, P . . Z]	Version of preprogrammed firmware
[0]	Delivered without preprogrammed firmware

Table 47: Production version codes

<YY>	Description
[16 . . 99]	Production year: 2016 to 2099

Table 48: Year codes

<WW>	Description
[1 . . 52]	Week of production

Table 49: Week codes

<LL>	Description
[AA . . ZZ]	Wafer production lot identifier

Table 50: Lot codes

<CC>	Description
R7	7" Reel
R	13" Reel

Table 51: Container codes

10.5 Product options

The following tables define nPM1300 product options.

Order code	MOQ ¹	Comment
nPM1300-CAAA-R	7000 pcs	
nPM1300-CAAA-R7	1500 pcs	
nPM1300-QEAA-R	4000 pcs	
nPM1300-QEAA-R7	1500 pcs	

Table 52: nPM1300 order codes

Order code	Description
nPM1300-EK	Evaluation kit
nPM-FG	Fuel gauge board

Table 53: Development tools order code

¹ Minimum Ordering Quantity

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